IDEA League

Master of Science in Applied Geophysics
Research Thesis

From Particles to Waves

Towards coupling Dynamic Snow Modelling and Spectral Elements to detect Avalanche Breakoff in DAS Data.

Salome Anna Bachmann

Monday August 3, 2026

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Master of Science Thesis

for the degree of Master of Science in Applied Geophysics
by

Salome Anna Bachmann

Monday August 3, 2026

IDEA LEAGUE
JOINT MASTER'S IN APPLIED GEOPHYSICS

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Abstract

Avalanches are the most fatal natural hazard in Switzerland (SLF, 2025), and while processes like crack propagation, the transition between crack propagation regimes and the influence of snow and terrain parameters have been studied, there is little link between crack propagation and its seismic signature. In this master thesis, spectral element modelling in SALVUS will be used to simulate crack propagation in snow, as measured by distributed acoustic sensing (DAS). The spectral element modelling results will encompass simple simulations of sub-Rayleigh and Supershear propagation of uniform source wavelets in snow to determine the different types of waves in snow. Furthermore, the results from material point method simulations will be used to obtain seismic moment tensors and source wavelet shapes in an attempt to make the crack propagation model more physical.

The simple model results show that sub-Rayleigh and Supershear crack propagation exhibit distinct seismic signatures. In the sub-Rayleigh case, there is a precursor to the main energy front, which propagates at rupture speed. The precursor has been identified to be the S0 wave because of its speed and the plate wave speed, which is consistent with Herwijnen and Schweizer, 2011. Supershear crack propagation produces a single wavefront because the S0 wave speed is close to the rupture speed. Additionally, sub-Rayleigh regimes have lower energy than Supershear propagation regimes. Both regimes produce distinct rupture arrest signatures: a unidirectional fan in the Supershear case and a symmetric V-shape in the sub-Rayleigh case, which could aid in identifying the propagation regime from DAS field data. Because of the lower amplitude, the sub-Rayleigh onset of a crack that transitions to Supershear, as described by Trottet et al., 2022, may fall below the detection threshold of a DAS cable at the snow bottom, meaning field data could be misinterpreted as purely Supershear propagation.

The material-point method guided results further support the effects observed in the simple model. The main wavefront, which travels at sub-Rayleigh speed, shows similar signatures. The S0-precursor in the space-time plots can also be observed. The same is true for the v-shaped rupture arrest signatures in space-time, which were observed in the simple sub-Rayleigh model. Furthermore, the MPM-guided results show a quasi-static strain field, which is absent from the simple model. This quasi-static strain field is a precursor to the main wavefront and could be used for early warning of avalanche nucleation. The amplitude of this quasi-static strain field is higher than the S0 precursor, which means that it might be more easily detected in DAS field data. The two-wavefront structure and sub-Rayleigh arrest signature are confirmed in the five-layer MPM geometry, validating the simple model signatures as geometry-independent.

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Nomenclature

DAS	Distributed acoustic sensing
SALVUS	Spectral-element wave propagation software framework
SE	Spectral element
FE	Finite element
FEM	Finite element method
MPM	Material point method
DEM	Discrete element method
CZM	Cohesive zone model
PST	Propagation saw test
STF	Source time function
S_0	Symmetric Lamb-wave mode
A_0	Antisymmetric Lamb-wave mode
WL	Weak layer
ABC	Absorbing boundary conditions
FK	Frequency-wavenumber
DC	Direct Current. In Seismics, this stands for the asymmetry of a signal.
SNR	Signal to noise ratio
ρ	Density [$\mathrm{kg\,m^{-3}}$]
σ	Stress tensor [Pa]
u	Displacement field [m]

v Velocity field [$\mathrm{m\,s^{-1}}$]

f Body-force density [$\mathrm{N\,m^{-3}}$]

C Fourth-order stiffness tensor [Pa]

$\lambda,\ \mu$ Lamé parameters [Pa]

E Young's modulus [Pa]

$v_p,\ v_s$ P-wave and S-wave speeds [$\mathrm{m\,s^{-1}}$]

G Computational domain

∂G Boundary of the domain

$\hat{n}$ Outward unit normal vector on ∂G

w Test function in the weak form

$\bar{\tau}(x)$ Prescribed boundary traction

$\bar{u}(x),\ \bar{v}(x)$ Initial displacement and initial velocity

ϕ_j Basis (shape) function with index j

F_j Expansion coefficient of basis function ϕ_j

N Number of basis functions per expansion

M Number of quadrature points

W_i Quadrature weight at point i

x Physical coordinate [m]

Reference (element-local) coordinate

t_o Subfault onset time [s], e.g. $t_o = (x - x_{\mathrm{start}})/v_{\mathrm{rupture}}$

x_{start} Rupture starting position [m]

v_{rupture} Rupture propagation speed [$\mathrm{m\,s^{-1}}$]

ϵ Strain tensor (or strain quantity, context-dependent)

$\dot{\epsilon}$ Strain-rate quantity [$\mathrm{s^{-1}}$]

M_{ij} Moment tensor component [N m]

M_0 Scalar seismic moment [N m]

V_p Particle (or source-cell) volume [$\mathrm{m^3}$]

r_{mode} Mode-mix ratio (0: pure mode I, 1: pure mode II)

$\Delta_n,\ \Delta_t$ Normal and tangential separations in CZM [m]

δ_n, δ_t Characteristic normal and tangential opening scales [m]

$\delta_{n,c}$, $\delta_{s,c}$, δ_c Critical opening distances [m]

$\sigma_{\max}$, $\tau_{\max}$ Peak normal and shear cohesive tractions [Pa]

G_I, G_{II} Mode-I and mode-II fracture energies [J m^{-2}]

l_{ch} Characteristic Irwin length [m]

l_{cz} Cohesive-zone length [m]

$\bar{\epsilon}_{\mathrm{DAS}}$ Gauge-averaged DAS strain

$\dot{\bar{\epsilon}}_{\mathrm{DAS}}$ Gauge-averaged DAS strain rate

l Gauge length [m]

s_c Center position of DAS channel along the fiber [m]

$\mathbf{e}(s)$ Unit vector tangent to the fiber at arc-length position s

$w(s)$, w_k Continuous and discrete spatial weighting kernels

N_g Number of grid samples within the gauge window

k Wavenumber [rad m^{-1}]

ω Angular frequency [rad s^{-1}]

p, q Vertical wavenumber terms for P- and S-wave potentials

α, β Positive decay constants for evanescent wave terms

ν Poisson's ratio

M_{shear}, M_{elastic} Shear and elastic moduli [Pa]

c_p, c_s P-wave and S-wave speeds in plate theory [m s^{-1}]

c_p^{plate} Plate wave speed [m s^{-1}]

v_{ph}, v_g Phase velocity and group velocity [m s^{-1}]

h Slab half-thickness [m]

ϕ, ψ Scalar P-wave potential and vector S-wave potential

ξ Normalized opening in CZM

T_{peak} Peak cohesive traction [Pa]

ϕ_n, ϕ_t Normal and tangential work of separation [J m^{-2}]

f_M Ricker wavelet peak frequency [Hz]

α_{Mach} Mach cone angle

ϕ_{laser}	DAS laser phase delay [rad]
$\mathbf{F}$	Deformation gradient tensor

Chapter 1

Introduction

1.1 Avalanches & Avalanche Breakoff

Snow avalanches are a major natural hazard in mountainous regions, they endanger human life as well as infrastructure (Schweizer et al., 2003). The Swiss Institute for Snow and Avalanche Research has reported an average of 22 fatalities in Switzerland, which are caused by avalanches every year, with the number of non-fatal accidents being tenfold higher (SLF, 2025). These data show that avalanches are a significant threat, which is why it is crucial to understand how avalanches form and propagate in order to protect people and infrastructure better.

Avalanches are rapid masses of snow that move down mountain slopes due to gravity (Ancey, 2001). Snow avalanches can either be loose snow, which starts from a point in a cohesionless surface, wet snow, or snow slab avalanches (Schweizer et al., 2003). In this Master Thesis, the focus will be on dry snow slab avalanches, so mostly coherent masses of snow which contain no liquid water. Snow slab avalanches consist of a cohesive slab of snow (Schweizer et al., 2003) that is released because of loading at the surface, which triggers crack propagation in a weak layer (Gaume et al., 2017). Weak layers are characterized by the grain shape of snow, which leads to lower cohesion of those layers (Föhn et al., 1998).

Because avalanches are a major threat to human life in mountainous regions, which is why detecting the breakoff of the snow slab is important for hazard mitigation. Detecting the breakoff of the slab means that warnings can be issued in lower parts of the mountain for potential avalanches.

1.2 Seismic Methods for Avalanche Detection

In this master's thesis, the aim is to create an accurate spectral element model (SE) for crack propagation in snow. This is achieved through two approaches: pure spectral element (SE) simulations, which have simple source-time functions in order to set up the model and to understand the underlying physics. These are then combined with simulations which are driven by moment tensor source-time functions (STFs) which are extracted from MPM simulations. This helps towards understanding the effects of processes associated with crack propagation in snow on seismic signals, such as the ones measured by distributed acoustic sensing.

In this thesis, a fibre-optic cable installed on a slope is treated as a continuous seismic sensor in a numerical simulation, so that displacement can be observed along its full length rather than at only a few isolated stations (Parker et al., 2014). This is possible because DAS sends optical pulses through the fibre and records the backscattered light from every position along the cable (Shatalin et al., 1998). Ground motion near the cable, for example weak-layer failure or slab breakoff, changes the fibre strain, which appears as phase variations in the backscattered signal (Hartog, 2017).The phase variations are converted into dynamic strain along the cable (Hartog, 2017). Strain is the first spatial derivative of deformation, so deformation is the integral with respect to space of strain (Trabattoni et al., 2023).

Similar methods are already used to measure signals associated with avalanches. A pilot early-warning system using distributed acoustic sensing has been implemented in Norway, where avalanches are detected by their seismic signature (Turquet et al., 2024), meaning when the avalanche is already moving downhill. Such sensors are also located in Vallée de la Sionne in the Swiss Alps (Paitz et al., 2023). The detectability of avalanches by DAS has also been studied in the Swiss Alps by Paitz et al., 2023 and Edme et al., 2023, but the detectability of the slab breakoff itself has not been studied. The same is true for the seismic signatures associated with crack propagation leading to slab breakoff. This is important because the time between initial crack propagation and slab breakoff is crucial for detectability and warning, as it relates the time of the initial crack to the time where the slab breaks off depending on snow and terrain parameters.

1.3 Snowpack & Crack Propagation Modelling

A multitude of models have been previously used to study different processes in snow. The SNOWPACK and CROCUS models used in the Swiss and French Alps model the structure and evolution of snow (Brun et al., 1992; Bartelt and Lehning, 2002). These models were developed to understand how snow evolves under different conditions, for example a changing temperature gradient or water content and what consequences this has for the formation of weak layers where failure likely occurs and avalanches are triggered (Brun et al., 1992; Bartelt and Lehning, 2002). SNOWPACK and CROCUS are comprehensive models that take into account multiple governing equations and processes and the consequences they have for the structure of snow. While it is important to understand the influences of for example temperature gradient on the snow structure and weak layer formation and stability as a consequence, this is not the focus of the thesis. This is because these models are too complex and comprehensive for the purpose of trying to detect slab breakoff in DAS data as they model the entire snow structure. Furthermore, SNOWPACK and CROCUS do not model the seismic signal itself, which is what is needed to detect and understand crack propagation in DAS. Using SNOWPACK or CROCUS to investigate how different snowpack compositions and factors influence the DAS signal is beyond the scope of this thesis.

To model crack propagation in snow, multiple numerical modeling methods have been used. Cundall and Strack, 1979 developed the discrete element method (DEM) for granular media. This model discretizes the domain as spheres and disks, which interact with each other based on Newton's second law (Cundall and Strack, 1979). The interactions of particles are modeled by the governing equations and the resulting forces and displacements (Cundall and Strack, 1979). This method has been used by Bobillier et al., 2020; Bobillier et al., 2024; Mulak and Gaume, 2019 and many others to model crack propagation and its influencing factors. The main limitation of DEM is that the particle contacts and the corresponding equations need to be recalculated every time a grain contact is made or broken (Cundall and Strack, 1979). This means that for large, slope-scale simulations, this method becomes increasingly computationally expensive. In addition to the high computation cost, DEM also cannot account for anticrack propagation, the closure of cracks in this propagation mode is accounted for by manually removing elements from the mesh, which is not very efficient (Gaume et al., 2019).

The same is true for spectral finite element modelling (FE, SE or FEM), which is done using SALVUS in this Thesis. FEM has been used to model wave propagation in snow since the 1970s, a review of the different approaches has been published by Podolskiy et al., 2013. It is important to note that the FE approach used in this thesis is concerned with wave propagation in snow and not the mechanics of snow itself, so it is fundamentally different from other methods. The finite element modelling approach used in this thesis is described more in detail in 2.1. FE can also not account for anticrack propagation, similarly to DEM, mesh elements would need to be removed to account for anticrack propagation.

This problem is solved by using the material point method (MPM). In this method, the material is discretized in a set of Lagrangian points, which are used to track mass, deformation and momentum (Bardenhagen et al., 2000). Because the points are not fixed, an Eulerian background mesh is needed to calculate spatial derivatives (Bardenhagen et al., 2000). This method is not only more computationally efficient than DEM on the slope-scale, it can also account for anticrack behavior and strain softening of snow (Gaume et al., 2019). This is why the material point method will also be used in the thesis to investigate slab breakoff in DAS data on a large scale. The material point method has been used by Bergfeld et al., 2025, who modeled slope-scale experiments to understand the transition of crack propagation modes, Trottet et al., 2022 who modeled propagation saw tests to observe transitions in crack propagation speeds, as well as Gaume et al., 2019 and Gaume et al., 2018.

While MPM is very good at accounting for anticrack propagation, FE is more computationally efficient and accurate (Lian et al., 2012). This is relevant especially for large-scale simulations, such as the ones done in this Thesis. This is because in MPM, the discretization through particles causes the displacement of each particle to be tracked and the contact at each particle to be calculated (Zhang et al., 2023). Furthermore, MPM does not include boundary conditions in the way FE does (Coombs et al., 2025). Finite element is more efficient on the slope scale because the mesh is fixed, as described by (Zienkiewicz et al., 2025). This means that the degrees of freedom and thus the number of calculations which need to be done are much smaller than in the MPM case. This means that MPM is more accurate at representing snow as a medium with its porous character and anticrack behavior, but FE is more efficient at performing large-scale simulations. This is why data from MPM simulations will be used in FE on a slope scale in this thesis.

While it is important to understand the different modeling methods and their limitations, it is also important to understand the different types of crack propagation and what factors influence them. Understanding how fast a crack can propagate and which factors influence this is important for warning, as it relates the time of the initial crack to the time where the slab breaks off depending on snow and terrain parameters.

1.4 Crack Propagation in Snow

Two main types of crack propagation are believed to be important for weak layer failure: anticrack propagation and shear crack propagation (Bergfeld et al., 2025). There are three ways in which cracks can open up: mode I, where space is opened between the crack faces under tensile loading, and modes II and III, which represent parallel shearing and diagonal shearing (Heierli et al., 2003). In anticrack propagation, the weak layer consolidates under loading (C. A. Knight and N. C. Knight, 1972), while cracks propagate in a combination of modes I-III (Heierli et al., 2003). This means that anticrack propagation is different from modes I-III because it is a collapse rather than shearing or opening. A combination of the different crack propagation regimes has been observed by multiple studies such as Gaume et al., 2015; Mulak and Gaume, 2019. Gaume et al., 2018 modeled anticrack propagation induced by a propagation saw test (PST) using MPM, which showed that the scale of the simulations influences the propagation direction of the cracks. For smaller scale simulations, cracks propagate from top to bottom of a slope, whereas for slope-scale simulations, cracks propagate bottom to top, which means that remote triggering of avalanches is possible (Gaume et al., 2018). This is an important consideration for the thesis because it influences where the slab breakoff would be observed in terms of DAS cable location. Other models such as Mulak and Gaume, 2019; Gaume et al., 2019 also show that the crack propagation speed of anticracks is influenced by slope angle and curvature: concave slope areas have higher propagation speeds than convex areas of the slope (Gaume et al., 2019). Furthermore, there is a directionality in propagation speed, simulations by Gaume et al., 2019 show that cracks propagate twice as fast upslope than in the downslope direction, which is especially important when thinking about remote triggering. The slab breaks off when the slope angle exceeds the friction angle of the material (Gaume et al., 2019). Slope angle also influences which mode of crack propagation is more dominant (Rosendahl and Weißgraeber, 2020). In steeper terrains, shear modes (II and III) dominate, whereas in flatter terrain, mode I dominates the crack propagation (Rosendahl and Weißgraeber, 2020). The critical force, which is for example exerted by a skier, needed for weak layer failure is also dependent on slope angle, it decreases with increasing slope angle (Rosendahl and Weißgraeber, 2020). Snow parameters also have an influence on this critical force, as shown by simulations by Rosendahl and Weißgraeber, 2020: the thicker the slab or the higher the slab tensile strength, the better the force by the skier can be distributed on the weak layer and the more force is needed for failure.

Multiple experiments and numerical models have shown that cracks can propagate faster than the elastic wave speed in snow, so-called Supershear crack propagation (Trottet et al., 2022; Bobillier et al., 2024; Bergfeld et al., 2025). This is also linked to the crack propagation modes described above. Sub-Rayleigh crack propagation is usually a mixed-mode anticrack, whereas Supershear cracks propagate as pure mode II (Rosendahl and Weißgraeber, 2020; Trottet et al., 2022). Understanding what causes the transition from sub-Rayleigh to Supershear crack propagation is of high importance for the thesis, because it can give an estimate of whether Supershear crack propagation will occur under certain conditions, which leads to faster propagation of cracks and less time between loading and slab breakoff. In MPM models of PST, Trottet et al., 2022 observe that the occurrence of Supershear cracks is dependent on slope angle: in steeper terrains, Supershear propagation was observed. This can be explained by slab bending, because in steep terrains, bending of the slab is not limited to the collapse depth of the weak layer (Trottet et al., 2022). In in steeper terrain, the slab can bend more because of the terrain angle, the additional bending increases tensile stress in the slab, which leads to faster crack propagation (Trottet et al., 2022).These observations were also made by Bobillier et al., 2024 as well as Bergfeld et al., 2025. The release area of an avalanche is also tied to propagation speed: at Supershear propagation speed, there is less crack arrest and therefore larger release areas (Trottet et al., 2022; Bergfeld et al., 2025). This is crucial because larger release areas also mean a larger snow volume, which has more potential for damage.

Independent of crack propagation mode, different factors influence the speed and distance at which cracks can propagate in the weak layer before failure occurs (Gaume et al., 2018; Gaume et al., 2017; Gaume et al., 2015). As discussed above, slope angle and stiffness are important factors. Other factors include weak layer parameters: the weaker the weak layer,meaning the lower its Young's Modulus, the slower crack propagation, the thicker the weak layer, the lower the crack propagation distance before failure (Gaume et al., 2015). The depth of the slab also has an influence on propagation distance and speed: the thicker the slab, the larger the propagation distance and velocity (Gaume et al., 2015). How fast a slab will break off is also dependent on the critical crack length needed for failure (Gaume et al., 2017). If the snowpack is more stable, which is the case when the slab has a high elastic modulus or the weak layer is thicker, the critical crack length is higher (Gaume et al., 2017) because the snow can sustain longer cracks before it fails. A very important factor in material strength is loading rate: if loading is slow, sintering can occur (Mulak and Gaume, 2019). This means that new snow contacts can form, which can strengthen the material (Mulak and Gaume, 2019). This is in contrast to rapid loading, which happens for example due to a skier (Mulak and Gaume, 2019).

1.5 Aim of the Thesis

Because each of the variables described such as weak layer thickness, critical crack length and so on, alter the velocity, geometry, and energy release of the failure in the snowpack, they directly control the characteristics of the resulting seismic signature. In order to understand how crack propagation and slab breakoff would look in DAS data, numerical modeling of the wave propagation is done. This is done in SALVUS, which is a package that uses spectral elements to model seismic wavefields (Afanasiev et al., 2017).

The different snow processes described have different implications for the seismic wavefield, which will be modeled. Firstly, the loading which occurs before crack nucleation changes the stress state of the snowpack. The weak layer collapse and anticrack propagation produce volumetric compaction in the snowpack, this produces a seismic signal. The crack propagation, whether sub-Rayleigh or Supershear, creates a moving rupture front, which can be likened to a moving source of seismic waves, the cracks itself also create a seismic signal. Lastly, the slab breakoff and sliding creates a seismic signal because a large mass of snow moves downhill, which likely creates the largest seismic signal of all of the processes mentioned. The signals of the different crack propagation regimes described will be modeled in the thesis, which will help to understand how the different processes look in DAS data and whether they can be distinguished from each other.

Based on the above information, there are multiple considerations necessary in order to model DAS data of crack propagation accurately. The scale of the models is an important factor, because as mentioned, it influences the direction of crack propagation. This influences at which location along or relative to the DAS cable slab breakoff will occur, which changes the measured data. Furthermore, multiple parameters influence the time passed between initial cracking and slab breakoff, these are important to consider when interpreting data. The different modes of crack propagation are also important to consider, it is possible that a transition between sub-Rayleigh and Supershear crack propagation can be observed in the data. The loading rate can also be considered, because slower loading leads to stabilizing processes such as sintering.

This thesis will focus on the modelling of crack propagation, as recorded by DAS. Sub-Rayleigh as well as Supershear crack propagation will be modelled using different methods and models. The aim is to determine what the different cases of crack propagation will look like in DAS data. This is then relevant for detecting actual avalanches in real-world data. The main questions are what crack propagation in different regimes looks like, how it differs when different modeling methods are used, and how the results can be explained in terms of wave physics.

The main questions investigated in this thesis are:

1. What FE modelling framework is required to simulate DAS measurements of crack propagation in snow, and what are the key assumptions and limitations?
2. What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?
3. To what extent do the results change if source wavelets from a MPM simulation of a PST are injected?
4. How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?

Chapter 2

Theory

2.1 Spectral Element Wave Propagation Modeling

The model described in 3.2.1 is implemented using the finite-element (FE) method in SALVUS. The following section describes the physics and workflow used by SALVUS. Understanding the physics behind the modeling is relevant for understanding and interpreting the results, but also for appreciating the limitations of the model.

To discretize a domain, the domain is divided into a finite number of elements (Zienkiewicz et al., 2025). The number of mesh elements per wavelength is user-input, but needs to be small enough to satisfy the condition for the maximum wave frequency that needs to be resolved accurately. The boundaries of these elements describe the mesh, the mesh has a finite number of grid-points (edge points of the elements) (Zienkiewicz et al., 2025). The governing equations are determined at every element node, so for rectangular elements with nodes 1,2,3 and 4, there is the system $r^1 = \begin{pmatrix} r_1^1 \\ r_2^1 \\ r_3^1 \\ r_4^1 \end{pmatrix}$ for force and $u^1 = \begin{pmatrix} u_1^1 \\ u_2^1 \\ u_3^1 \\ u_4^1 \end{pmatrix}$ for displacement (Zienkiewicz et al., 2025). The residual or characteristic relationship of linear elastic elements is of the form $r^1 = f^1 - K^1 u^1$, where f are the nodal forces distributing load on the nodes, r are the residual forces and K is the stiffness matrix of a specific element (Zienkiewicz et al., 2025). In FE, the approximation of the wavefield by basis functions leads to a representation over the whole domain and not just at a set of points (Afanasiev et al., 2019). This means integrals can be calculated over surfaces or even volumes (Afanasiev et al., 2019).

Snow is a porous medium (Mellor, 1977), which is approximated as a solid in this Thesis. This is because SALVUS meshes need an isotropic, solid material and cannot account for pore spaces. Higher frequencies in the resulting data would be attenuated much more quickly in an actual porous medium (Koushik et al., 2024). The medium assumed here is an effective medium, because the wavelengths of interest are much larger than the pore spaces in snow, this is why such simplifications do not really have an effect on the results. The presence of air or water in snow is not neglected completely in the SALVUS simulations implemented however, because the material properties taken from Guillemot et al., 2021; Trottet et al., 2022 are based on field measurements of snow. This means that the velocities measured by Guillemot et al., 2021 are not of pure snow, but snow that contains air or water.

The governing equation considered in this case is the elastic wave equation, because real-life crack propagation emits P- and S-waves. The wave equation for a heterogenous elastic solid in the n-dimensional domain G is

$$\rho \partial_t^2 u = \nabla \cdot \sigma + f \tag{2.1}$$

(Afanasiev et al., 2019; Aki and Richards, 1981), with ρ as density of the material, u as displacement, σ as stress, and f as the force. Stress can be related to displacement using Hooke's law: $\sigma = C : \nabla u$, where C is a 4th order tensor which characterizes the stiffness of the medium (Afanasiev et al., 2019). The symbol : denotes contraction over adjacent indices (so contracting indices 1 and 3 with 2, for example). Combining 2.1 with Hooke's law, this leads to

$$\rho \partial_t^2 u = \nabla \cdot \frac{C}{\nabla u} + f, \tag{2.2}$$

which is the strong form of the linear elastic wave equation.

Boundary conditions, such as free surfaces, as well as initial conditions must be stated explicitly, they have the form $\sigma \cdot \hat{n}|_{\partial G} = \bar{\tau}(x), u|_{t=0} = \bar{u}(x), v|_{t=0} = \bar{v}(x)$ (Afanasiev et al., 2019). In this case, $\bar{\tau}(x), \bar{u}(x), \bar{v}(x)$ are the spatially variable distributions of traction, displacement and velocity (Afanasiev et al., 2019). The outward pointing normal vector on the boundary of domain G is described by $\hat{n}|_{\partial G}$.

FE methods use the weak form of the governing equations (Zienkiewicz et al., 2025). To obtain the weak form of the linear elastic wave equation in 2.1, the whole equation needs to be multiplied with a test function w, which is not time-dependent (Afanasiev et al., 2019). The weak form enforces the solution of the governing equation across the domain, instead of solving the equation everywhere, an integration over time leads to

$$\int_G \rho w \cdot \partial_t^2 u \, d^n x = \int_G w \cdot (\nabla \cdot \sigma) \, d^n x + \int_G w \cdot f \, d^n x \tag{2.3}$$

(Afanasiev et al., 2019). The divergence theorem relates the flux of a vector field through a closed surface to the divergence of the field in an enclosed volume: $\iiint_V (\nabla \cdot F) dV = \oiint_s (F \cdot \hat{n}) dS$, where F is the vector field. When this is applied to the first term on the right in 2.3, this leads to

$$\int_G \rho w \cdot \partial_t^2 u \, d^n x = \int_{\partial G} w \cdot (\sigma \cdot \hat{n}) \, d^{n-1} x - \int_G (\nabla w : \sigma) \, d^n x + \int_G w \cdot f \, d^n x \tag{2.4}$$

(Afanasiev et al., 2019). The first integral on the left side of 2.4 is the boundary integral, which changes depending on the boundary conditions implemented. The integral on the right-hand side is the mass term, the second on the left-hand side is the stiffness term, and the last on the left-hand side is the external forcing term, all in 2.4. Equation 2.4 can be compacted to

$$(\rho \partial_t^2 u, w)_G - \langle \sigma \cdot \hat{n}, w \rangle_{\partial G} + (\sigma, \nabla w)_G = (f, w)_G, \tag{2.5}$$

using the notations $(a, b)_G = \int_g a \cdot b d^n x, \langle a, b \rangle_{\partial G} = \int_{\partial G} a \cdot b d^{n-1} x$ as well as Hooke's law (Afanasiev et al., 2019). This can be even more compacted by describing a mathematical model I with M for mass term, B for boundary term, S for stiffness and F for forcing term:

$$I_I(M) - I_I(B) + I_I(S) = I_I(F) = \sum_{P \in \{M,B,S,F\}} I_I(P) = 0. \tag{2.6}$$

Physically, this means that the entire model can be described as a combination of the different terms, which is very useful, especially if one of the terms is negligible (Afanasiev et al., 2019). This means that the same codebase can handle acoustic, elastic and viscoelastic media

(Afanasiev et al., 2019), because different terms of equation 2.6 can be set to zero for different materials.

The dependence on the reference coordinate system is described by E, which is a model containing the physical coordinate to reference coordinate transformation, in other words, the determinant of the Jacobian matrix and its inverse (Afanasiev et al., 2019). Using basis B of the polynomial functions of the approximation V_h, which are piecewise per-element because of possible discontinuities, the abstract form in 2.6 becomes

$$\sum_{P \in \{M,B,S,F\}} I_I(P, B(E)) = 0 \tag{2.7}$$

(Afanasiev et al., 2019). This means that fields such as displacement and velocity are calculated using the abstract approximation in equation 2.7. From these fields, the data from the model in section 3.2.1 is obtained and plotted.

2.2 Cohesive Zone Physics

The cohesive zone model (CZM) framework utilized in this study is based on the mixed-mode formulation by Mahajan and Joshi, 2008. This model relies on a potential function to describe the relationship between tractions and separations under mixed-mode conditions. With $\Delta = (\Delta_n, \Delta_t)$ denoting normal and tangential separations, the opening potential is defined as

$$\phi(\Delta) = \phi_n \exp\left(-\frac{\Delta_n}{\delta_n}\right) \left\{ \left[1 + \frac{\Delta_n}{\delta_n}\right] + q \left[1 - \exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right)\right] \right\},$$

from which normal and tangential tractions are derived. Peak tractions occur at $\Delta_n = \delta_n$ and $|\Delta_t| = \delta_t/\sqrt{2}$. The normal and shear work of separation relate to the peak strengths via

$$\phi_n = e\,\sigma_{\max}\,\delta_n, \qquad \phi_t = \sqrt{\tfrac{e}{2}}\,\tau_{\max}\,\delta_t.$$

The critical openings for pure normal (Mode I) and shear (Mode II) modes are computed from the corresponding fracture energies and peak tractions as

$$\delta_{n,c} = \frac{2G_I}{\sigma_n}, \qquad \delta_{s,c} = \frac{2G_{II}}{\tau_s}.$$

For mixed-mode behavior, these are combined into a single mixed-mode critical opening:

$$\delta_c = \sqrt{(1 - r_{\text{mode}})\,\delta_{n,c}^2 + r_{\text{mode}}\,\delta_{s,c}^2},$$

where $r_{\text{mode}} \in [0, 1]$ controls the mode mix (where 0 represents pure Mode I and 1 represents pure Mode II).

To describe the cohesive weakening, a linear softening law is adopted. The traction T depends on a normalized opening parameter ξ, defined as the ratio of current separation to the critical opening. The traction follows

$$T(\xi) = T_{\text{peak}}\,(1 - \xi) \quad \text{for } 0 \le \xi \le 1, \qquad T(\xi) = 0 \text{ for } \xi > 1,$$

where T_{peak} is the peak cohesive traction.

The spatial extent over which this cohesive weakening is active ahead of the crack tip is related to the characteristic Irwin length for shear, calculated as $l_{ch} = \frac{E_{\text{eff}} G_{II}}{\tau_s^2}$ (Kirchner et al., 2002).

The cohesive zone length is theoretically estimated as $l_{cz} = \frac{\pi}{8} l_{ch}$, this is the zone where damage occurs before the crack (Mahajan and Joshi, 2008). Additionally, the bulk elastic behavior of the surrounding medium is defined by Lamé parameters, which can be derived from seismic wave speeds (v_p, v_s) and density (ρ) following Aki and Richards, 1981:

$$\mu = \rho v_s^2, \qquad \lambda = \rho(v_p^2 - 2v_s^2).$$

Young's modulus is subsequently computed as $E = \dfrac{\mu(3\lambda + 2\mu)}{\lambda + \mu}$ (Aki and Richards, 1981).

2.3 Wave Propagation Physics

The snow layer in this thesis acts as a thin layer bounded by a free surface above (snow-air interface) and a rigid boundary at the bottom (either domain boundary or bedrock interface). This geometry supports the appearance of Lamb waves: Lamb waves are elastic guided waves that occur in plates (Su and Ye, 2009) and their amplitude is shaped by layer boundaries (Lamb, 1917). They are guided by the boundaries of the layer or media in which they propagate and have a symmetric (S0) and an antiymmetric (A0) mode (Su and Ye, 2009). The symmetric mode dominates the in-plate motion, while the antisymmetric mode dominates the vertical (out of plane motion) (Su and Ye, 2009). A sketch of the propagation of these two modes, taken from Carta et al., 2023, can be seen in appendix G. Both modes are observed in the simulation results shown in 4.1, and to interpret and understand the results correctly, their properties such as speeds, dispersion relations and excitation need to be understood. This section derives the dispersion relations of both modes and explains the depth-dependent amplitude structure that determines how each mode decays between the crack plane and the snow bottom.

2.3.1 Potentials and Wave Equations

Starting from the equation of motion for a homogeneous elastic solid used in SALVUS,

$$\mu\nabla^2 u + (\lambda + \mu)\nabla(\nabla \cdot u) = \rho \frac{\partial^2 u}{\partial t^2}, \tag{2.8}$$

the displacement field is decomposed into scalar (ϕ, P-wave) and vector (ψ, S-wave) potentials via the Helmholtz decomposition (J. Achenbach, 1973):

$$u_x = \frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial y}, \qquad u_y = \frac{\partial \phi}{\partial y} + \frac{\partial \psi}{\partial x}. \tag{2.9}$$

Substituting into equation 2.8 decouples the problem into separate wave equations for ϕ and ψ, each propagating at the P-wave speed c_p and S-wave speed c_s respectively. Assuming harmonic plane wave solutions of the form $\phi = f(y)\, e^{i(kx-\omega t)}$ and $\psi = g(y)\, e^{i(kx-\omega t)}$, the depth functions satisfy

$$\frac{d^2 f}{dy^2} + p^2 f = 0, \qquad \frac{d^2 g}{dy^2} + q^2 g = 0, \tag{2.10}$$

with $p^2 = \omega^2/c_p^2 - k^2$ and $q^2 = \omega^2/c_s^2 - k^2$. The general solutions are sinusoidal in y (depth) when $p^2, q^2 > 0$ (propagating through the layer thickness), and hyperbolic when p^2 or $q^2 < 0$ (evanescent decay with depth). The evanescent case applies to wave components whose horizontal phase velocity is slower than the bulk wave speeds, which is the A0 mode at low frequency-thickness product, like it is the case in this thesis and explains why the A0 wave train decays more rapidly between crack depth and snow bottom than the faster S0 mode (Graff, 1975).

2.3.2 Boundary Conditions and the Lamb Dispersion Relations

At the snow-air interface, the large impedance contrast produces a stress-free boundary: normal stress $\sigma_{zz} = 0$ and shear stress $\sigma_{xz} = 0$ at the top and bottom faces of the slab (Igel, 2016). Applying these conditions to the general potentials yields two types of solutions depending on the symmetry of the displacement field across the slab midplane, so across the horizontal axis of the slab.

For the symmetric mode (S0), horizontal displacement is even in z and vertical displacement is odd. Setting the determinant of the resulting 2×2 system to zero gives the S0 dispersion relation:

$$\frac{\tan(qh)}{\tan(ph)} = -\frac{4k^2pq}{(q^2 - k^2)^2}, \tag{2.11}$$

where h is the half-thickness of the slab (Su and Ye, 2009).

For the antisymmetric mode (A0), the displacement symmetries are reversed: vertical displacement is even in z and horizontal displacement is odd. The A0 dispersion relation is:

$$\frac{\tan(ph)}{\tan(qh)} = \frac{4k^2pq}{(q^2 - k^2)^2}. \tag{2.12}$$

Both relations give phase velocity as function of frequency, meaning the two modes are dispersive: different frequency components propagate at different speeds (Aki and Richards, 1981). At low frequency-thickness product ($f \cdot h \ll 1$, as in this case, A0 is strongly dispersive and slow, while S0 is nearly non-dispersive and converges to the plate wave speed with $c_p^{\text{plate}} = 2c_s\sqrt{1 - \frac{v_s^2}{v_p^2}}$ (Rose, 2014).

2.3.3 Mode Symmetry and Depth Dependence

Because the crack source in the slab middle or weak layer acts as an asymmetric load applied below the slab midplane, it couples energy into both modes simultaneously (Rose, 2014; J. Achenbach, 1973). The two modes have distinct depth profiles: S0 carries horizontal strain or velocity that is nearly uniform through the slab thickness, while A0 horizontal strain / velocity is proportional to distance from the neutral axis and is therefore maximum at the plate boundaries (top and bottom of snow in this case)(Graff, 1975).

2.4 Static and Dynamic Components of Seismic Sources

A moment tensor source, such as the sources used in this thesis, has two contributions to the wavefield (Aki and Richards, 1981). There is a dynamic component, which propagates away from the source as waves, and the static component, this component represents the permanent deformation field after the sources have stopped firing (Freund and Freud, 1998; Zhu and Rivera, 2002; Archambeau, 1968).

The static displacement field of a crack with half length a, in an elastic medium decays with $\frac{1}{r^2}$ with distance r from the crack tip, and produces a permanent strain in the surrounding medium proportional to the total slip δ and the shear modulus μ (Freund and Freud, 1998):

$$\varepsilon_{xx}^{\text{static}} \propto \frac{\mu\,\delta\,a}{r^2}. \tag{2.13}$$

This strain accumulates as the crack grows in length, but does not propagate away from the crack location, which means it stays in the near-field (Freund and Freud, 1998). This occurs because the static field represents the new equilibrium stress state of the medium around the opened crack, rather than a transient disturbance propagating through it.

The control on whether a static field occurs is the shape of the STF. The direct current (DC) component of the STF, which is its integral over time, has to be non-zero in order for there to be a static component (Aki and Richards, 1981; Zhu and Rivera, 2002). For a Ricker wavelet source for example, the integral over time is zero because the positive and negative lobes are equal in area and they cancel out. For more irregular shaped source wavelets, as described by Kagan, 1987; Aki and Richards, 1981, like for example the MPM STFs described in the next section, this is not true and their integral is non-zero.

2.5 From Particles to Waves: Translating MPM to SE

The material point simulations of the PST performed by (Gaume et al., 2018; Trottet et al., 2022) result in the recorded stresses in the model according to Gaume et al., 2019. The output of the model is particle velocity, the stress tensor, strain and the J-value, which distinguishes between elastic and plastic deformation. The J-value is used as a filter to remove any irreversible or non-elastic deformation such as the weak layer collapse. To accurately construct the equivalent seismic source, only the elastic stress field outside the inelastic plastic zone is retained for calculating the moment tensors. The inelastic deformation serves as the only physical driver for the elastic wavefield in the bulk medium. This also ensures that any signals emitted from the crack are not taken into account, because inputting them in SALVUS would mean that waves emitted by the crack are used as sources.

To translate the continuum source to something which can be input to SALVUS, the relationship between stress and strain must be defined. The deformation of the material is described by the infinitesimal strain tensor ϵ_{ij}, which is linearly related to the Cauchy stress tensor σ_{ij} via the fourth-order stiffness tensor C_{ijkl} (Holzapfel, 2010). This relationship, known as generalized Hooke's law, is expressed as:

$$\sigma_{ij} = C_{ijkl}\epsilon_{ij}. \tag{2.14}$$

For an isotropic, linear elastic material, such as the snow in this case, the stiffness tensor simplifies and can be described by the Lamé-parameters. In the xx-direction this is:

$$\sigma_{xx} = (\lambda + 2\mu)\epsilon_{xx} + \lambda\epsilon_{yy} \tag{2.15}$$

(Aki and Richards, 1981). When scaling this with the particle volume, the individual moment tensor components can be obtained using

$$M_{ij} = \Delta\sigma_{ij} vol_p. \tag{2.16}$$

The scaling with the particle volume is done because in a small material volume, such as a snow particle, the stress drop ($\Delta\sigma_{ij}$) is the same as a small seismic point source (Backus and Mulcahy, 1976). This is why multiplying the stress tensor with the particle volume yields the moment tensor component.

These moment tensor components can then be input into SALVUS as custom source time functions. The waveforms are obtained via the amplitude of the different moment tensor components at different times. Each moment tensor (in all three directions of the 2D model) is injected as a point source. This means that every point source represents the localized movement of a few particles in the snowpack.

The visual representation of this workflow can be seen in the methods section in figure 3.4.

2.6 Simulation of DAS Data

To compare the results obtained in the SALVUS simulations to field DAS data, the numerical outputs have to be transformed. This is because field DAS measurements, the output are the fields averaged over the gauge length (Lindsey and Martin, 2021) rather than output at every point along the cable. The backscattered delay of a laser pulse along a straight cable is proportional to the integrated strain tensor components along the gauge length (Noe, 2025). The relationship between the deformation tensor F and the phase delay of the laser pulse is described by:

$$\phi_{laser}(t) \propto \int_{-\frac{1}{2}}^{\frac{1}{2}} |I + F(s,t)e(s)|ds, \tag{2.17}$$

according to Fichtner et al., 2022. In this equation, I is the identity matrix and e is the local unit tangent vector at the arc length s along the DAS fibre in 3D (Noe, 2025).

Because the deformations of interest are small, meaning that $F << I$, equation 2.17 can be linearized. This leads to

$$\phi_{laser} \propto \int_{-\frac{1}{2}}^{\frac{1}{2}} e^T(s)\epsilon(s,t)e(s)ds, \tag{2.18}$$

where the strain tensor is $\epsilon = \frac{1}{2}(F + F^T)$ (Fichtner et al., 2022).

Noe, 2025 state that DAS interrogators scale phase delay measurements to convert them to strain, which leads to the vanishing of the proportionality. Equation 2.18 becomes

$$\bar{\epsilon}_{DAS}(s_c,t) = \frac{1}{l}\int_{s_c-\frac{1}{2}}^{s_c+\frac{1}{2}} e^T(x)\epsilon(s,t)e(s)ds \tag{2.19}$$

(Noe, 2025). This means that the measured strain is averaged over the gauge length l (Noe, 2025). Physically, this means that the strain field recorded by DAS is smoothed because of the averaging over the gauge length.

2.6.1 Noise Floor Assessment

To assess the signal-to-noise ratio in the DAS data simulated in this thesis, the amplitude spectral density of the Silixa iDAS is used. To use this for comparison with the values obtained in the space-time plots generated as described in the previous subsection, the value from the factsheet (Silixa, 2022) needs to be converted from the frequency- to the time domain:

$$RMS_{noise} = \frac{2p\epsilon}{\sqrt{Hz}}\sqrt{\Delta f}, \tag{2.20}$$

where Δf is the frequency bandwidth of the DAS signal (Parker et al., 2014). This can then be compared to peak simulated strain (from the colorbar in the results) via the signal-to-noise ratio:

$$SNR = \frac{\epsilon_{peak}}{RMS_{noise}}. \tag{2.21}$$

If the signal-to-noise ratio SNR is 1, the signal is easily detectable, if it is ≤ 1, then the signal will be buried in the noise (Gu et al., 2024).

Chapter 3

Methodology

3.1 Model Setups

3.1.1 Simple Model

A 2-layer domain spanning 400m in the x-direction and 3m in the y-direction was set up. The domain consists of a 1.5m snow layer and a 1.5m air layer, the parameters for this medium are listed in appendix D, table D as well as in the sketch of the domain in figure 3.1. The medium extent of 400m was chosen to ensure that sources can be placed far away from the absorbing boundaries, but to still have a source line which would be physical as an avalanche release zone in terms of size. Furthermore, a large domain combined with a long simulation time ensures that both the near- and the far-field of the wave can be observed, which is important for interpretation of field DAS data, because the DAS cable might be located in either field. In the simple case, a series of SALVUS moment tensor sources are set up with a certain distance, they span from 30m to 270m with a spacing of 10cm, which results in a rupture length of 240m and 2400 sources. The small source spacing was chosen to mimic a moving, continuous source (Madariaga, 1976), like a crack in snow, accurately. The model domain setup can be seen in figure 3.1. The air layer is shown in light blue, the snow layer in dark blue. Because the sources have 10cm spacing, the source extent instead of individual sources is shown in red. The time delay with which the sources fire is dependent on the Sub-Rayleigh or Supershear case which is implemented, the bounds for the crack propagation velocity are taken from Trottet et al., 2022, they are 0.6 or 1.6 times the shear wave velocity for sub-Rayleigh and Supershear cases respectively.

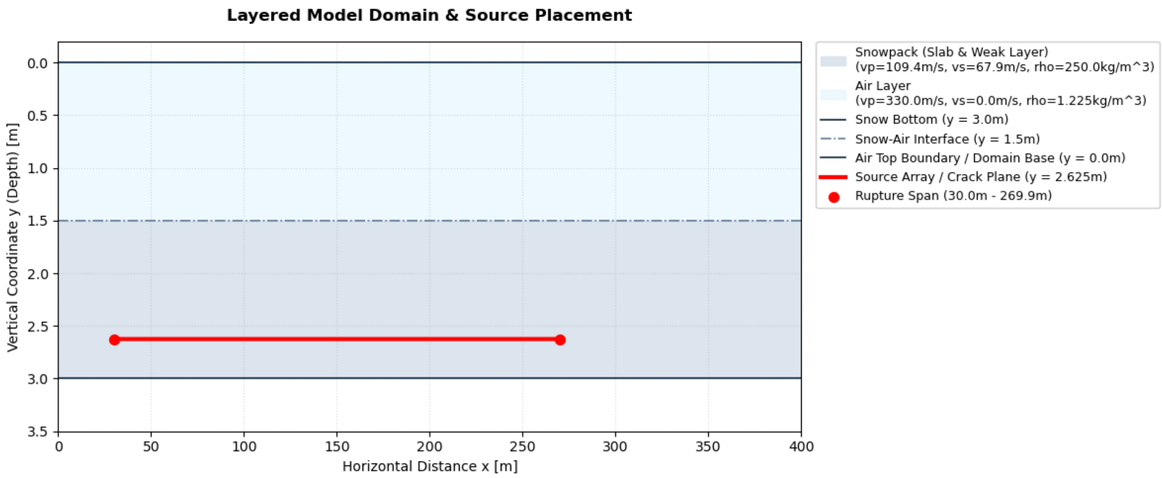


Figure 3.1: Layered model setup. Air layer in light blue, snow layer in darker blue. Source line extent in red. Model properties used can be seen in legend, but also appendix D

The bedrock or soil is not considered in both models. This is because preliminary tests (Bachmann, 2026) of a 3-layered medium have shown that it acts as a reflector of waves. This is because of the high impedance contrast between snow and bedrock, which causes waves to be reflected almost entirely at the interface (Aki and Richards, 1981). For computational efficiency, the bedrock layer was thus neglected and the bottom of the snow was made a regular boundary, in other words, reflecting, to mimic the presence of a layer below the snow. A reflecting bottom boundary is physically equivalent to a rigid half-space with infinite impedance contrast, which is a reasonable approximation given that snow P-wave speeds (109 m/s) are substantially lower than typical bedrock values (1500-3000 m/s), the same is true for the contrast between the S-waves in the medium as well (Aki and Richards, 1981). The top as well as both side boundaries of the model are set to be absorbing boundaries, which means that waves that reach these boundaries are absorbed and not reflected back into the model. This is done so that the model accurately represents the physical behavior of wave propagation in a much larger domain, which is the reality of field DAS data. The absorbing boundaries were set to be 3.5 times the maximum wavelength present in the model, this is according to the numerical requirements of the solver (Afanasiev et al., 2017).

3.1.2 MPM Model

To represent the results of the more complex source time functions obtained from the material point method simulations, the same layered model as Trottet et al., 2022 was implemented. This model is more complex than the two-layer case described in order to have results that represent avalanche physics better. The material properties, taken from Trottet et al., 2022, are listed in appendix D, table D. The p- and S-wave velocities are computed from the material properties via the shear and elastic moduli:

$$M_{shear} = \frac{E}{2(1+\nu)}, \quad M_{elastic} = \frac{E}{(1-2\nu)(1+\nu)} \tag{3.1}$$

(Aki and Richards, 1981). From these, the wave velocities can be derived:

$$v_p = \sqrt{\frac{M_{elastic}}{\rho}}, \quad v_s = \sqrt{\frac{M_{shear}}{\rho}} \tag{3.2}$$

(Aki and Richards, 1981).

The parameters obtained by equations 3.2 are vp=127.10 m/s, vs=67.94m/s for the slab, in the weak layer the wave speeds calculated are vp=116.02m/s, vs=62.02m/s.

The 5-layer model consists of the bed layer, a weak layer and a snow slab layer. In order to represent a physical snowpack with a weak layer buried in snow (Schweizer et al., 2003), the weak layer here is in the middle of the slab layer. Contrary to real-world snow conditions (Bartelt and Lehning, 2002), the slab above and below the weak layer used in the simulation has uniform medium properties. To account for the absorbing boundaries which are implemented at the top and sides, the model domain was extended. At the sides, this was done by extending the model extent to accommodate the abcs. At the top however, an air layer was added which accounts for the abcs in thickness. This was done to keep the original thickness of the snow slab layer described by Trottet et al., 2022, but still have a boundary which leads to no reflections from the top of the domain. In other words, adding an air layer above the slab allows the snow-air free-surface boundary condition to remain at the correct physical height while the absorbing boundaries occupy the air layer above, which mimics a domain with infinite extent in y while also avoiding any interaction between the ABC and the elastic snow medium (Afanasiev et al., 2017). Although the bed (rock) layer was omitted in the simple model for computational efficiency, it was kept in the MPM domain setup. This was done to facilitate comparison between the output of the MPM model and the MPM-guided SE model. The setup can be seen in figure 3.2.

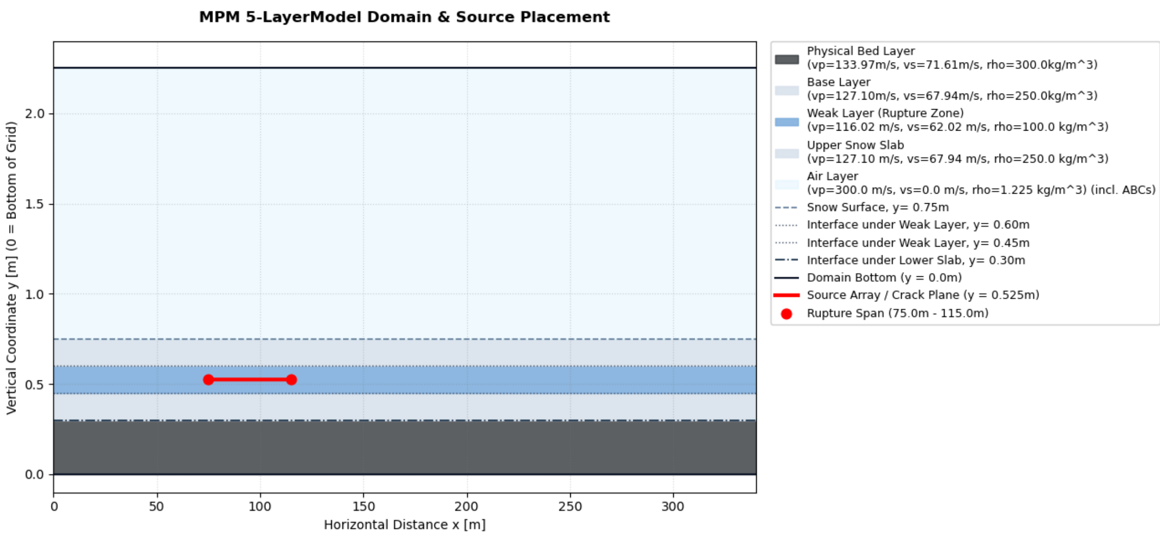


Figure 3.2: Layered model setup for MPM. Air layer in light blue, snow slab in light grey, weak layer dark grey and bedrock in blue grey. Source line extent in red. Model properties used can be seen in legend, but also appendix D

3.2 Source Generation for Simple and CZM Model

All sources generated use SALVUS' custom source time function option. For the simple model and the cohesive zone model, this is done by writing a file with the source wavelets in them. The source wavelets have ricker shape with peak frequency f_m according to Aki and Richards, 1981; Graff, 1975; *Ricker Wavelet* 2021:

$$f(t) = (1 - 2\pi^2 f_M^2 t^2) e^{-\pi^2 f_M^2 t^2}. \tag{3.3}$$

The ricker wavelet was chosen over other source time functions in SALVUS because its frequency content is controlled at a single value, the central frequency (Aki and Richards, 1981).

The workflow of the code, including how the source speed is set up can be seen in the flowchart in 3.3.

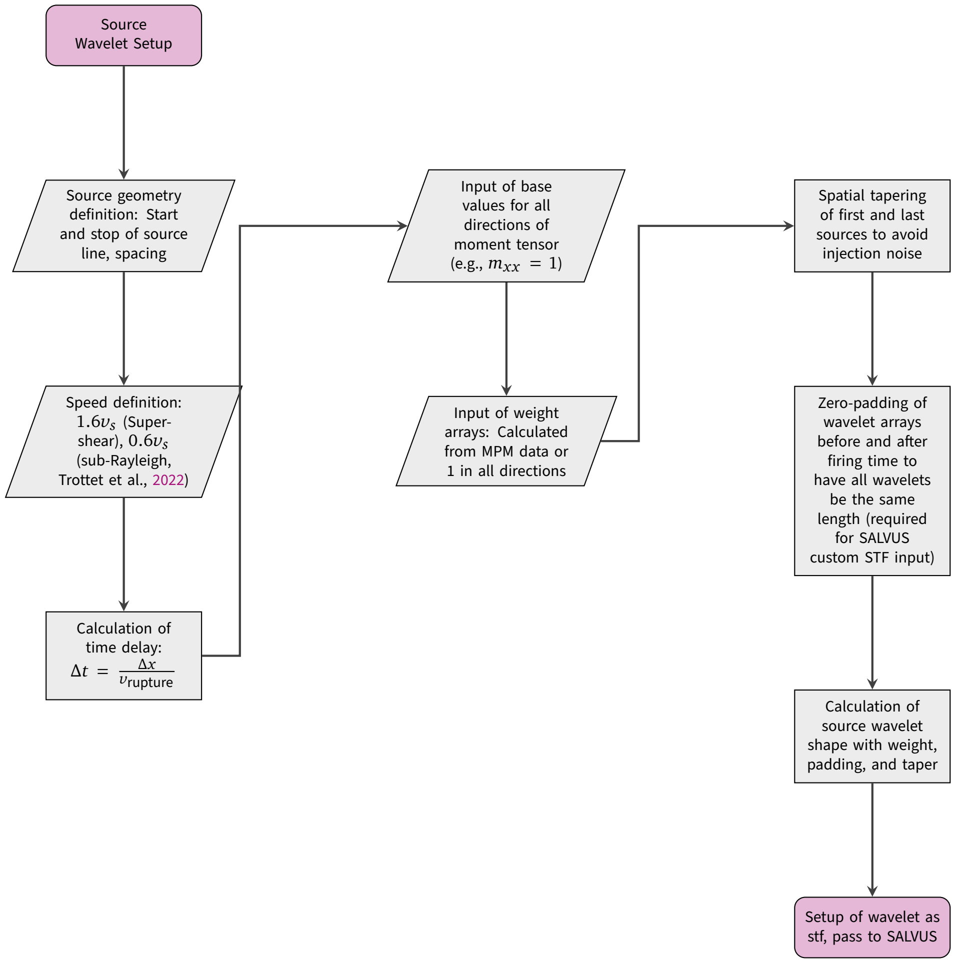


Figure 3.3: Operational workflow for the SALVUS source time function (STF) setup.

3.2.1 Cohesive Zone Model

A graphical representation of the model setup and the equations used can be found in appendix B.

The rupture model used in the simulations is implemented as a mixed-mode Cohesive Zone Model (CZM) following Mahajan and Joshi, 2008. This model was chosen because there is a narrow band ahead of the fracture zone, the cohesive zone, where the material is already weakened (Mahajan and Joshi, 2008). This means that prior to the failure of a point in the material, there is already weakening of the material. This seems to be reasonable for representing crack propagation in snow, since failure occurs as a consequence of loading and is not spontaneous. The computational domain extends 400 m in the horizontal (x) direction and 3 m in the vertical (y) direction and contains a 1.5 m snow layer over a 1.5 m air layer, this is the same as in figure 3.1. Material parameters for the snow layer are taken from Guillemot et al., 2021. The mixed-mode behavior is implemented because in reality, crack propagation can involve both normal and shear displacement.

Each activated subfault is represented as a point source in the SALVUS source array. The kinematic moment formulation serves as the numerical link to the underlying cohesive zone model through a strict energy balance. Within the CZM, the total strain energy released during fracture is split between the fracture energy (G_I or G_{II}) required to break down the cohesive zone, and the remaining energy radiated as elastic waves. By prescribing x_{slip} across a finite grid cell width Δx, the resulting scalar seismic moment M_0 mathematically captures the radiated energy that propagates away from the process zone into the bulk medium as a seismic wavefield. The scalar seismic moment per subfault is

$$M_0 = \mu\,\delta_c\,\Delta x, \tag{3.4}$$

where μ is the shear modulus of the weak layer, δ_c is the critical slip distance, and Δx is the subfault spacing.

To take the mixed-mode mechanics of the process zone into account, the individual components of the moment tensor are scaled by the mode-mix ratio r_{mode}, which can be user input:

$$m_{xx} = 0, \quad m_{xy} = M_0\,r_{\text{mode}}, \quad m_{yy} = -(1 - r_{\text{mode}})M_0.$$

Consequently, at the extremes of $r_{\text{mode}} = 0$ or 1, the unaligned component of the moment tensor drops to zero, perfectly matching the radiation patterns of pure opening or pure shear failure. All subfault sources are driven by an identical central frequency, source wavelet, and amplitude scaling factor, thereby reconstructing the discrete subfault array into a highly coherent, continuously moving source that mimics a smooth, physical crack tip rupture throughout the wave propagation mesh.

Key parameters and their provenance used in the simulations are summarized in appendix D, table D.

3.3 Source Generation for MPM

For the MPM results that are translated to FE, the moment tensor components and source wavelet shapes are obtained as described in section 2.5. These are then loaded and translated into individual components, which can be passed to SALVUS. The time delay previously described is already contained in the MPM data. A visual representation for the workflow with the MPM source wavelets is:

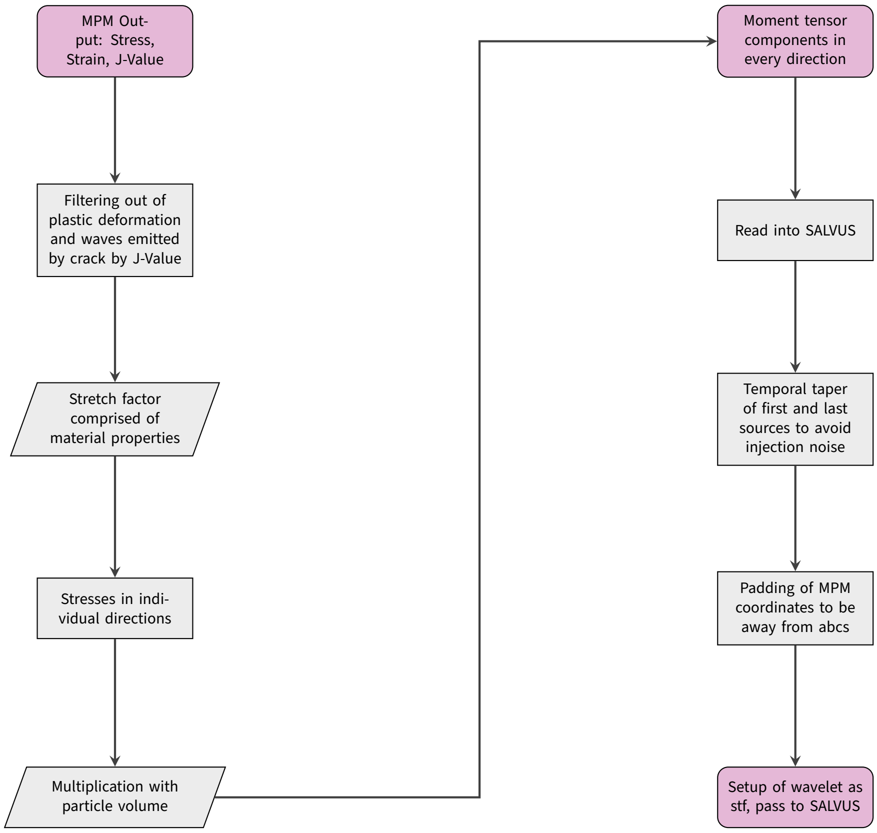
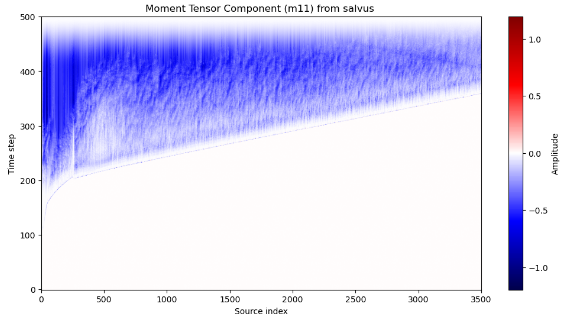
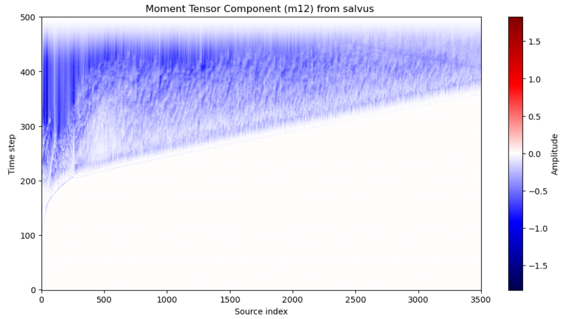


Figure 3.4: Workflow for conversion of MPM data and input as SALVUS STF.

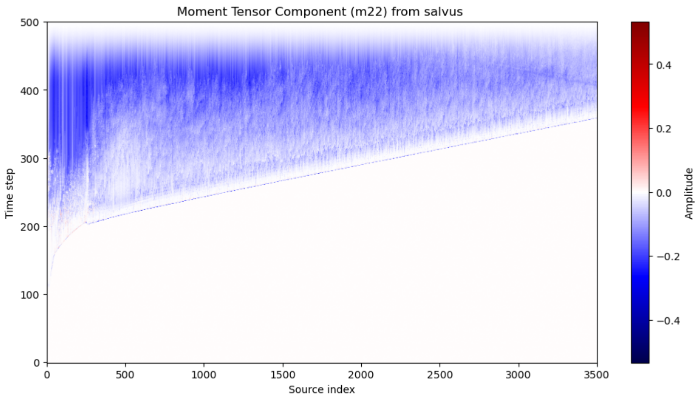
The directionality and amplitude of the moment tensor components extracted over all sources can be seen in figure 3.5. The colorbar shows the amplitude of the individual components. This is important because the source components show the predominant directions of the sources, which is important for the interpretation of the wavefield. The diagonal boundary separating zero (white, lower triangle) from active values (upper region) marks the crack propagation front. The moment tensor components are injected into SALVUS as custom source time functions, plotted as a function of source index (horizontal axis, corresponding to spatial position along the crack) and MPM timestep (vertical axis).



(a) Moment tensor component M_{11} (horizontal stress).



(b) Moment tensor component M_{12} (shear stress).



(c) Moment tensor component M_{22} (vertical stress).

Figure 3.5: Moment tensor components injected into SALVUS as custom source time functions, plotted as a function of source index (horizontal axis, corresponding to spatial position along the crack) and MPM timestep (vertical axis). The colorbar shows amplitude (blue: negative amplitude, red: positive amplitude). The diagonal boundary separating zero (blue, lower triangle) from active values (upper region) marks the crack propagation front.

From all figures 3.5a, 3.5b and 3.5c it can be seen that the moment tensor has a predominantly negative direction that does not bounce back to neutral. To further investigate this, the cumulative integral of the moment tensor integrated in respect to space was calculated:

$$I_{ij}^{\text{tot}}(t) = \int_0^t M_{ij}^{\text{tot}}(\tau)\,\mathrm{d}\tau, \tag{3.5}$$

which if nonzero, means that there is a residual, non-reversible deformation. This results in figure 3.6, which supports the trend seen in figure 3.5, that there is a negative permanent deformation in respect to y after the sources have stopped firing.

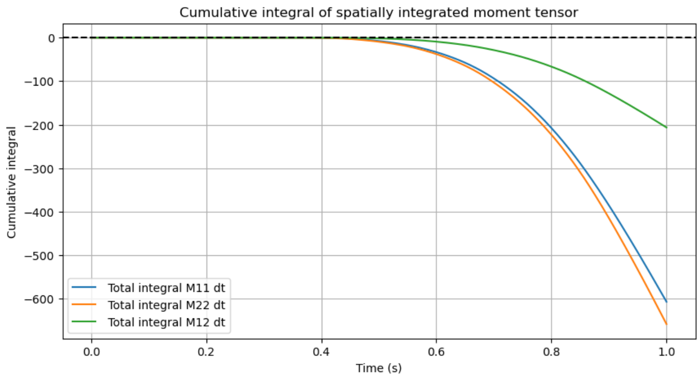


Figure 3.6: Cumulative integral of moment tensor components. Blue is m11, green is m12 and orange is m22. X-axis shows time, y-axis shows magnitude of cumulative integral.

3.4 Simulation of DAS Data

The theory in section 2.6 is used to simulate DAS data from the outputs given by SALVUS. This is done by extracting the displacement output. The spatial derivative of displacement is taken to obtain strain: $\epsilon_x = \frac{\partial u_x}{\partial x}$, the temporal derivative of strain results in strain rate($\frac{\partial^2 u_x}{\partial x \partial t}$).

Strain and strain rate are averaged over a 10.4 m gauge length, which was chosen based on the avalanche warning system installed by Turquet et al., 2024, which has a gauge length of 10.4m. This is combined with the Pyber library by Noe, 2025. The Pyber gauge length physically averages the data over the input gauge length, which is the same as what is described in section 2.6. This is done by calculating weights for the points within the gauge length window, all weighted points in a window are then added up (Noe, 2025). To replicate smoothing caused by the DAS laser pulse, Pyber uses a Gaussian window, which calculates the weights. A Gaussian window is used instead of a rectangular (boxcar) window to more accurately represent the Gaussian shape of the interrogator laser pulse, which produces a weighted rather than uniform average over the gauge length (Noe, 2025).

Chapter 4

Results

4.1 Simple Model Results

The following results shown are for the Supershear and sub-Rayleigh cases of the custom STFs. The sources are spaced 0.1 m apart between 30 m and 270 m and have a Ricker wavelet in all directions. The moment tensors have components 1 in xx (11) and xy (12) and -1 in yy (22). These moment tensor directions were chosen to mimic a load applied to the snow, such as a skier passing over the snow, which results in compression (negative yy). The forces applied by a skier on the surface also extend in xy and xx because of the motion of the skis on the snow, to simplify the input, all components were kept at 1. The simple, two layer model was set up to test the custom STF option in SALVUS and to verify how it works. The results are shown here because it is important to understand the physics underlying crack propagation in snow before moving on to the more complex case. Since the focus of this thesis is to model crack propagation as recorded by DAS data, strain and strain rate plots are shown first. The velocity results are shown to verify observations made in the strain and strain rate plots.

4.1.1 Sub-Rayleigh Strain and Strain Rate Plot

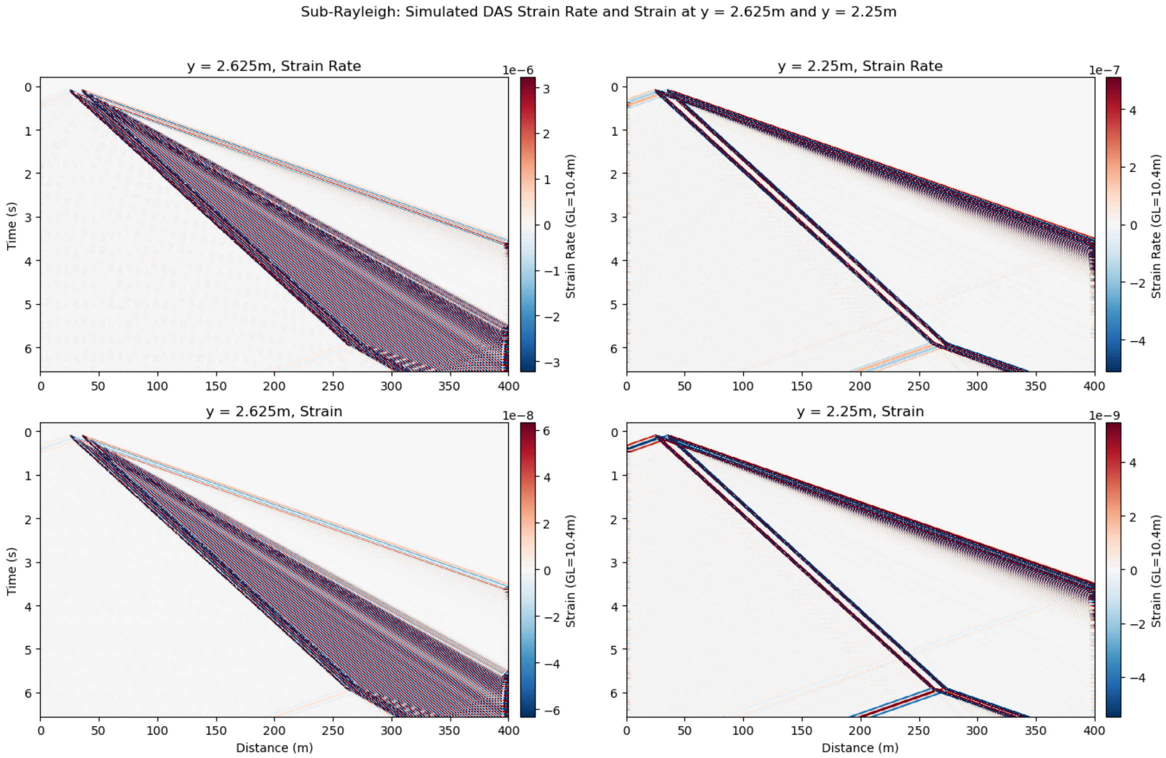


Figure 4.1: Sub-Rayleigh strain and strain rate (horizontal component) averaged over 10.4 m gauge length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left side, snow bottom on right side. Colorbar shows magnitude of strain or strain rate, colorbars scaled to minimum and maximum of dataset. In the numerical domain, y is defined as vertical elevation measured upward from the base of the air layer (y=0.0m) to the bottom snow surface (y=3.0m). Therefore, deeper positions correspond to smaller y values. The receiver line at y=2.625m (Crack Depth) is higher in elevation than y=2.25m (Snow Bottom), which correctly places the crack above the snow base.

Both the strain and the strain rate at crack depth look very similar to each other, the same is true for the snow bottom. At crack depth, both the strain and strain rate in the left column of figure 4.1 show one strong main front, which is preceded by a preliminary front. The main front with higher amplitude has a slope which corresponds to the prescribed rupture speed, whereas the precursor is faster and has lower amplitude (fainter color). The same is true for the snow bottom, but here, the main front is thinner than the preliminary front. The main front still has higher amplitude than the preliminary front, but the preliminary front weakens in amplitude with increasing time.

At crack depth, the main front broadens with increasing time, which cannot be observed for the main front measured at the snow bottom on the right of figure 4.1. At both depths, for strain and strain rate, the two fronts have visibly different slopes, which corresponds to different apparent propagation speeds.

At both depths, a continuous wavefront extends from 0s starting at 30m to 6s ending at 270m, this corresponds to the source coordinates defined and also to the time delays prescribed to the sources.

In all plots, the strain and strain rate fields show alternating polarity in both space and time. At a fixed receiver position, the temporal sequence of arrivals is: a positive strain (red, extension) arriving first, followed by negative strain (compression), then returning to positive. This oscillatory pattern of alternating tension and compression is the expected signature of a propagating elastic wave, in which the material at a fixed point cycles between extension and compression as successive wave peaks and troughs pass (Aki and Richards, 1981). The same alternating polarity pattern is observed along the spatial axis (x-axis) at any fixed time: adjacent points along the cable are in opposite phases of the oscillation. The oscillatory pattern can also be seen in the single traces in appendix K.

The amplitude of both strain and strain rate is larger at crack depth than at the snow bottom. The colorbar range at the snow bottom is approximately one order of magnitude smaller than at crack depth, indicating strong amplitude decay over the 0.375m of vertical separation between the two receiver depths.

At x=270m, t=6s in all sub-Rayleigh plots in figure 4.1, the main wavefront stops, but there is a new wavefront propagating from this point. This front has a change in slope, which means a different propagation speed, this is the rupture arrest front. This new wavefront is bidirectional, it propagates from the arrest point in positive as well as negative x with the same slope.

In the simulated DAS at gauge length 10.4m, a doubling of the wavefront is observed. This dual-band signature is heavily dependent on the chosen gauge length, as shown in appendix H, reducing the gauge length to 1.0 m resolves the signal into a single, continuous wavefront. The 10.4m results are shown here because they correspond with DAS cables used for avalanche detection (Turquet et al., 2024).

4.1.2 Supershear Strain and Strain Rate Plot

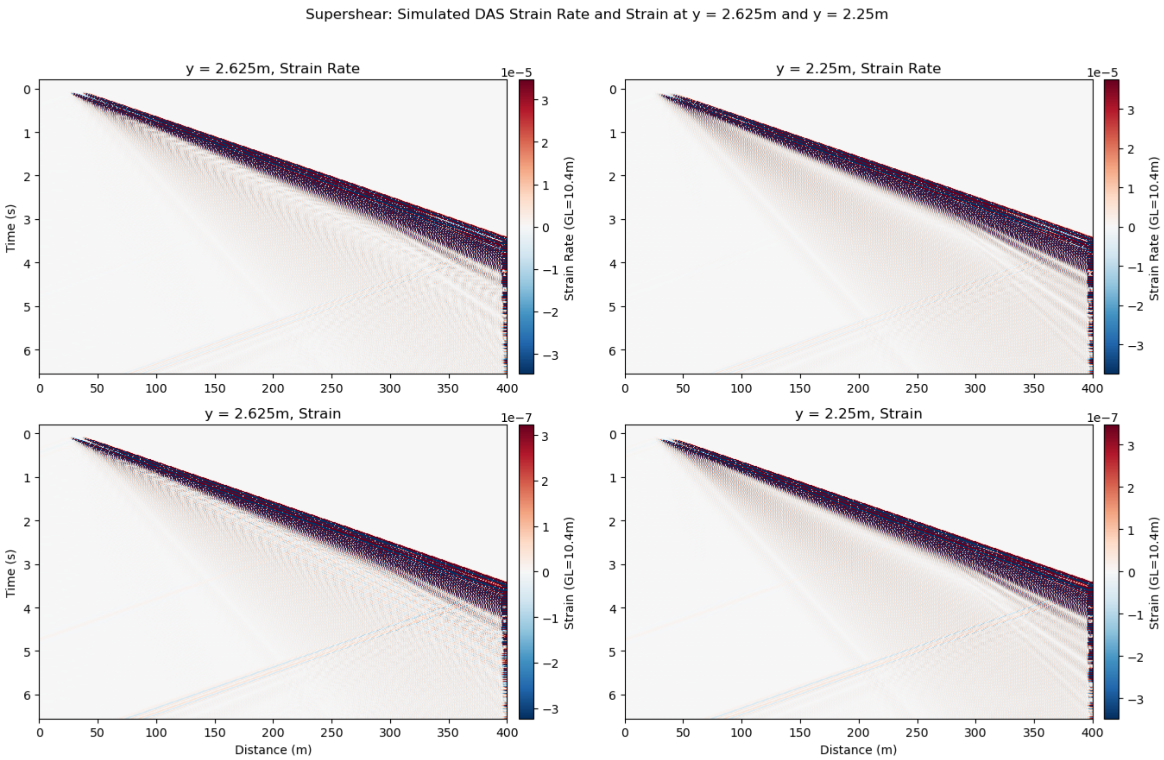


Figure 4.2: Supershear strain and strain rate (horizontal component) averaged over 10.4 m gauge length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left side, snow bottom on right side. Colorbar shows magnitude of strain or strain rate, colorbars unscaled.

In the Supershear case, strain and strain rate at both receiver depths in figure 4.2 show a single dominant wavefront with comparable amplitude, with no significant reduction between crack depth and snow bottom. This main wavefront has strong amplitude at first, but experiences significant reduction in amplitude after about 0.5s have passed, which is stronger the more time passes. This corresponds to loss of energy of the different frequencies contained in the wavefront.

The strain and strain rates observed have opposite polarity at adjacent time steps (positive followed by negative). This is again consistent with the signature of an oscillatory wave passing, as described in the sub-Rayleigh case. In the strain rate plots, the polarity reversal can be seen a bit more clearly than in the strain plots in the bottom row of figure 4.2.

In figure 4.2, there is no visual difference in the amplitude of strain and strain rate between the crack depth and snow bottom. However, the peak strain rate in the Supershear case is approximately one order of magnitude larger than in the sub-Rayleigh case, as indicated by the colorbar scales in figures 4.1 and4.2.

Similarly to the sub-Rayleigh plots in figure 4.1, the Supershear plots also show wavefronts with different amplitudes and slopes at the rupture arrest at 270m. Here, this is characterized as a fan of distinct wave trains coming from the rupture arrest point.

4.1.3 Sub-Rayleigh Velocity Plots

The following plots show the horizontal (v_x) and vertical (v_y) particle velocities at two receiver depths: $y = 2.25$ m (snow bottom) and $y = 2.625$ m (crack depth). Particle velocity rather than displacement is shown to better align with standard seismological field observables. In the field, standard geophones directly measure particle velocity. In contrast, DAS systems intrinsically measure axial strain or strain rate averaged over a physical gauge length (Lindsey and Martin, 2021). Comparing DAS data to traditional point sensors typically requires converting the measured strain rate into an equivalent particle velocity, a process that relies on assumptions about apparent wave speeds. Deriving displacement would require a further step of time integration, which is often avoided in practice due to the amplification of low-frequency noise and baseline drift. Therefore, particle velocity serves as the most robust and common physical quantity for comparing models to field data (Sayed et al., 2024). Presenting both components is highly relevant for instrumentation: the horizontal velocity (v_x) is fundamentally linked to the longitudinal strain fields measured by a horizontal DAS cable, while the vertical velocity (v_y) represents the direct signal that a traditional vertical geophone, which is the standard instrument used in snow and avalanche monitoring, would record.

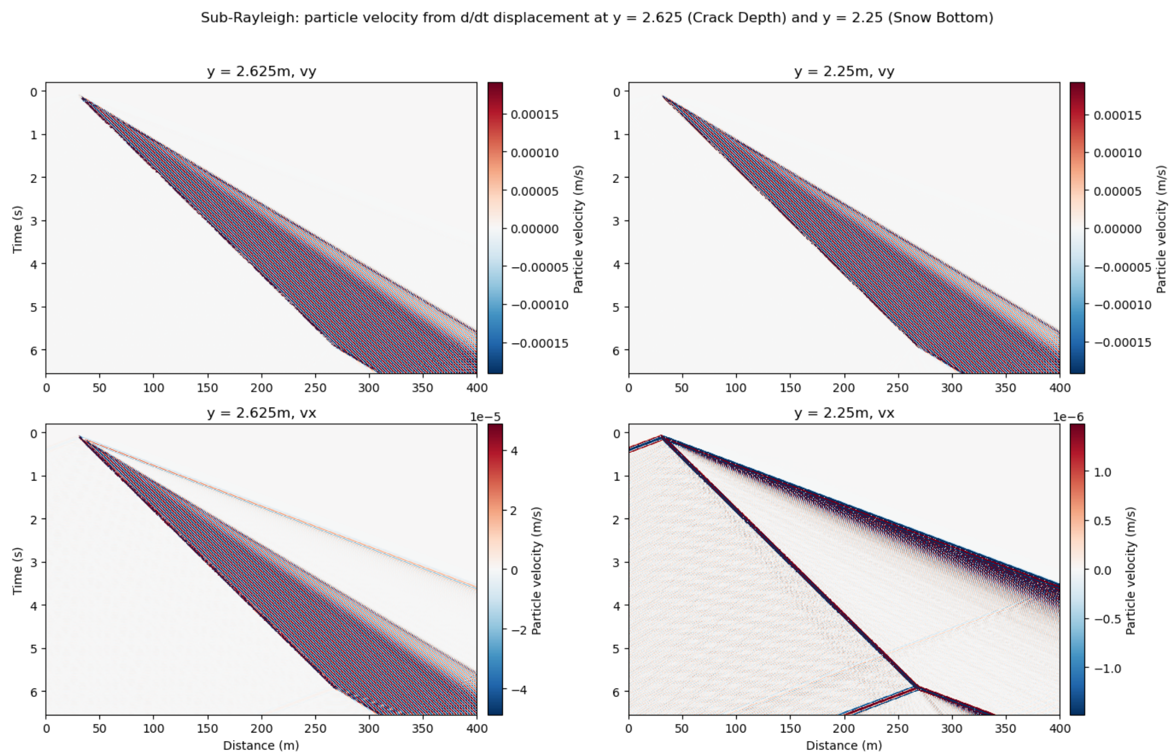


Figure 4.3: Sub-Rayleigh Particle velocity: vertical velocity in top row, horizontal in bottom row. Crack depth in left, snow bottom in right column. Colorbar is unscaled, showing velocity (m/s).

The sub-Rayleigh velocity plots in figure 4.3 all show a distinct wavefront starting at 0s at 30m and ending at 270m. This main wave train ends at 6s, but sends out waves with a different speed (different angle) in direction of the source propagation. This corresponds to the arrest of the rupture which was already observed in the strain and strain rate plots. The horizontal velocity at the snow bottom also shows a backward-propagating train from the crack tip.

Both vx plots (bottom row in figure 4.3), show the main wavefront with waves travelling ahead of it, as time and distance progresses, the main wavefront gets wider. This means that they have different slopes, which corresponds to different propagation velocities. In all but the plot of vx at the snow bottom, the first couple of waves in the main wavefront are weaker in velocity. In vx at the crack depth at the bottom left of figure 4.3, the precursor is weaker in color and thus amplitude of velocity than the main wavefront. At the snow bottom, which is shown on the bottom right, the precursor has a high amplitude of velocity initially, which then decays.

In the main wavefront observed, it's very noticeable that the particle velocity changes polarity at different time- and space-steps, which is consistent with an oscillatory wave passing: peaks of the wave are followed by troughs, which leads to velocities of opposite polarities. This is also noticeable in the cases that have multiple wavefronts, although it's less pronounced.

In all panels of figure 4.3, the main wavefront between 30m and 270m does not decay in amplitude with increasing time. It is noticeable however that in all but the horizontal velocity at the snow bottom, the main wavefront broadens with time, which is caused by the radiation of S-waves from the source, which constructively interfere in the slab to form the A0 Lamb waves.

4.1.4 Supershear Velocity Plots

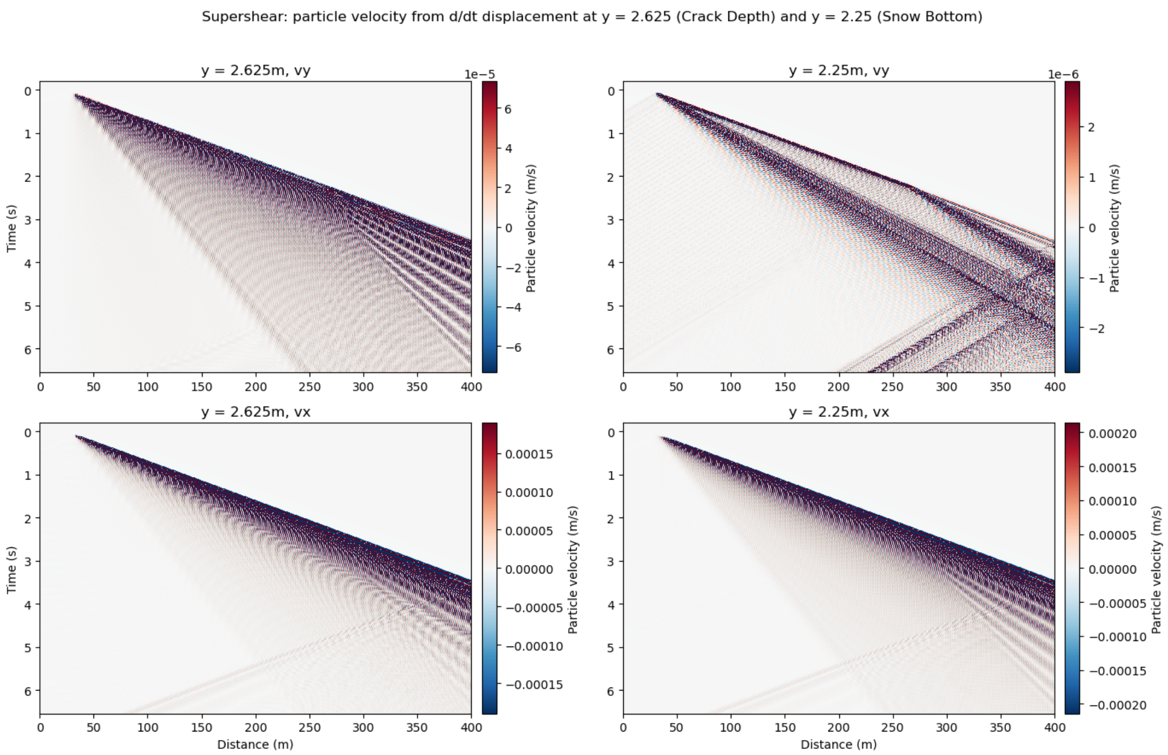


Figure 4.4: Supershear Particle velocity: vertical velocity in top row, horizontal in bottom row. Crack depth in left, snow bottom in right column. Colorbar is unscaled, showing velocity (m/s).

In all of the Supershear velocity plots in figure 4.4, a single dominant wavefront extends from 30m to the end of the medium at 400m. The slope of this front corresponds to the prescribed Supershear rupture velocity. This wavefront broadens with increasing propagation time, where it also loses amplitude (weaker color). At the crack arrest at 270m, the end of the source array, the wavefront generates a fan of outward-radiating waves consistent with stopping-phase radiation from the crack arrest point (Madariaga, 1977).

In the vy panel at the snow bottom (top right in figure 4.4), there is a clear distinction between two wavefronts, which cannot be seen in all other Supershear velocity space-time plots. The first wavefront with high amplitude corresponds in slope and therefore propagation speed to the wavefront in all other plots, it also has a fanning out of waves from 270m, so the rupture arrest. From the crack onset at 30m, there is a second wavefront which also is of a higher amplitude. This wavefront has a shallower slope and therefore a lower propagation speed than the main wavefront and does not emit a fan at the rupture arrest point.

The oscillatory wave processes described in the previous sections can also be observed in the Supershear velocity panels in figure 4.4. While the pattern of opposite polarity is the same in all cases, the slope of the wavefronts change. In all plots, after the first wavefront, the waves seem to slow, which means their slope flattens.

4.1.5 Frequency-Wavenumber (f-k) Plot

The f-k-plot in figure 4.5 shows the energy (colorbar) density at frequencies (y-axis) and wavenumbers (x-axis) present in the results. This energy was taken from the horizontal velocity results at the snow bottom because the horizontal direction corresponds to the data measured by a horizontal DAS cable buried in snow. The plots shown are normalized to the 99th percentile of the energy present, this is done so that outliers with large energies are scaled and do not skew the color scale.

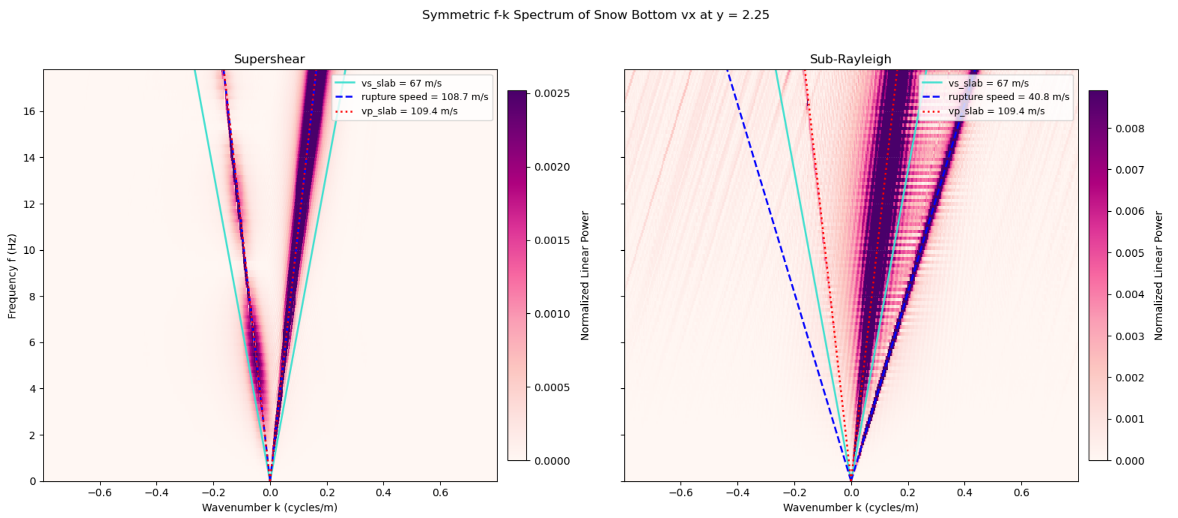


Figure 4.5: F-k plot for the Supershear (left) and sub-Rayleigh (right) case. Slab P-wave velocity in red, S-wave velocity in turquoise and respective rupture speed in blue. Colorbar shows energy normalized to the 99th percentile.

The Supershear and sub-Rayleigh f-k plots in figure 4.5 show qualitatively different energy distributions, consistent with the two distinct propagation regimes. The Supershear case shows a concentration of energy around the rupture speed (blue line) and the vp (red line) of the slab on both sides of the wavenumber spectrum. It is important to note here that these speeds are very close in value, so the lines cannot be distinguished well in the plots. The sub-Rayleigh case also shows two concentrations of energy. A very broad band at the vp of the slab and a concentration of energy at the rupture speed on the postive wavenumber side. In all cases, the energy is more concentrated around the vp and or rupture speed for lower frequencies, at higher frequencies, the concentration of energy starts to deviate from speeds present in the medium in figure 4.5.

In the Supershear case, the dark spots of concentrated energy are more broad on the positive side compared to the negative side, where they are narrower and less smeared out. This asymmetry reflects the directionality of the source: the crack propagates in positive x and more energy is released in that direction. In the sub-Rayleigh case on the right side of figure 4.5, there is no concentration of energy at negative wavenumbers. Rather than a physical absence of backward propagation, this asymmetry in sub-Rayleigh is a simulation effect: due to the slower rupture speed, the backward-propagating arrest wave develops later in the time window and quickly encounters the absorbing boundary. As a result, the backward-traveling wavefield is truncated before sufficient energy can enter the time-space window analyzed by the f-k transform.

Figure 4.6 shows the same frequency-wavenumber spectrum. The Lamb-wave dispersion curves obtained in section 2.3 for a 1.5m thick slab are indicated by the orange (antisymmetric, A0) and the green line (symmetric, S0). These are plotted separately because they fall close to the vp of the slab in both cases and the lines would not be very well recognizable.

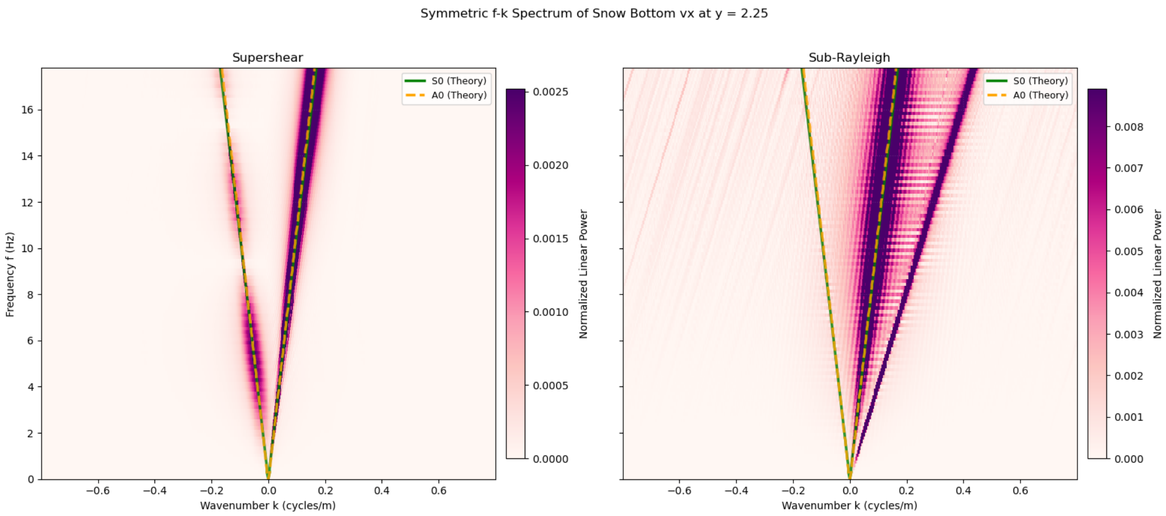


Figure 4.6: F-k plot for the Supershear (left) and sub-Rayleigh (right) case. Lamb wave dispersion curves calculated from 2.3. A0 mode shown in orange, S0 mode in green. Colorbar shows energy normalized to the 99th percentile.

Both the Supershear and the sub-Rayleigh f-k plots show energy concentrated at the A0 and the S0 speeds on the positive wavenumber side, which fall close to the p-wave velocity of the slab in both cases. The lines indicating the dispersion curves follow the dark concentration of energy in both cases. In the supershear case on the left side of figure 4.6, there is also energy concentrated around the negative A0 and S0 speeds. In the Supershear case, the speeds calculated from the dispersion curves in 2.3 also fall close to the rupture velocity.

4.2 CZM Results

The results of the cohesive zone model are shown here for a pure mode I, Supershear crack. Unlike the simple model, the CZM model has only one moment tensor component for a pure mode I crack. This is because of the way the mode-mix is accounted for, as described in the methods section 3.2.1. Nevertheless, the results look almost identical to the simple model case, this is why only one result is shown here. The setup of this model was still valuable for the thesis because it helped in understanding crack propagation and modelling in snow.

4.2.1 CZM Strain and Strain Rate Plots

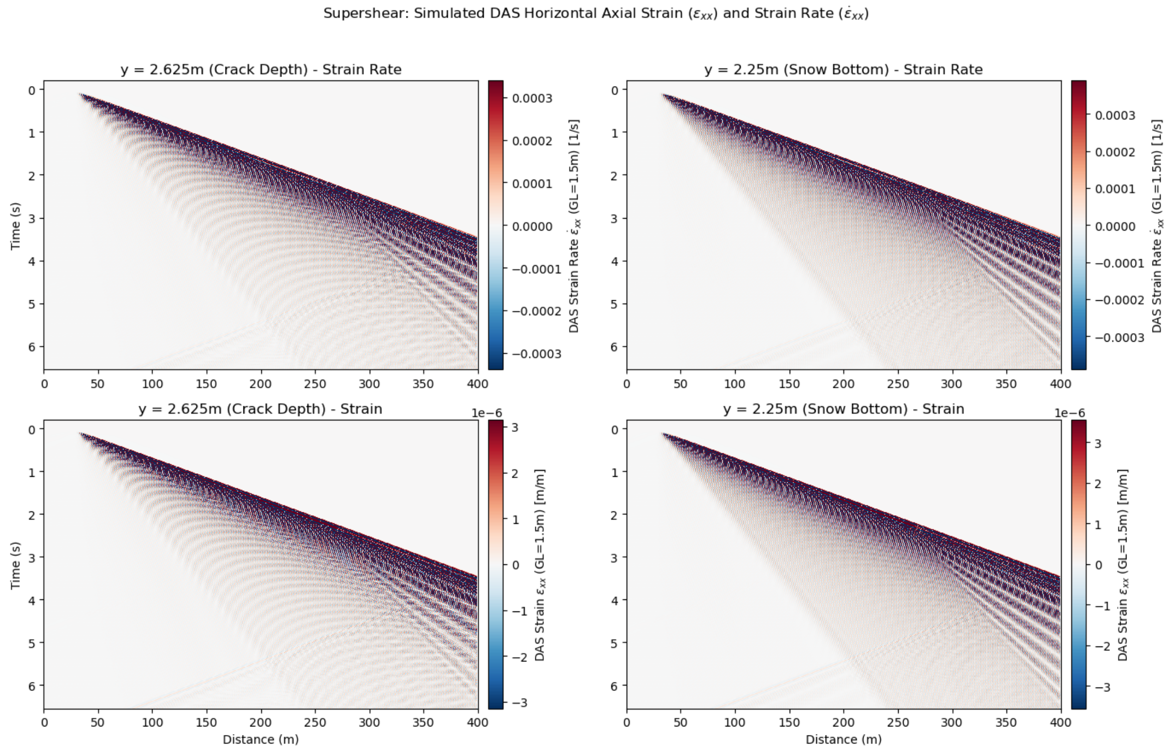


Figure 4.7: CZM Supershear strain and strain rate (horizontal component) averaged over 10.4 m gauge length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left side, snow bottom on right side. Colorbar shows magnitude of strain or strain rate, colorbars unscaled.

The strain rate and strain panels produced by the cohesive zone model (figure 4.7) are visually indistinguishable from the simple model results at the same prescribed rupture speed (figure 4.2). The wavefront slopes, depth dependence, rupture arrest signature and polarity sequence are all consistent between the two source models. The CZM results are therefore not discussed further, as the simple model captures the same wave physics at this level of analysis, and the CZM adds no observable difference in the DAS signatures for the parameters used here. The only difference is in the amplitude, which can be seen from the colorbar units, but this is likely caused by physics.

4.2.2 CZM Velocity Plots

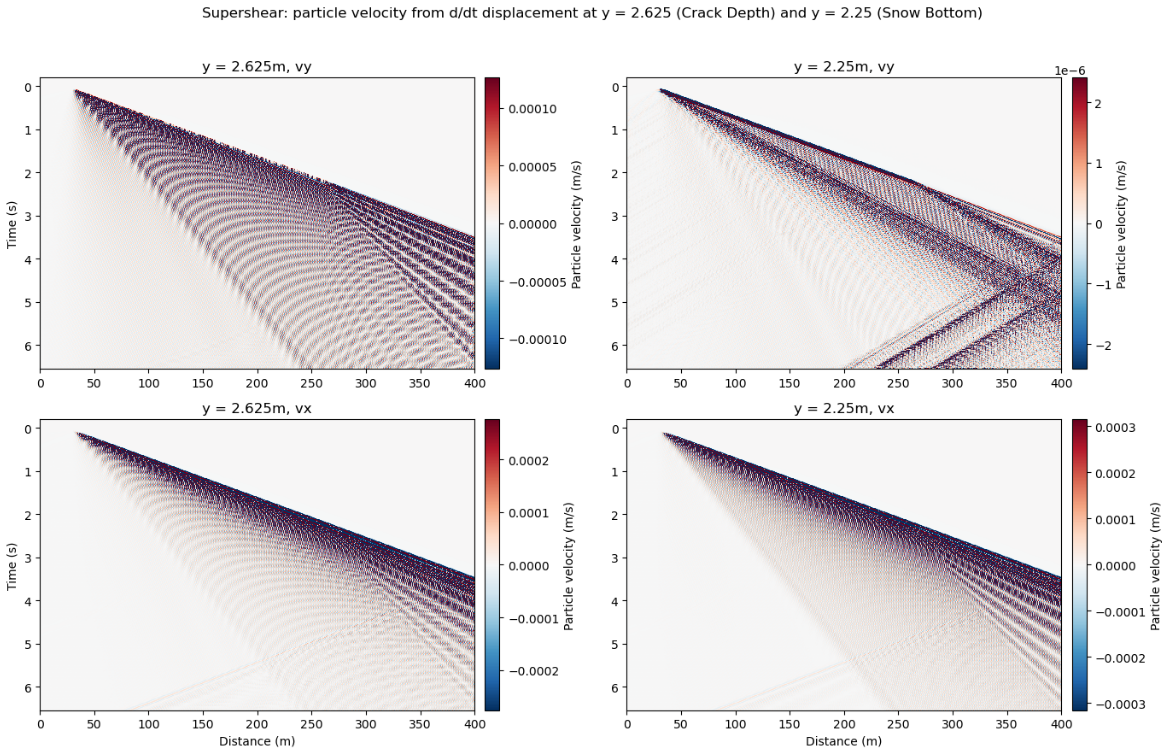


Figure 4.8: CZM Supershear Particle velocity: vertical velocity in top row, horizontal in bottom row. Crack depth in left, snow bottom in right column. Colorbar is unscaled, showing velocity (m/s).

The same is true for the velocity plots in figure 4.8, which are identical to the Supershear velocity plots in figure 4.4.

While the CZM model does not alter the DAS signature observed compared to a model with simpler crack physics, this finding might be important in demonstrating that a DAS cable with 10.4m gauge length is not sensitive to small-scale processes. Such processes, like the damage zone and the mixed-mode moment physics described in section 2.2, allthough accounted for in the model, might not be recorded by the cable, which justifies the use of the simpler source physics case in this thesis.

4.3 MPM Guided Results

The results shown here are from the MPM-guided simulation. The sources are spaced 0.01m apart from 75m to 125m. Unlike the results in section 4.1, which are for a simple, 2 layer model with prescribed rupture velocities (sub-Rayleigh and Supershear case), the results here are for a 5-layer model with a rupture velocity coming from the MPM simulations. The results shown here are filtered using a low-pass filter with 50Hz cutoff. Because both strain and velocity are derived from displacement, the displacement field extracted is shown in appendix I. The displacement data itself is unfiltered, but because strain, strain rate and velocity are derivatives, the differentiation acts as a high-pass filter which amplifies noise (Igel, 2016). Displacement is shown in the appendix to show that there are little numerical artefacts and noise in the extracted data. This means that filtering strain, strain rate and velocity should not be a concern because the noise is simply from the derivative, the input is a clean field.

4.3.1 MPM Strain and Strain Rate Plots

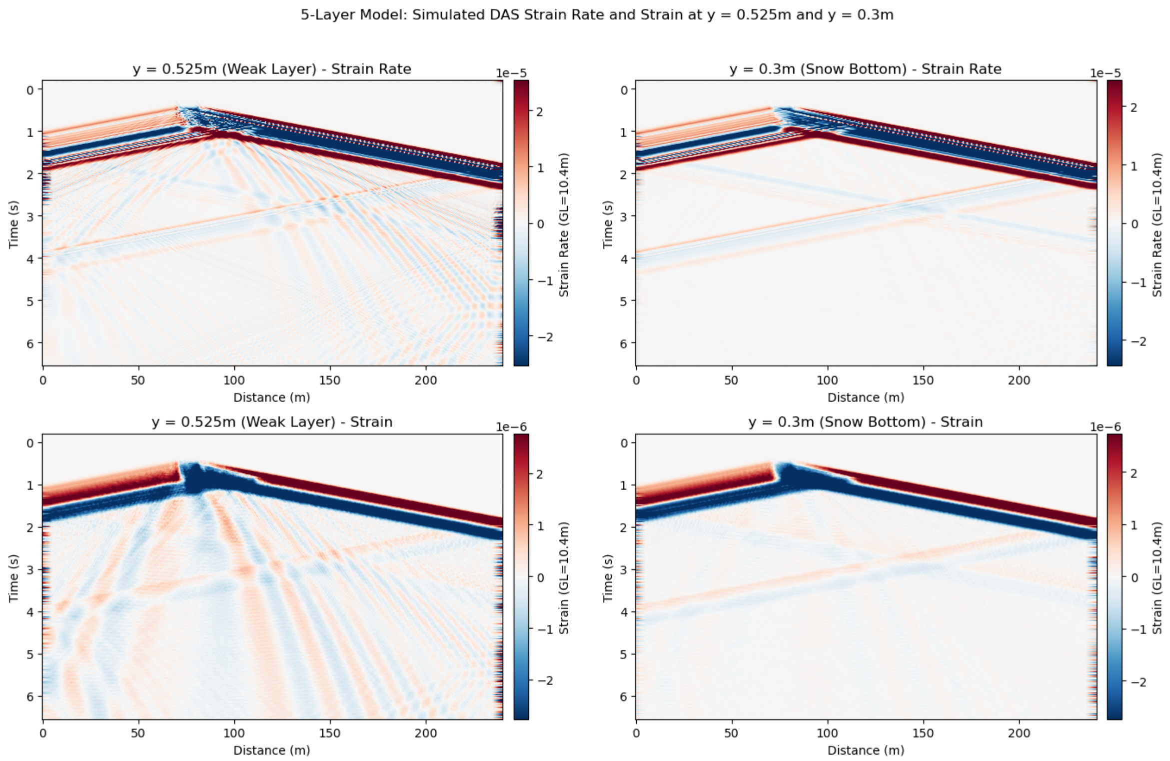


Figure 4.9: Strain and strain rate (horizontal component) averaged over 10.4m gauge length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left, snow bottom on the right. Colorbar shows magnitude of strain or strain rate.

The strain as well as the strain rate plots in figure 4.9 all show a distinct v-shape initiating from the crack onset point at x=75m. The left arm of the V corresponds to backward-propagating body waves radiated from the crack initiation point, while the right arm corresponds to the propagating crack traveling from the initiation point onwards.

Both strain panels (bottom column) show a front of positive strain (red, on top) arriving first, which is followed by a negative strain front. The extension (positive strain) followed by compression as time increases could indicate P-waves or the S0 Lamb wave mode. At both the crack depth as well as the snow bottom, in positive x-direction, the red front widens in time until about x=125m, when there is a jump and the front has about same width as the blue front. This location coincides with the crack arrest location. In the negative x-direction, both the positive and negative front have the same width from the initiation point on to the domain boundary.

The same positive and negative fronts from the crack onset point can be observed in the strain rate plots in the top row of figure 4.9. Contrary to the strain panels, where the fronts are solid colors, in the strain rate panels, these are much more heterogenous. The forward propagating positive front is much narrower in strain rate than in strain, the same is true for the backward propagating negative front. The wedge-shape to the crack arrest point observed in strain can still be seen in strain rate however. In the strain rate plots there also more than just two wavefronts in the v-shape coming from the crack onset. The main positive and negative fronts are still visible, but there is another positve front under the blue front in the positive directon. In the negative direction, the same is true: under the blue wavefront there is multiple positive and negative layered wavefronts. The oscillatory nature of the strain and strain rate values measured can also be seen in the traces in appendix K, figure K.3.

In the strain plot at crack depth as well as in the strain rate at the same depth, there are more wavefronts additional to the v-shape from the crack onset. Those fronts fan out from the crack initiation point at 75m as well, but they have a much steeper slope than the main front. In contrast to the main front, they are weaker in color meaning amplitude. These fronts are oscillatory, meaning a front of positive polarity (red) is followed by one of negative polarity, which is consistent with the processes described in the simple case.

These weaker amplitude fronts broaden with increasing time, which is consistent with dispersive wave propagation. Their amplitude does not decrease however. In the strain rate at crack depth (y=0.525m, weak layer, shown on top right of figure 4.9), the colors of the second wavefront are weaker than in the strain panel at the same depth. This means that they have lower amplitude relative to the main wavefront in the strain rate panel.

The main wavefront observed in figure 4.9 in all panels does not broaden over time, it also shows no fading in color, which would indicate a decrease in amplitude.

In all panels, there are two red-blue wavefronts coming from the domain side boundaries. These are especially noticeable in the snow bottom strain and strain rate plot (right side of figure 4.9), because the main front is the only wavefront present there. The wavefronts coming from the side boundaries of the domain have the same slope as the main wavefronts (wave train coming from the right boundary has the same slope as backward-propagating front from crack), but they are much weaker in amplitude.

When comparing the strain rate plots (top column) to the strain plots, although the main structures are the same, the colorbar scale in the strain rate plots is one order of magnitude (one power of 10) larger than that of the strain plots. This does not depend on depth (snow bottom has the same amplitude as crack depth) but on the type of field shown.

4.3.2 MPM Velocity Plot

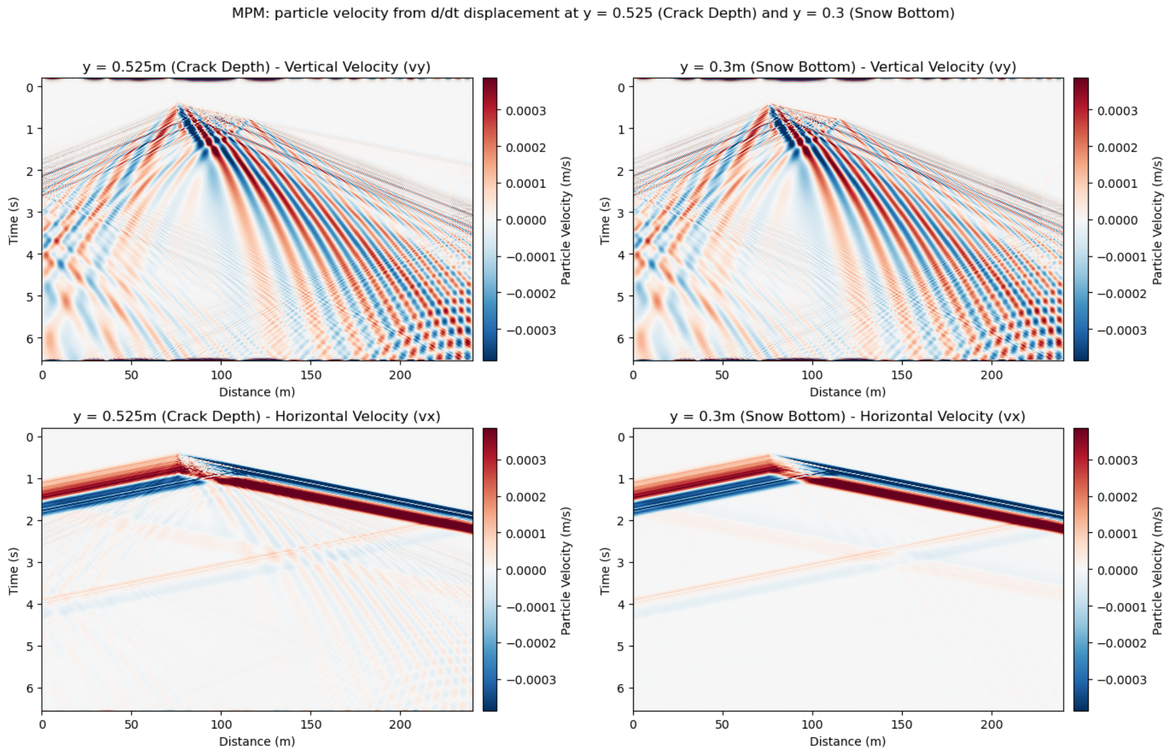


Figure 4.10: Particle velocity: vertical velocity in top row, horizontal velocity (vx) in bottom row. Crack depth on left, snow bottom on the right. Colorbar shows magnitude of particle velocity (m/s).

The velocity plots in figure 4.10 show that both cases look pretty similar to each other independent of depth. This means that the vertical velocity in the top row, looks almost identical at crack depth compared to the snow bottom. The vertical velocity plots can be distinguished from the horizontal velocity plots however.

The horizontal velocity in the bottom two panels of figure 4.10 both show a flat v-shape from the crack origin at 75m, similar to that observed in the strain and strain rate plots in figure 4.3.1. The forward propagating front is characterized by a wave train of negative velocity (blue), which is the first arrival, followed by a train of positive energy (red) which arrives shortly after. In the direction backwards of propagation, this order is flipped: a positive front arrives followed by a negative front. In the horizontal velocity plots in figure 4.10, the vertex of oscillatory fronts seems to be shifted. The first front, which is blue in positive direction of propagation and red in negative direction, is at x=75m, which is the crack onset. The vertex of the second v-shape, which has opposite polarity (red in positive, blue in negative direction), is not right below the vertex of the first v-shape. This can also be seen in the zoomed-in section in figure 4.11, especially in the vy plots. It is shifted in positive direction of propagation, which indicates that it comes from the crack arrest point at x=125m.

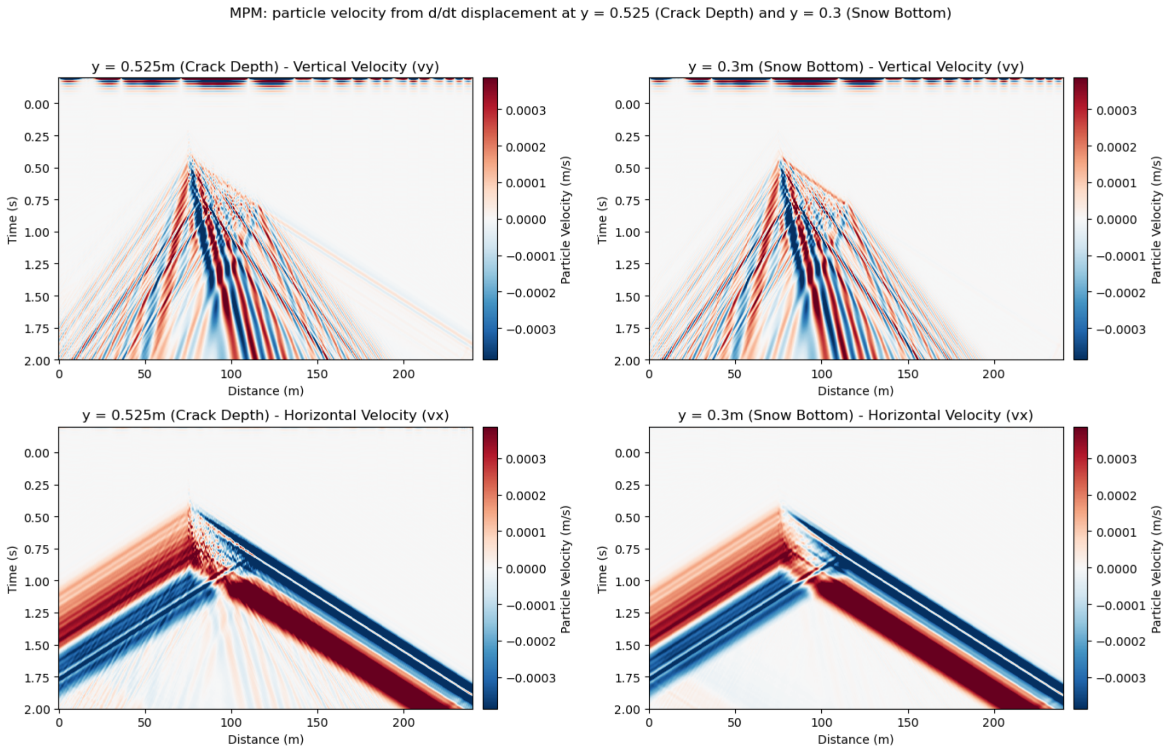


Figure 4.11: Particle velocity, zoomed into first arrival: vertical velocity in top row, horizontal velocity (vx) in bottom row. Crack depth on left, snow bottom on the right. Colorbar shows magnitude of particle velocity (m/s).

From the same vertex, in the vx plots at crack depth (bottom left in 4.10 and 4.11), fronts with weaker amplitude originate. These wavefronts fan out from around x=125m and they are layered in polarity (positive followed by negative). They also widen as time increases, which could indicate dispersion. Such secondary wavefronts are not present in the vx plot at the snow bottom.

In the vertical velocity plots in the top row of figure 4.10, the secondary wavefronts are very strong and can be seen clearly at both depths. They originate in a v-shape from the crack initiation point at x=75m. These wavefronts are very strong in amplitude (saturated colors), and show the propagation of a periodic oscillatory wave. The wavefronts in both the positive and negative direction of propagation widen as time increases and their amplitude decreases, which is indicative of gemetric spreading and dispersion. When looking at the later times in the top panels of figure 4.10, the blue and red velocities seem to curve towards the origin of the plot (towards x=0). This is another indicator for dispersion of higher frequencies.

Although the wavefronts from x=75 m are most noticeable, there are V-shaped wavefronts which also originate at x=125m which is the crack arrest point. These are mainly noticeable at earlier times, for example between 1s and 2s, because at later times, there are multiple other wavefronts interfering with them. The second v of waves from the crack arrest starts later in time than the one from the crack initiation point, which is intuitive because the crack onset and arrest are not at the same time. The slope of the individual components from the arrest wavefront is similar to the one of the onset.

From the crack initiation point at x=75m, in the vertical velocity panels at the top of figure 4.10, there is a second wavefront which also propagates in the positive and the negative direction. The wave trains in this front are much thinner than in the front that was described previously, they also have a flatter slope, which means lower speed. The slope of this wavefront is comparable to the one in the vx plots below them. This wavefront also has layered polarity, but here the bands of one polarity are much thinner than in the vx plots or the other wavefronts in the vy plots.

In the area where the crack propagates, so between 75m and 125m, at the early times in vy in the top panels of figure 4.10, of which a zoomed-in section can be seen in figure 4.11, the slope of the thinner layer wavefront is different than in the rest of the wavefront. This area has a slope which is steeper than the rest of the wavefront, it is especially noticeable in vy at the snow bottom, where a red line of positive velocity seems to connect the crack onset and arrest point. The slope of this line is likely the crack propagation speed.

In the vertical velocity plot at crack depth, one very faint wave from the crack onset point can also be seen. This wave train has a different slope than all other wavefronts in the plot, but it is very weak in amplitude (color).

4.3.3 MPM f-k Plot

The data used for the f-k plot is the horizontal velocity data, for the reasons described in 4.1.

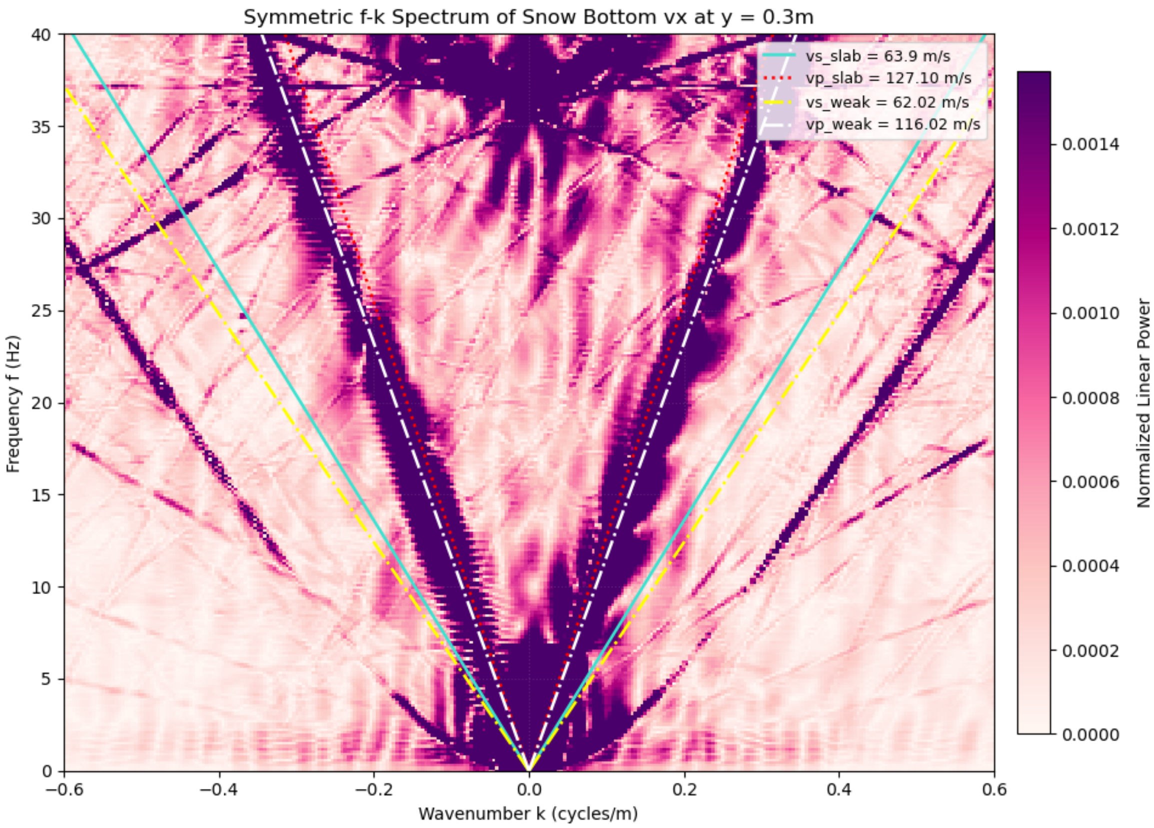


Figure 4.12: F-k plot for the MPM-guided simulation. Slab P-wave velocity in red, slab S-wave velocity in turquoise. Weak layer P-wave velocity shown by the white layer, S-wave velocity of WL shown by yellow line. Colorbar shows energy normalized to the 99th percentile to avoid outliers.

The frequency-wavenumber plot in figure 4.12 shows a symmetric distribution of energy around the origin. A lot of energy is concentrated around the positive and negative slab and weak layer P-wave velocity, which are indicated by the red (slab vp) and white (weak layer vp) lines.

The concentration of energy is not strictly at those two speeds, the dark spot of concentrated energy is broader and more spread out and therefore not contained by the two lines.

In the f-k plot in figure 4.12, there is no energy concentrated at the S-wave speeds of the slab, which is indicated by the turquoise line, or the vs of the weak layer, shown by the yellow line.

Additional to the energy concentrated around the P-wave speeds, there are bands of concentrated energy on the positive and negative side of the wavenumbers. These energy bands are stronger in the positive direction and they have a shallower slope than the S-wave speeds present in the medium.

Figure 4.13 shows the same frequency-wavenumber spectrum. The Lamb-wave dispersion curves obtained in section 2.3 for a 0.15m thick slab are indicated by the orange (antisymmetric, A0) and the green line (symmetric, S0). The reason for not taking the full slab thickness for 0.3m for the calculaation is the fact that there is a weak layer with different material properties in the middle of the two 0.15m thick slab layers. The dispersion curves are plotted separately because they fall close to the vp of the slab in both cases and the lines would not be very well recognizable.

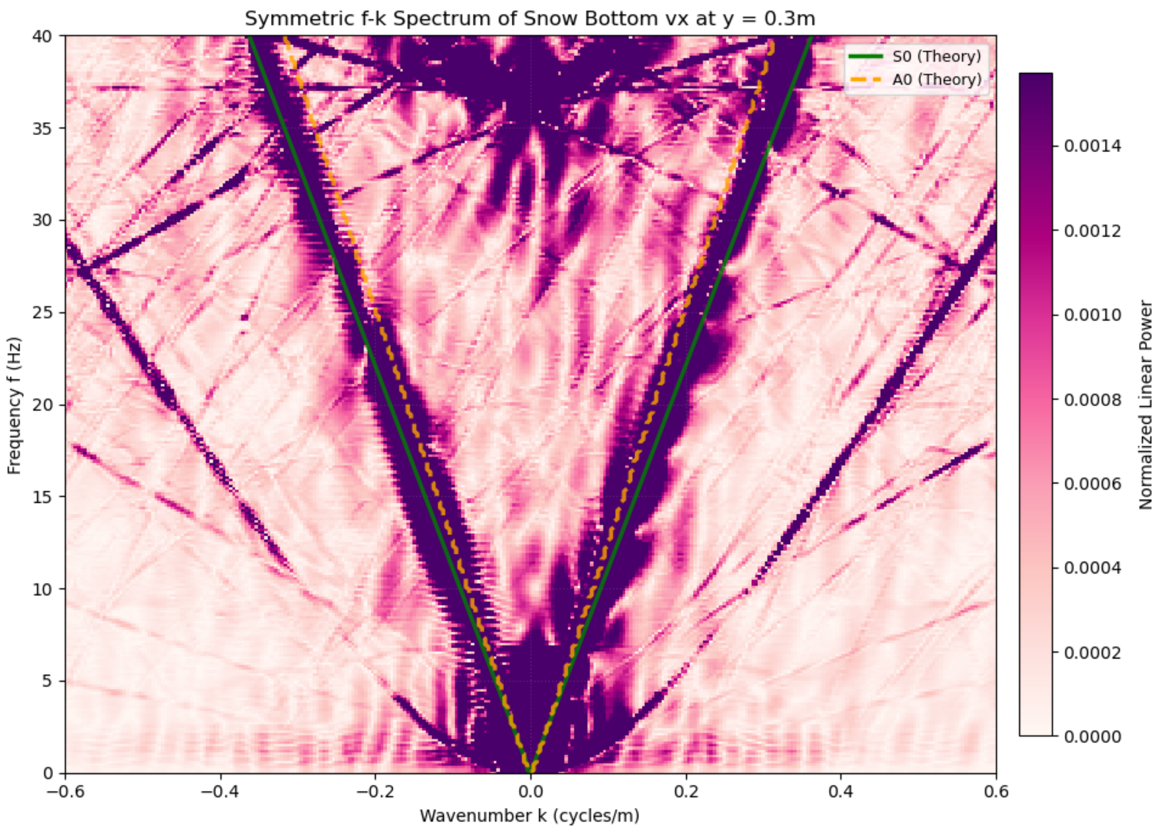


Figure 4.13: F-k plot for the MPM-guided simulation. Lamb wave dispersion curves calculated from 2.3. A0 mode shown in orange, S0 mode in green. Colorbar shows energy normalized to the 99th percentile.

Figure 4.13 shows that the main energy observed in the f-k plot is concentrated around the symmetric (green) and antisymmetric (orange) Lamb wave speeds. This is true for the positive and negative side of the plot.

The A0 mode has a steeper slope than the S0 mode, it is closer to the origin of the symmetric concentration of energy in both cases. The two dispersion curves are almost overlapping each other up until frequencies of about 7Hz, when they start to diverge and the A0 mode in orange becomes steeper and the two curves start to diverge.

The symmetric S0 mode, shown by the green line in figure 4.13 is close to the center of the energy concentration structure, but gets closer to the outer edge as frequencies increase.

In the f-k plot in figure 4.12, the p-wave velocities of the slab and the weak layer (red and white line), are close to the location where the Lamb wave dispersion curves are located in figure 4.13.

Chapter 5

Discussion

5.1 Simple Model

5.1.1 Strain and Strain Rate Space-Time Plots

The main wavefronts which can be observed in figures 4.1 and 4.2 are the crack fronts. This is because when taking the slope of these two wavefronts, it corresponds to the rupture velocity in the Supershear and the sub-Rayleigh case. The prescribed versus the calculated speed from the figures can be seen in table 5.1. Both cases produce strain and strain rate outputs that look very different, this is important when relating the results here to field measurements. The different behaviors mean that in theory, from the strain and strain rate space-time output, the transition point between sub-Rayleigh and Supershear propagation regimes, as described by Trottet et al., 2022; Bergfeld et al., 2025; Siron et al., 2023, could be observed because the strain and strain rate measurement would change.

Case	Prescribed Speed	Speed from Results
Sub-Rayleigh	0.6vs = 40.8m/s	$\frac{270-30m}{6s} = 40m/s$
Supershear	1.6vs = 108.7m/s	$\frac{270-30m}{2.3s} = 104m/s$

Table 5.1: Simulation input rupture speeds versus speeds calculated from slopes in figures 4.1, 4.2.

Two-wavefront Structure in Sub-Rayleigh

In the sub-Rayleigh strain and strain rate plots, two main wavefronts can be observed. One, the main wavefront, which is broadest in the strain rate and strain plots at crack depth in figure 4.1, corresponds in slope to the sub-Rayleigh rupture speed, as discussed above. The second wavefront is faster and arrives sooner than the main wavefront and only appears to be emitted from the first source. To confirm that this precursor is a physical wave type rather than a simulation artifact, simulations with larger source spacing were run and showed this changes when source spacing is increased, for example from 0.1m to 10m, as shown in appendix E, figure E.2. This leads to the conclusion that the precursor to the main wavefront disappears because of interference. All other sources but the first and the last one have interference from the sources before and after it, which means that this wave type, although emitted, is cancelled out because of interference (Madariaga, 1976). When the source spacing is increased, the fronts of the individual sources travel further apart in space and they interfere less.

This precursor in the sub-Rayleigh strain and strain rate plots has a slope which corresponds to a speed around the p-wave speed in the snow, which is also close to the speed of the symmetric Lamb wave mode (S0). The reason for this is discussed in the velocity section 5.1.2 because it is easier to derive from the velocity output.

Wavefront Identification and Polarity Sequence

In both the sub-Rayleigh (figure 4.1) and Supershear (figure 4.2) cases, all panels show a consistent polarity sequence: positive strain (extension, red) arrives first, followed immediately by negative strain (compression, blue), then a return to positive. This extension-first, compression-second sequence is the expected signature of a compressional P-wave or S0 Lamb mode arrival at a receiver ahead of the source, and is well established in elastic wave propagation theory (Aki and Richards, 1981). When the crack-tip stress field expands into the undisturbed medium, it first pushes material away from the source (extension), and the elastic restoring force of the medium then pulls it back (compression) before the wavefield equilibrates. This can be seen for both cases in the traces of the first 5 recievers in appendix K. Both the Supershear (figure K.2) and the sub-Rayleigh (K.1) strain and strain rate traces show a small positive strain, followed by a negative strain, which then oscillates in the positive direction again, confirming the expected S0 behavior. The same sequence has been reported in field measurements of avalanches with single-component geophones where p-wave precursors have been observed in seismograms (Herwijnen and Schweizer, 2011).

After this leading extensional arrival comes the dense alternating polarity pattern of the A0 flexural wave train. As derived in section 2.3, the horizontal strain of the A0 mode varies as a cosine in space and time, cycling between maximum tension and maximum compression over each half-wavelength. The detailed arguments of why the A0 Lamb wave mode is present in the results is discussed in the velocity section 5.1.2.

Rupture Arrest Structure

As observed in all plots in section 4.1, the stopping of the crack at x=270m produces a different wave radiation pattern compared to the crack itself. In the Supershear case in figures 4.2, 4.4, this manifests as a fan of waves coming from the crack end point, whereas in the sub-Rayleigh case, there seems to be a v-shaped coming from the tip of the crack.

The fan in the Supershear case is consistent with stopping-phase radiation from a crack that has outrun its own shear wave cone (Madariaga, 1977). Because the rupture speed exceeds the S-wave speed, the accumulated wave energy can only release forward into the undisturbed medium ahead of the arrest point, producing a unidirectional fan, which also explains why there is no backward radiation in the Supershear case.

The v-shaped observed in the sub-Rayleigh case in figures 4.1, 4.3, can be explained by the fact that the crack never outruns its radiated waves. This means that the rupture arrest wavefront can propagate in all directions, unlike the Supershear case (Freund and Freud, 1998). The slopes of the arms of the v are steeper than the one of the rupture front itself, which means that the stopping phases travel at faster speeds than the rupture. This means that the waves emitted by the arrest could for example travel at the highest medium speed (the p-wave or S0). In both the sub-Rayleigh strain and strain rate as well as velocity plots, the backward-propagating wavefront is strongest at the snow bottom. This can be attributed to the reflection of the backward-propagating waves by the rigid bottom boundary (Graff, 1975).

The distinct rupture arrest signature in the sub-Rayleigh versus Supershear case has implications for DAS measurements because it could mean that the propagation regime of the crack could be determined from the crack arrest itself. The presence of a rupture arrest front also has implications for avalanche propagation. As Gaume et al., 2019; Gaume et al., 2018 state, the application of a load or disturbance somewhere on a slope can cause remote triggering of avalanches. Furthermore, Burridge, 1973; Trottet et al., 2022 have stated that one crack can trigger another (daughter crack nucleating from mother crack). This is important in relation to the rupture arrest because the wavefronts emitted by the arrest could trigger daughter cracks or even trigger avalanches remotely.

Dispersion and Broadening of Wavefronts

In the Supershear strain and strain rate plots, the main wavefront weakens with increasing time. The same is true for the precursor in the sub-Rayleigh strain and strain rate plots at the snow bottom. The weakening of the wavefront behind the crack tip with increasing time is observed clearly in the Supershear case, because the crack travels faster than the guided waves (Dunham and Archuleta, 2005). A Mach-front is created, where all the waves travel together in a "shock-front" (Dunham and Archuleta, 2005), but once this front passes, there is attenuation of energy. This is confirmed by the f-k plots in figures 4.5 and 4.6, where in the supershear case, all speeds and dispersion curves fall close to each other. In the sub-Rayleigh case on the other hand, the guided waves are faster than the crack (Aki and Richards, 1981), there is reflection of the waves between the top and bottom of the snow, which creates the oscillations which can be seen in figure 4.1.

In the sub-Rayleigh case, the precursor wave also broadens and weakens with time (right column of figure 4.1). At the snow bottom, they are further away from their source (which is at crack depth). Because the S0 precursor is only emitted by the first and last source, its first arrival is strongest. After that, it loses energy because there is no source emitting this type of wave anymore, and the S0 precursor loses energy (Aki and Richards, 1981), which can be seen by the colors fading. This wavefront does not broaden, it only starts fading with increasing time.

The broadening of the wavefront with increasing time can be explained by dispersion. By the reflections of the top and bottom snow boundary, the waves present start interfering with each other, which can lead to constructive or destructive interference (Aki and Richards, 1981). This means that waves can start travelling at different phase velocities, because of the interference Graff, 1975. The phase and group velocities of waves are described as

$$v_{\text{ph}} = \frac{\omega}{k}, \quad v_{\text{g}} = \frac{d\omega}{dk}. \tag{5.1}$$

The interference leads to the broadening of the wavefront with increasing time, described by the envelope spreading law (Aki and Richards, 1981)

$$\Delta x(t) = \Delta x_0 \sqrt{1 + \left(\frac{\frac{d^2\omega}{dk^2} t}{(\Delta x_0)^2} \right)^2}. \tag{5.2}$$

As time goes on, there are more reflections present, so there is more potential for interference and thus dispersion.

Amplitude Decay and Depth Dependence of Strain and Strain Rate

The weakening of the wavefronts arriving after the main wavefront which has a slope consistent with the rupture speed, especially observed in the Supershear case in figure 4.2 and the precursor in 4.1 can be explained by depth decay and geometric spreading from the source. The crack tip acts as a moving two-dimensional point source, travelling along a line in a 2D elastic medium, the peak amplitude A of the radiated wavefield decays inversely with the square root of the radial distance r ($\frac{1}{\sqrt{r}}$) from the source plane (J. Achenbach, 1973). Wave components whose horizontal phase velocities are below the bulk wave speeds of the snow become evanescent in the vertical direction, decaying exponentially with depth from the crack plane (Graff, 1975). This explains why higher-frequency content is attenuated more rapidly between crack depth and snow bottom. The decay to distance relationship changes in 3D compared to 2D (J. Achenbach, 1973), which is an important consideration for future, more complex simulations.

The sub-Rayleigh strain and strain rate plots in figure 4.1 look very different for crack depth versus snow bottom. At crack depth, the main wavefront corresponding to the rupture is broad, while the S0 precursor is faint in energy and narrow. At the snow bottom on the other hand, the precursor is wider and decays in amplitude and the main wavefront is very narrow but high in energy. As discussed, the S0 precursor widens at crack depth because of geometric spreading (J. D. Achenbach, 1975). At crack depth, this precursor is narrow because it is impulsive and non-dispersive (Rose, 2014). The main wavefront at crack depth is wide because it is measured directly at the source, this means that the full field of the source is captured (Rose, 2014), this includes the near-field of the source and the dispersive A0 coda emitted by the crack. At the snow bottom, the coda has dispersed, which leaves only the crack front which is resolved as a sharp line because it is the signal with the highest energy.

Magnitude of Wavefronts

In the Supershear case (figure 4.2), the front driven by the crack tip and the S0 precursor wavefront merge into a single broad band because the rupture speed is within a few metres per second of the p-wave speed and the S0 speed. The strain and strain rate panels at both depths look essentially identical with no change in amplitude of strain or strain rate from crack depth to snow bottom. This contrasts sharply with the sub-Rayleigh case 4.1, where the amplitude drops significantly between the two depths.

A consequence of this difference in depth dependence of magnitude of strain is its ratio between the two regimes. In the Supershear strain rate plots, the colorbar minimum and maximum is approximately one order of magnitude larger than in the sub-Rayleigh panels. Supershear ruptures are more efficient radiators of seismic energy to the free surface because the crack tip is at the same location as wave front rather than trailing behind it, concentrating energy into a Mach cone that hits the snow surface with maximum constructive interference (Dunham and Archuleta, 2005; Rosakis, 2002). In the sub-Rayleigh regime, the energy is spread over a broader spatial and temporal footprint because the rupture front and the precursor waves propagate at different speeds, reducing the peak strain rate at any point in space or time. This amplitude difference has a direct implication for field measurements: Supershear avalanche releases would be substantially more detectable by DAS than sub-Rayleigh events of the same moment release, and consequently, a DAS cable that can just barely detect a Supershear event would be unable to detect a sub-Rayleigh event of equal size at the same distance. The same is implied for strain and strain rate at different depths in sub-Rayleigh propagation regimes. Because the magnitude of strain (rate) in figure 4.1 is one order lower at the snow bottom than at crack depth, the cable location relative to the nucleation is important. Since cables are usually installed before the first snowfall (Edme et al., 2023; Paitz et al., 2023), they would be located at the snow bottom rather than at the depth of the crack. This means that the sub-Rayleigh crack propagation might be too small in magnitude to be recorded by the DAS cable. Since Bergfeld et al., 2025; Siron et al., 2023; Trottet et al., 2022; Gaume et al., 2018 have shown that there can be a transition between sub-Rayleigh and Supershear crack propagation, this means that the sub-Rayleigh onset of the crack might not be recorded in DAS because of its smaller magnitude.

5.1.2 Velocity Space-Time Plots

Like in the strain and strain rate plots, the main wavefronts in the Supershear and sub-Rayleigh case in figures 4.4, 4.3 correspond in slope to the rupture speed of each case. Like in strain (rate), the velocity output is vastly different for sub-Rayleigh and Supershear, which means a transition between propagation regimes can not only be observed in the strain output, but also in the velocity output.

Similarity to Strain and Strain Rate Plots

The sub-Rayleigh and Supershear velocity plots (Figures 4.3 and 4.4) closely mirror the corresponding horizontal strain and strain rate fields (figures 4.1 and 4.2). This similarity is expected, as horizontal strain ($\epsilon_{xx} = \frac{\partial u_x}{\partial x}$), strain rate ($\dot{\epsilon}_{xx} = \frac{\partial^2 u_x}{\partial x \partial t}$), and horizontal velocity ($v_x = \frac{\partial u_x}{\partial t}$) are all direct derivatives of the same displacement field (u_x). The only key visual difference occurs in the vertical velocity (v_y) at the snow bottom (Figure 4.4, top right), as the strain plots represent purely horizontal components.

Because both metrics share identical wavefield signatures, such as wavefront broadening from dispersion, amplitude decay with depth, and precursor waves, these shared features will not be re-analyzed here. To avoid redundancy, this section focuses exclusively on observations unique to particle velocity, while the underlying physical mechanisms are detailed in the strain discussion.

Two-Wavefront Structure in Sub-Rayleigh

The precursor wave which can be observed in the vx plots in the sub-Rayleigh case in figure 4.3 has a much faster speed than the rupture itself. This wave also only occurs in the sub-Rayleigh case in the vx plots. As for strain, simulations with larger source spacing (for example 10m spacing instead of 0.1m spacing, see appendix E, figure E.1) have shown that this precursor disappears if source spacing is increased. This leads to the same conclusion as discussed in the strain case that this is because of interference between the sources (Madariaga, 1976).

The precursor wavefront is the symmetric Lamb wave mode (S0), whose propagation speed is limited by the plate wave speed:

$$c_p^{plate} = 2c_s\sqrt{1 - \frac{v_s^2}{v_p^2}} \tag{5.3}$$

(Rose, 2014). When using the material parameters used in the simulation, this is just short of the p-wave speed of the slab. The speed of the S0 wave converges to the plate wave speed for low frequencies (long wavelength limit) (Rose, 2014), which is why this wave is believed to be the S0 mode which is excited.

This two-wavefront structure with a fast S0/P-wave precursor followed by a slower crack-front signal is the signature of sub-Rayleigh rupture. The spatial separation between the two fronts grows linearly in time as

$$\Delta x(t) = (v_p - v_r)\,t, \tag{5.4}$$

where v_p is the P-wave speed and v_r is the rupture speed. This linear growth is directly measurable from the space-time plots in figure 4.3 and is consistent with the theoretical prediction for an expanding crack in a homogeneous elastic medium (Freund and Freud, 1998). The origin of the precursor is the redistribution of the stress field ahead of the crack tip, which propagates at the fastest available wave speed in the medium regardless of the crack tip speed (Aki and Richards, 1981), so in this case, the p-wave or the S0 wave speed.Because the p-wave speed is very close to the S0 speed, this result is consistent with field observations of avalanches using seismic geophone arrays by Herwijnen and Schweizer, 2011, a p-wave precursor to the main avalanche release was also observed. While Herwijnen and Schweizer, 2011 did not observe the propagation regime, the p-wave or S0 precursor could be an important warning signal for sub-Rayleigh crack propagation leading to avalanche release.

The precursor is visible in sub-Rayleigh space-time panels because the separation $\Delta x(t)$ grows fast enough to resolve the two fronts within the observation window. In the Supershear case, the rupture front, S0 and the pressure wave all have velocities within a couple of metres per second of each other. Nevertheless, in the vertical velocity at the snow bottom in 4.4, two more distinct wavefronts can be seen, one which could be the S0 wave. This can be explained by the fact that vy is the dominant component of the A0 mode (Lamb, 1917), which at the snow bottom has sufficient geometric spreading to separate from the S0 precursor (Rose, 2014). This means that the slower front is likely the A0/main wavefront from the crack, while the steeper one is the S0 mode. The presence of the S0 mode as a precursor even in the Supershear case is confirmed by the traces of the first 5 recievers, shown in figure K.2, appendix K, as discussed in the strain discussion. In all Supershear velocity plots, at the end of the crack, there is a fanning out of waves which can be observed, which could be a combination of the stopping phases of the different wave types. This behaviour at arrest is consistent with the stopping-phase radiation described by Madariaga (1977), who showed that abrupt arrest of a crack generates a burst of high-frequency energy that propagates at the bulk wave speeds of the medium, so the p-wave speed, which is close to the S0 in this case, independent of the rupture speed before the arrest.

Lamb Wave Identification

The polarity reversal of the velocities in both sub-Rayleigh and Supershear cases is caused by oscillatory waves. The presence of these wave types can be proven mathematically; the detailed derivation of this is shown in section 2.3.

The animations of the velocity wavefield in appendix F show the propagation intensity of the velocity wavefield. When comparing the horizontal velocity vx in figure F.1 with the wavefield intensity of the two Lamb wave modes, for example in Carta et al., 2023, which can be found in appendix G.1, it becomes evident that the main mode which can be observed is the antisymmetric (A0) mode. This is because at source depth, the velocity field has a negative polarity. Above and below the sources, there are narrower bands of negative polarity (blue in figure F.1, red in the reference figure G.1) beside bands of positive velocity. This can be explained by the fundamental theory of Lamb waves (Lamb, 1917; Rose, 2014). Because the moment tensor used here has a distinct out of plane component (negative myy), the primary mode which is excited is A0 (Rose, 2014). As discussed, the S0 mode can also be seen, but is a precursor rather than the main wavefront and carries less energy. This precursor can also be seen in figure F.1.

The dominance of A0 over S0 follows from source-mode coupling theory. For a moment tensor source applied normal to the slab midplane (so normal to the horizontal of the slab), the coupling coefficient to A0 scales as the bending moment integrated over the source region, while coupling to S0 scales as the in-plane force resultant (Rose, 2014; J. Achenbach, 1973). Because the force applied has a negative y-component (Myy of the moment tensor) and is normal to the slab horizontal axis, this maximizes A0 excitation. Because the force is only in the negative direction, it is asymmetric, so while A0 excitation is maximized, S0 coupling is minimized at low $f \cdot h$. This theoretical expectation is confirmed by the simulations: S0 appears primarily as a precursor in v_x (in-plane, extensional motion), consistent with the mode shapes described by Lamb (1917) and the coupling analysis of Rose (2014). This identification is strengthened by the animation stills in appendix F, especially vx in figure F.1. The velocity field shows a polarity reversal around the crack, meaning that for example, the panel above the crack has positive polarity whereas the one below has negative polarity. Alongside of the comparison with figure G.1 in appendix G, this is proof for the A0 mode because S0 would have the same polarities on either side of the crack (Lamb, 1917). This can be seen in the animations as well, where the stills show bands of opposite polarity with symmetry around the crack before the main wavefront, which is the precursor. The presence of the two Lamb wave modes is also confirmed by the dispersion curves in figure 4.6, which aligns with the concentration of energy in the f-k plot.

5.1.3 Discussion of f-k-Plot

Similarly to the results for the velocity and strain plots, the f-k-plots shown in section 4.1.5 show that the energy is concentrated around the p-wave speed (red line in figure 4.5) and the rupture speed (blue line in figure 4.5). In the Supershear case, these two speeds are very close together, which leads to one energy band, whereas in the sub-Rayleigh case, they are more separated and two energy bands are observed. The S0 wave speed, as discussed previously, is also very close to the p-wave speed, which also has an effect on the concentration of energy in those areas, this can be seen when comparinf figure 4.5 to figure 4.6 with the dispersion curves.

Wavefront & Propagation Speed Identification

In the sub-Rayleigh f-k plot in figure 4.5, the energy between the two dominant slopes (between the rupture speed and the P-wave speed lines) is not random noise. It corresponds to the dispersive A0 Lamb mode, which at different frequencies propagates at different phase velocities between v_r and v_p. This becomes clear from figure 4.6, where the A0 mode is shown by the orange line, which falls into that wedge. The fan of diagonal stripes observed in the sub-Rayleigh v_y waterfall plots maps onto this wedge in f-k space, as shown in figure 4.6: each stripe is a different frequency component of A0 arriving at a different time at a given location. This dispersive fan is a direct encoding of the A0 dispersion curve of the snow slab at $f \cdot h \approx 6$ Hz m (Rose, 2014; Lamb, 1917). In principle, the shape of this dispersive wedge could be inverted to recover the elastic properties of the slab (thickness, shear modulus) using the same surface-wave dispersion inversion methods applied in geophysical prospecting (Yilmaz, 2001).

In the sub-Rayleigh case, the f-k plot in figures 4.5 and 4.6 shows a concentration of energy at the v_p of the slab, the A0 and the S0 mode, as well as the rupture speed, consistent with the observations made in the velocity and strain plots.

This is also the case in the Supershear f-k plots and the corresponding velocity and strain plots. Because the rupture speed is very close to the P-wave, A0 and S0 speed, as seen in figures 4.5 and 4.6, in the medium, energy is concentrated around these. The narrowness of the energy concentration in the Supershear case compared to sub-Rayleigh reflects the absence of the dispersive two-front fan: the Mach cone geometry radiates energy preferentially at the cone angle $\alpha_{Mach} = \sin^{-1}(c_s/v_r)$ (Dunham and Archuleta, 2005; Rosakis, 2002), which at $v_r \approx v_p$ collapses to a nearly flat front, concentrating all radiated energy into a single, tightly bounded slope in f-k space. Nevertheless, Lamb waves are also present in the Supershear case, as the dispersion curves in figure 4.6 show.

The strong contrast in how the energy is concentrated between the sub-Rayleigh and Supershear case in 4.5 can be explained by the number of locations the energy is concentrated. As described above, the Supershear regime has energy at essentially one slope, whereas the sub-Rayleigh case fills a wedge between two slopes. This explains why the energy bands in the Supershear case appear much narrower and more concentrated compared to sub-Rayleigh.

Dispersion Effects

The observation that at higher frequencies in figure 4.5, the lines of the energy present do not match as well anymore with the concentration of energy, can be explained by numerical as well as physical dispersion. As Komatitsch and Tromp, 1999 describe, as frequency increases, the wavelength decreases. This means that higher frequencies are not resolved as well by a spatial grid with a certain spacing Yilmaz, 2001, this starts bending the energy in the f-k plot. This could explain the mismatch between the physical velocity lines and the concentrations of energy.

The horizontal striping visible in the sub-Rayleigh f-k plot at constant frequency (for example, radiating from the vp-line) is a temporal aliasing artifact arising from the finite output sampling interval of the simulation, not a physical feature. This can be distinguished from the physical dispersion fan by the fact that the stripes are perfectly horizontal (constant frequency, for all wavenumbers), whereas physical A0 dispersion produces curved or oblique features in f-k space.

5.1.4 Implications for DAS measurements & Synthesis of Results

The results in section 4.1 and their discussion have shown that there are multiple implications for the measurement and interpretation of DAS data, even in a simple physics case.

In all sub-Rayleigh results, the presence of a precursor wavefront, which is the S0 wave, is shown. This is consistent with field observations made by Herwijnen and Schweizer, 2011, but the precursor is also weaker in amplitude compared to the main wavefront and decays in amplitude with time. This is important because it shows that the precursor wave might not be a reliable "early warning" because it might not be detected on account of its low magnitude. The main wavefront travels at crack propagation speed and does not undergo significant amplitude decay.

In all Supershear results, there is only one wavefront because the S0 and p-waves have a similar speed to the rupture. The precursor can be identified by looking at the vertical velocity at the snow bottom as well as the rupture arrest and the strain traces. In the Supershear case, the wavefront is dispersive and decays in amplitude with depth, it also broadens with increasing propagation time. This is because the rupture and the precursors travel together as a Mach front, which is attenuated (Dunham and Archuleta, 2005). The broadening can be explained by dispersion, so the change in phase velocities caused by reflections and interference (Graff, 1975).

The presence of the arrest phase in both the Supershear and sub-Rayleigh propagation regime and their different manifestations could be important in further distinguishing propagation regimes. While sub-Rayleigh has a v-shaped arrest, Supershear propagation has a fan-like arrest front that only propagates in the positive direction. This could help identify the propagation regime of a crack in snow, which in turn is important for warning. The faster a crack propagates, the faster the critical area needed for release is reached and the avalanche will be released (Bergfeld et al., 2025; Trottet et al., 2022).

The Supershear velocities and strain (rate) has a higher amplitude than their sub-Rayleigh counterparts. This means that Supershear ruptures can be detected more easily in DAS measurements because they radiate more energy. In the sub-Rayleigh case, there is also significant difference in the amplitudes of strain, strain rate and velocity between the crack depth and the bottom of the snow, the bottom of the snow having tenfold lower amplitudes. This is crucial for DAS field measurements because the cable is usually located at the bottom of the snow. The implications for field measurements are that the sub-Rayleigh propagation preceding the Supershear transition (Trottet et al., 2022) might not be seen in DAS data because it has too low amplitude. This is important when interpreting field data because false conclusions of pure Supershear rupture might be drawn from this.

The gauge length used to simulate DAS measurements here is 10.4m, like in Turquet et al., 2024. This has implications for how the results look because some wave types might be smoothed over on account of their wavelength. The S0 mode which has a speed around $v_p \approx 109m/s$ and A0 is similar to the rupture speed, in sub-Rayleigh this is $v_r \approx 40m/s$. Calculating the wavelength from speed according to $\lambda = \frac{v}{f0}$ (Aki and Richards, 1981) means that the S0 mode has a larger wavelength. This means the larger the DAS gauge length, the more the A0 mode is suppressed because one DAS gauge length spans more A0 wavelengths. Additionally, when a wave is spatially narrower than the 10.4m gauge length, the fiber registers the wave passing both the start and end of the segment, creating a physical "double wavefront" or dipole artifact in the plot. To make sure the gauge length of 10.4m does not lead to misrepresentation of the results, data for a 1m DAS gauge length were simulated, these can be found in appendix H. The Supershear case shown in figure H.2 at 1m gauge length looks almost identical to the strain at 10.4m gauge length in figure 4.2, apart from the double wavefront. The same is true for the sub-Rayleigh strain at a smaller gauge length (H.1), compared to the Turquet et al., 2024 gauge length in the results figure 4.1. Apart from eliminating the double wavefront artifact, there is no major difference or benefit to using the smaller gauge length. This comparison shows that the observations made in the sections above are valid for small as well as large gauge lengths. This is important because it shows that crack propagation looks similar no matter the gauge length of the DAS cable used.

5.2 Material Point Method Guided Simulations

In the following sections, results from the material point method guided simulations will be discussed. Because many phenomena which are observed in the simple model are also observed in the MPM model, these will not be discussed in detail.

5.2.1 Strain and Strain Rate Space-Time Plots

Propagation Speeds & Wavefront Geometry

The v-shaped wavefronts which can be observed in figure 4.3.1 are likely emitted by the crack itself. The slope of these two arms in both sides of propagation does not change from the vertex, which is the crack onset, to the end of the domain. This indicates a constant propagation speed, which means unlike simulations and experiments by Siron et al., 2023; Bergfeld et al., 2025, there is no transition in crack propagation speed. When taking the slope of the positive strain wavefront which arrives first in figure 4.9, this results in: $v_{prop} = \frac{125-75}{1.2-0.5} = 57.14m/s$. Considering the medium velocities used (shown in table D but also in the f-k plot 4.12), this is sub-Rayleigh propagation in both the weak layer but also the snow slab.

The slope of the backward propagating arm in figure 4.9 is calculated according to the values in table 5.1, the same is true for the second wavefront which is observed as well as the forward-propagating wavefront before the crack arrest.

Wave type	Slope formula	Calculated speed
Between crack onset and arrest	$v_{prop} = \frac{125-75}{1.2-0.5}$	57 m/s
Backward propagating	$v_{prop} = \frac{75-0}{1-1.6}$	-125 m/s
Second wavefront	$v_{prop} = \frac{175-150}{4-3}$	25 m/s
Forward propagating after crack arrest	$v_{prop} = \frac{200-125}{1.8-1.2}$	125 m/s

Table 5.2: Speeds calculated from slopes in figure 4.9.

From the speeds calculated from the MPM strain results shown in table 5.2 it becomes more clear which wave types are observed.

The crack propagation speed of 57 m/s, which is in the sub-Rayleigh regime, is consistent with the conditions needed for transition between the sub-Rayleigh and Supershear regimes. Trottet et al., 2022 state that on flat terrain, the normal stress at the crack tip is higher than the shear stress in the same location. This drives anticrack propagation, which is in the sub-Rayleigh regime (Gaume et al., 2018; Trottet et al., 2022). For a transition to the Supershear regime, a slope would be needed which drives a transition from a mixed-mode anticrack propagation to pure mode II crack propagation (Trottet et al., 2022). The moment tensor components which were used as STFs shown in figure 3.5 all show negative amplitudes. For a mode II crack, which is in-plane shear, the component shown in 3.5b would have to have an alternating sign (red and blue), alongside m11 and m22 shown in 3.5a and 3.5c very small or zero. Since all three components have a strong negative amplitude, a mixed-mode anticrack is more likely, which would lead to sub-Rayleigh crack propagation.

The speeds of the forward and backwards-propagating waves emitted by the crack is the same in the positive as well as the negative direction. This means that this type of wave is emitted on both sides of the crack. Its speed is between the p-wave speed of the weak layer and the p-wave speed of the snow slab (table D). This is consistent with the results discussed in the simple model section 5.1, where a sub-Rayleigh crack propagation regime emits an S0 precursor, which is also consistent with field observations of Herwijnen and Schweizer, 2011. This is confirmed by the traces in appendix K, figure K.3. These traces show oscillations of strain rate values between $-1 * 10^{-5}$ and $0 * 10^{-5}$. The pattern of oscillation between extension and compression discussed in section 5.1.2 associated with the S0 wave mode is also present in this case. The values of strain and strain rate are predominantly negative however, this is because of the quasi-static strain field, which will be discussed in the next section.

The third wavefront which is observed in figure 4.9, has a much lower propagation speed than all other wavefronts. This wavefront also broadens with time and the waves curve downward, towards the origin of the coordinate system. This wavefront could be the dispersive A0 Lamb wave mode, because at low frequency thickness product, and at the dominant frequencies present in the model, has a group velocity which is significantly below the S-wave speed of the slab (Rose, 2014).

Wavefield Structure & Quasi-Static Strain Field

The most striking feature identified in the strain and strain rate plots is the thick bands of positive and negative polarity coming from the crack onset. In the positive direction of propagation, the first arrival is a positive strain or strain rate, followed by a negative (blue) band of negative strain (rate). In the negative direction of propagation, this order is reversed, the negative polarity arrives first. This band changes when the crack has stopped: at x=125m, the positive wavefront suddenly jumps to be wider. In the region of the crack, so from 75m to 125m, it was a narrow wedge whereas the blue region was very broad.

The positive wedge of energy is the quasi-static strain field, as described by Freund and Freud, 1998. It builds up progressively between $x = 75$ m and $x = 125$ m as the crack propagates, then transitions abruptly to a broad negative region (blue) after the crack arrests at $x = 125$ m. This red wedge is narrow at the crack onset ($t \approx 0.5$ s, $x = 75$ m) and widens to its maximum at the crack arrest ($t \approx 1.2$ s, $x = 125$ m), tracing the spatial footprint of the growing crack. The abrupt widening of the positive front at $x = 125$ m marks the moment of arrest, where the accumulated quasi-static compression is released as the crack stops, and the strain field relaxes into its post-crack configuration. This arises because the moment tensor STF has a non-zero DC-component and there is permanent deformation, which is described in section 2.4. This means that the weak layer undergoes permanent deformation after the crack has passed. The deformation resulting is collapse, which is consistent with an anticrack as described by Heierli et al., 2008; Gaume et al., 2018; Trottet et al., 2022, which can trigger avalanches. This is also supported by the moment tensor components shown in figure 3.5, where all components are predominantly negative. The region before crack propagation (white) can be clearly distinguished from the region where the crack is propagating or has propagated, which have a negative value. This means that there are no restoring forces after the crack has propagated, which means that the deformation stays in place. This is further supported by the cumulative integral of all moment tensor components shown in figure 3.6. All three cumulative integrals approach a negative value, which means that the displacement caused by them is also negative. In other words, because of the asymmetric shape of the moment tensor source time functions used in the MPM simulation, which do not integrate to 0 over time, there is a net displacement. Since all components result in a negative displacement, this results in an anticrack, or weak layer collapse.

The quasi-static wavefield is contained within a v-shaped wavefront originating from the crack onset at x=75m. It shows high amplitude lines of positive and negative strain which are alternating. In the positive direction ($x > 75m$), the first dynamic arrival is positive strain or strain rate (red) followed immediately by negative strain (rate), consistent with the extension-first, compression-second polarity sequence of P-wave or S0 Lamb mode radiation (Aki and Richards, 1981). This can also be seen in the traces close to the crack onset, shown in figure K.3. In the negative direction ($x < 75m$), this order is reversed because the measurement is on the opposite side of the source: the compressive push from the crack initiation reaches a point to the left as a compression-first arrival. The slope of the forward-propagating arm gives the crack propagation speed ($v_r \approx 57m/s$, in the region of the crack, and $v \approx 125m/s$ after the crack has propagated, seen in table 5.2). The slope of the backward-propagating arm gives the P-wave or S0 speed ($v \approx 125m/s$), consistent with the precursor identified in the simple model results and with field observations of Herwijnen and Schweizer, 2011.

The second dispersive wave train observed in the crack depth panels on the left side of figure 4.10 is the third feature which was identified. This front manifests as lower amplitude lines propagating from the crack onset. They alternate in polarity and have a steeper slope, which means lower apparent velocity (see table 5.2). These are the same alternating polarity bands identified in the simple model as the dispersive A0 Lamb wave mode. Their identification as A0 is confirmed in the velocity discussion, where the polarity reversal across the slab midplane is directly visible. Unlike the dynamic v-shaped wavefronts, these secondary bands broaden with increasing time and are absent from the snow bottom panel (right column), consistent with the evanescent depth decay of the A0 mode at this frequency-thickness product (Rose, 2014), which was also identified in the simple model discussion.

Curving and Broadening of Wavefronts

The secondary wavefronts observed in the crack depth plots, which are shown in the left column of figure 4.9, spread out with time. This spreading, as in the simple model discussion, can be explained by dispersion. This is consistent with the dispersive A0 Lamb wave mode discussed in section 5.1. The same dispersion mechanism applies here: reflections between the slab boundaries produce interference between frequency components traveling at different phase velocities, broadening and curving the wave train envelope with time (Graff, 1975).

5.2.2 Velocity Space-Time Plots

Similarity to Strain (Rate) Plots

As discussed in section 5.1, velocity, strain and strain rate are all derivatives of displacement. Since the horizontal component of strain and strain rate is shown and discussed for the MPM model, the velocity discussion section will only go into detail about phenomena which can be seen better or exclusively in the velocity plots.

Propagation Speeds

The propagation speeds of the wave types observed in the strain and strain rate plots shown in table 5.2 are consistent with those observed in the velocity plots in figure 4.10. The very saturated wavefronts in the horizontal component correspond to the same slope as the S0/P-wave precursor observed already in the strain and strain rate plots. The same is true for the very faint wavefront observed in the vy plot at crack depth (top left panel of figure 4.10), which also has the same slope as the S0 wave. This means that the precursor can also be observed in the velocity data.

The sub-Rayleigh crack propagation speed which was calculated from the strain and strain rate plots is the same speed which can be calculated from the slope of the line connecting the crack onset and arrest in figure 4.10.

Quasi-Static Offset

The quasi-static offset is also present in the velocity plots, mainly the horizontal velocity plots in the bottom row of figure 4.10. Contrary to the strain and strain rate plots, the first arrival in the positive direction of propagation has negative velocity, followed by positive velocity, but it is also reversed in order in negative direction of propagation. This polarity reversal between velocity and strain for the same compressive anticrack source is expected, because a compressive source produces extensional strain ahead of the source but a negative initial velocity as material is displaced away from the crack before elastic restoring forces act (Aki and Richards, 1981). The quasi-static offset can also be seen in the animation of the velocity wavefield in appendix F, figure F.3, where the quasi-static offset manifests as broad areas of negative velocity followed by positive velocity. This is the kinematic signature of anticrack collapse: the material above the weak layer is permanently displaced in the negative direction as the weak layer fails under pressure. This is consistent with the anticrack model described by McClung, 1981, and the negative DC component of the moment tensor shown in 3.5.

Lamb Wave Identification

Like the strain and strain rate plots, all velocity plots except vx at the snow bottom, show secondary wavefronts of alternating polarity. This can also be seen very well in the animation of the vertical component of the wavefield, a still of which is shown in appendix F, figure F.4, where behind the crack location bands of red and blue (positive and negative vertical velocity) alternate. The same can be seen in the vx still, but here the bands are much more broad.

In both animation stills F.3 and F.4, the crack is located at 100m. In the animation of the vertical velocity component, the velocities not only alternate in polarity, but they also switch polarity on either side of the crack. The crack depth, which is in the middle of the weak layer for example has positive polarity above it and negative polarity below it, which is characteristic for the asymmetric Lamb wave mode A0. The detailed derivation of the Lamb wave equations can be found in section 2.3. This polarity pattern is also consistent with figure G.1 in appendix G, showing the propagation intensities of the symmetric and antisymmetric Lamb wave modes. The A0 Lamb wave mode dominates the MPM wavefield for the same reason as in the simple model: the compressive normal component $m_{22} < 0$ of the moment tensor (shown in figure 3.5c) is oriented normal to the slab and maximally couples to the A0 flexural mode at the low frequency-thickness product of this geometry (Rose, 2014). The MPM results therefore confirm the Lamb wave mode identification made in the simple model.

Rupture Arrest Structures

In the vertical velocity plots in figure 4.10, two secondary wavefronts can be observed. One coming from the crack onset, which is in the foreground and can be seen clearly, and the other coming from the crack arrest point at x=125m. This structure also has a v-shaped structure, which is consistent with the rupture arrest structure observed in the simple sub-Rayleigh plots.

Again, the crack never outruns the waves which are radiated from it, which means that the stopping phase can radiate in all directions (Freund and Freud, 1998). The crack arrest at x=125m produces a v-shaped stopping-phase signature consistent with sub-Rayleigh arrest, analogous to the arrest structure identified in the simple model (section 5.1). Both arms of the V propagate at the P-wave and S0 speed of the slab (125 m/s, measured in table 5.2), confirming that the stopping phase radiates in all directions at the bulk wave speed, independent of the pre-arrest rupture speed (Madariaga, 1977).

5.2.3 F-k Plot

The f-k plot confirms the structures which have been observed in the previous sections. The f-k spectrum of the MPM-guided simulation (figure 4.12 and 4.13) shows two pairs of symmetric energy bands, one on each side of the wavenumber axis.

Wavefront & Propagation Regime Identification

The steep inner bands in figure 4.12, aligned with the P-wave speeds of the slab and weak layer ($v_p \approx 127m/s$ and $v_p^{\mathrm{wl}} \approx 116m/s$, shown by the red and white lines), correspond to the S0/P-wave precursor radiating from the crack initiation and arrest points at the fastest available wave speed. This can be seen when comparing figure 4.12 to 4.13, where the Lamb wave dispersion curves obtained in section 2.3 for a 0.15m slab are plotted in orange (A0) and green (S0). The S0 line falls in the center of the energy concentrated in figure 4.13. This is not just because the p-wave speeds in the slab and in the weak layer are close in value, but also because the crack initiates in the weak layer and radiates into the slab. These bands appear on both the positive and negative wavenumber sides because the precursor radiates in both directions from the initiation point at $x = 75m$ consistent with the forward and backward propagating arms of the v after the crack locations, measured at approximately 125m/s in table 5.2 and with the precursor identified in the simple model sub-Rayleigh results. Because the slab and weak layer P-wave speeds are nearly identical in the MPM material (table D), which are also close to the A0 and S0 velocities in the slab, the combined energy band appears broader than either line alone. The presence of the S0 precursor is further confirmed by the oscillatory pattern of the strain trace shown in figure K.3. This is consistent with the same precursor observed in field measurements by Herwijnen and Schweizer (2011), where a fast P/S0-wave arrival preceded the main crack signal.

The shallower outer bands, with a slope corresponding to $v_{rupture} \approx 57m/s$ from table 5.2, are the crack propagation front itself. The crack radiates energy both forward (in the direction of propagation) and backward (as each source fires and sends energy behind the advancing tip), producing bands on both the positive and negative wavenumber sides. The positive side is more concentrated and stronger in amplitude, reflecting the net forward propagation of the crack from $x = 75m$ to $x = 125m$. The apparent slope of these bands gives a rupture speed consistent with the direct measurement from the strain panels, providing an independent f-k confirmation of the sub-Rayleigh propagation regime.

Lamb Wave Identification

The distinct features associated with A0 dispersion observed in the simple model between the rupture speed and P-wave speed lines in figure 4.5, does not appear here. This is consistent with the shorter crack length in the MPM model (40m versus 240m in the simple model): a shorter source array produces less coherent beaming of A0 energy into a concentrated f-k ridge, spreading it instead into a diffuse background (Rose, 2014). This is consistent with the dispersion curves in figure 4.13, which shows that the A0 Lamb wave dispersion curve shown in orange falls along the slab p-wave velocity.

Symmetry of f-k plot

Unlike in the simple case f-k-plot, the spectrum shown in figure 4.12 is symmetric around positive and negative wavenumbers. Unlike in the simple model where the long crack (240m) produced strong directivity asymmetry between positive and negative k, the shorter MPM crack (40m) has weaker directivity because the directivity effect scales with crack length relative to wavelength (Madariaga, 1976).

5.3 Implications for DAS Measurements & Synthesis of Results

5.3.1 Signal to Noise Ratio and Field Detectability

To assess whether the signatures in the simulated data in this thesis are even discoverable by a DAS cable, a comparison with the noise spectral density of the Silixa iDAS instrument (Parker et al., 2014; Silixa, 2022) was done. This instrument was chosen because it was used for avalanche detection in the Vallée de la Sionne test site in the Swiss Alps by Paitz et al., 2023; Edme et al., 2023. For a gauge length of 10m, similar to the one used by Turquet et al., 2024 and this thesis, the amplitude spectral density, which describes how much noise is present per frequency bandwidth (Hartog, 2017), is $2p\epsilon per\sqrt{Hz}$ (Silixa, 2022). This noise floor is used to calculate the signal to noise ratio for the results in sections 4.1, 4.3 as described in section 2.6. The input values for equations 2.20 and 2.21 can be seen in table 5.3.

Parameter	Input Value
Amplitude Spectral Density	$2\ \mathrm{p}\epsilon/\sqrt{\mathrm{Hz}}$
Δf Simple Case	50 Hz
Δf MPM	30 Hz
ϵ_{peak} Simple, Sub-Rayleigh	3×10^{-6}
ϵ_{peak} Simple, Supershear	3×10^{-5}
ϵ_{peak} MPM	2×10^{-5}

Table 5.3: Input Parameters for signal-to-noise calculations.

From these values, the signal-to-noise ratios for the individual cases are shown in table 5.4.

Case	$\mathbf{RMS_{Noise}}$ (from Eq. 2.20)	SNR (from Eq. 2.21)
Simple Sub-Rayleigh	$0.0141\ \mathrm{n}\epsilon$	212'132
Simple Supershear	$0.0141\ \mathrm{n}\epsilon$	2'121'320
MPM	$0.0110\ \mathrm{n}\epsilon$	1'825'742

Table 5.4: Outputs from signal-to-noise calculations.

These calculations show that the DAS signals measured here can be detected against the noise floor of Silixa iDAS. However, these calculations do not account for coupling or environmental noise, which can be present in field data.

5.3.2 Synthesis of Results

The MPM-guided results have several implications for the interpretation and measurement of field DAS data. They extend and support the conclusions drawn from the simple model results.

Before comparing the MPM and simple model results directly, it is important to establish that the wave signatures identified in the two-layer simple model are not an artifact of the simplified geometry. To test this, the same sub-Rayleigh and Supershear prescribed rupture speeds used in the simple model were applied in the five-layer MPM geometry shown in figure 3.2 using Ricker source time functions, as described in section 3. The resulting strain rate and velocity panels (appendix J) show the same two-wavefront structure, the same dispersive A0 wave train, the same rupture arrest signatures, and the same amplitude depth dependence as the two-layer simple model results. This means that the five-layer geometry does not introduce additional wave types or change in the DAS signatures identified in section 5.1.

This confirms that, firstly, the physical wave signatures discussed throughout this thesis are robust to the choice of model geometry and are not specific to the simplified two-layer domain. Second, any differences observed between the simple model and the MPM-guided results can be attributed to the MPM source time functions and the more realistic physics rather than to the geometry change. This isolates the effect of the source physics on the DAS output. This separation is important for interpreting the additional features observed in the MPM results, specifically the quasi-static strain field and the different amplitude structure, as consequences of the MPM source rather than consequences of the five-layer geometry. This is why the two results can be compared to each other in the following section even though they have a different domain geometry.

The crack propagation speed measured from the MPM results ($v_r \approx 57 m/s$, table 5.2) is sub-Rayleigh for both the slab and weak layer material parameters. This is consistent with studies by Trottet et al., 2022; Siron et al., 2023; Bergfeld et al., 2025, which show that transition between sub-Rayleigh and Supershear crack propagation occurs at high slope angles and the normal to shear stress ratio at the crack tip. Because of the moment tensor input, which shows constant mixed-mode anticracks and no indication of pure mode II, which would be indicative of the transition (Trottet et al., 2022), a sub-Rayleigh regime is not only supported by the results, but also by the input. The DAS signatures observed in the MPM strain rate panels (figure 4.9) and velocity panels are therefore those of a sub-Rayleigh crack at physically realistic material parameters and STFs, rather than the simplified speeds and Ricker wavelets used in the simple model. It is important to note here that since the MPM-guided results are sub-Rayleigh, there is no verification for the effects observed in the simple case Supershear model. The two-wavefront structure, which is a fast S0/P precursor and a slower crack front, is present in the MPM results as well as in the simple model, confirming that this signature is not an artifact of the prescribed source geometry but reflects genuine wave physics of sub-Rayleigh crack propagation in snow. This is also confirmed by field data by Herwijnen and Schweizer, 2011.

In addition to the S0 precursor identified in the simple model, another effect which could be used for early warning was observed in the more complex simulations. The most significant additional effect observed in the MPM results is the quasi-static strain field, which is absent from the simple model but clearly visible in the MPM strain panels (figure 4.9, bottom row). A DAS cable records axial strain along the cable (Hartog, 2017), which means it would record both the dynamic wave radiation (visible in strain rate) and the cumulative static deformation (visible in strain) simultaneously. The quasi-static field builds up progressively as the crack propagates and persists after arrest, with no decay in amplitude, providing a sustained signal that does not decay with distance in the same way as the dynamic wavefield. In principle, this static component could be detectable as a slow, gradual change in DAS strain before the dynamic wavefront arrives, providing an early indicator of crack growth that precedes the impulsive dynamic signal (Freund and Freud, 1998; Heierli et al., 2008).This means that additional to the S0/P-wave precursor, which in some cases is weak in amplitude, there is an additional precursor in the quasi-static field, which could help in early warning of avalanche nucleation. Whether this signal would exceed the environmental noise, which is beyond the scope of this thesis and remains an open question for field validation, this also depends on the coupling of the instrument.

Additional to environmental noise, there is also instrument noise, which is caused by changes in the laser pulse or interference of the signal as it travels through the cable for example (Lindsey and Martin, 2021). This poses the question if the signatures discussed can even be discovered against the instrument noise, even when there is no environmental disturbance. An assessment for the instrumental noise floor of the Silixa iDAS system used by Paitz et al., 2023 is done in section 5.3.1. These calculations have shown that, neglecting environmental noise, the precursor, arrest and main wavefront signals should be discoverable over instrument noise.

The detectability of these signatures is dependent on the noise in the recorded data, but also on their amplitude. The amplitude of the MPM strain rate at the snow bottom (figure 4.9, lower right) is higher compared to the simple model sub-Rayleigh case. This is consistent with the thinner slab geometry of the MPM model (0.3m versus 1.5m in the simple model), which reduces the vertical distance over which evanescent amplitude decay acts between crack depth and snow bottom. The implication is that the detectability argument from the simple model that sub-Rayleigh signals at the snow bottom may be too weak to detect, applies less severely here because the slab is thinner. In a real snowpack with a thicker slab, the amplitude at the cable would be correspondingly lower, making the simple model's detectability concern more relevant for typical alpine snowpack geometries where slab thickness can exceed 1.5m (Schweizer et al., 2003; Bartelt and Lehning, 2002).

As in the simple model, the rupture speed can be identified from the slope of the space-time velocity, strain and strain rate plots as well as the frequency-wavenumber plot. While the precursor is not as clearly visible in the MPM-guided results compared to the simple case, it can also be detected via its rupture speed. In the MPM results, the time separation is smaller than in the simple model because $v_r \approx 57m/s$ is closer to $v_p \approx 125m/s$ than the simple model sub-Rayleigh speed of 40m/s is to $v_p = 109$m/s. The 10.4m gauge length used here is sufficient to resolve the two arrivals in the simulation, as shown in figure 4.9 via their slope, suggesting that a field DAS deployment with comparable gauge length to the one used by Turquet et al., 2024 could in principle measure the rupture speed of a sub-Rayleigh crack directly from the arrival time separation.

The last implication for DAS measurements of crack propagation in snow is only observed in the simple model. Since the Supershear velocities observed there have higher amplitudes than their sub-Rayleigh counterparts, sub-Rayleigh crack propagation regimes might not be observed because they have too low amplitude. This is especially important for transitional regimes, such as the one described by Trottet et al., 2022; Siron et al., 2023. If the crack propagation starts out as sub-Rayleigh, which sends out a weak signal and then transitions into Supershear with higher signal amplitude, a false conclusion of pure Supershear rupture might be drawn if the sub-Rayleigh regime is too weak to be measured.

5.3.3 Graphical Representation of DAS Signature Identification

All points made in the section above are represented graphically in the flowchart in figure 5.1. This figure was created to have a clear overview of the different signatures in DAS data and what they are associated with.

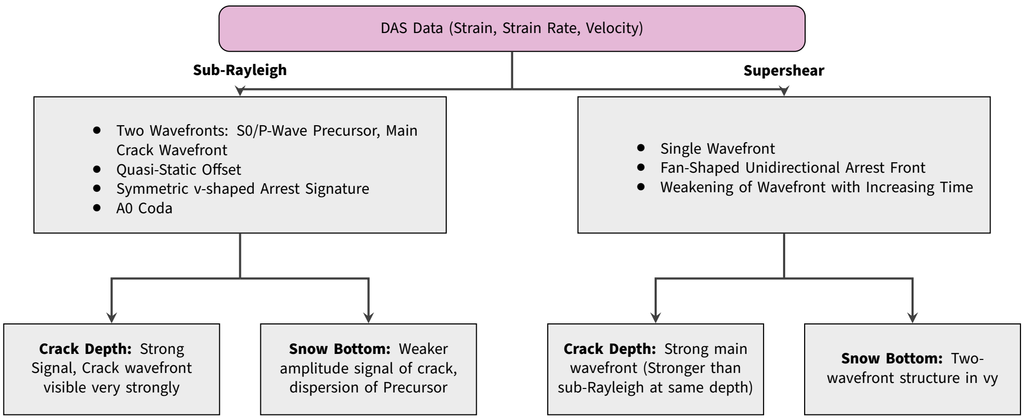


Figure 5.1: Graphical Representation to Identify Different Signatures in DAS

Chapter 6

Outlook

The results of this thesis establish a baseline for DAS signatures of sub-Rayleigh and Supershear crack propagation in a simplified geometry, the following additions would improve the physical realism and practical applicability of the model. Based on the results and methods presented in this thesis, there are many open lines of inquiry. Because the model described in the simple case is a basic two layer model, a next step could be to add multiple layers, especially because cracks in snow usually propagate in a weak layer (Schweizer et al., 2003). While this was done partially in the MPM-guided simulations, where a 5-layer model was implemented, the snow slab was still approximated as a homogeneous material with the same model properties above and below the weak layer, which is a major simplification. Here, there is also possibility to run the simulations described in this thesis for different snow parameters (density, v_p and v_s, elastic modulus), because Bartelt and Lehning, 2002; Mulak and Gaume, 2019; Bobillier et al., 2020 have all shown that snow properties not only change over time, but also have an influence on crack propagation. This would be useful to better understand DAS data from different times of the year or day, when there is the possibility of snow properties having changed.

In both simulation types run, there is also the possibility of adding topography or a slope angle to the simulation, because as indicated by Ancey, 2001; Trottet et al., 2022, avalanches are gravity driven and the crack propagation regimes can change greatly with the presence of topography, which also has an effect on the seismic signatures. This would also help identify how the signatures observed here change when the DAS cable is not on flat ground. All of the results shown in this thesis assume perfect coupling, which is not always the case in the field, especially when topography is present, this is another reason why adding topography is important.

Because of the possibility for transition between the different propagation regimes, meaning that the propagation speed changes, as described by Trottet et al., 2022, which is not only driven by topography and slope angle, the constant propagation speed prescribed in the simulations in this thesis are a major limitation. Another next step would be to have source propagation accelerate with increasing propagation distance, instead of having a constant propagation speed, which is not very plausible on a mountain slope. The MPM model which was used to obtain the moment tensors in this case had a slope angle of 45 degrees (Trottet et al., 2022). This means that there is no transition between modes in the data itself, but using these moment tensors obtained from data at a slope in flat terrain in SALVUS is also a major simplification.

Furthermore, the crack modeled in this thesis is linear and at constant depth. The moment tensor extraction from MPM used here averages the stress field over the weak layer thickness, which loses information about the vertical structure of the source. MPM simulations of weak layer cracking and collapse have shown that this is not the case (Trottet et al., 2022), so a next step would be to make the crack propagation modeled by seismic sources more physical. This could be done by having "jumps" of the crack into different depths of the snowpack or having multiple lines of sources firing at the same time to represent multiple smaller cracks. In the MPM case, a more refined approach would be extracting the moment tensor at individual MPM particle positions without averaging, preserving the depth-dependent stress distribution and potentially revealing differences in the Lamb wave mode excitation that are hidden by the averaging. This would also allow the crack depth to vary with propagation distance.

Another limitation is that the simple and CZM simulations cannot account for the physical collapse of the weak layer that characterizes anticrack propagation (Heierli et al., 2008; Heierli et al., 2003).This is because spectral element methods solve only the elastic wave equation in the surrounding medium and cannot represent inelastic failure of material elements without explicit element removal (Podolskiy et al., 2013). The prescribed moment tensor source in these models approximates the mechanical effect of the collapse but does not model the collapse kinematics themselves. In the MPM-guided simulations, this limitation is partially addressed: the inelastic collapse process is modelled fully within MPM, and its mechanical effect on the surrounding elastic medium is transferred to SALVUS via the extracted moment tensor time series. However, the elastic wave field generated by the collapse in the MPM simulation is not re-injected into SALVUS, only the quasi-static stress redistribution is captured through the moment tensors. Dynamic wave radiation generated within the MPM domain itself is therefore not included in the SALVUS simulation. The presence of these signals could also change the seismic signal. This could be verified by field tests of weak layer collapse measured by a DAS cable.

The quasi-static strain field identified in the MPM results could be further investigated as an early warning signal. The amplitude of this field relative to the DAS noise floor determines whether it would be detectable in a field deployment, and this depends on the specific DAS instrument, gauge length, and environmental noise. A field comparison using a PST experiment instrumented with both DAS and geophones would allow the quasi-static component to be identified in real data, validating the MPM simulation prediction. Such field experiments have been conducted for dynamic wave signatures (Bergfeld et al., 2025; Herwijnen and Schweizer, 2011) but the quasi-static DAS signature has not yet been specifically targeted.

In addition to the limitations described, the MPM data used to obtain the moment tensors resulted in a sub-Rayleigh rupture. While the results confirm the signatures observed in the simple model case, such validation was not obtained for the Supershear results or for a transitional regime. This would require MPM simulation data of those regimes, which was not available in this thesis. Here, there is a lot of potential for further validation of the signatures observed, because although they seem like useful precursors and identifiers in this data, more validation is needed.

The last step in making the model described here physically accurate is to implement the 2D model described here in 3D, which would help to understand slope DAS data even better. Such a 3D model could represent data recorded further away from the crack, in this thesis, the 2D geometry constrains the cable to be parallel and or co-planar to the crack, whereas in the field, cables may be oriented at arbitrary angles to the crack propagation direction. It would also be very interesting to compare different cable orientations relative to the crack to see if the wavefield data recorded changes.

Chapter 7

Conclusion

The aim of this thesis was to contribute to the understanding of seismic signatures of crack propagation and avalanche breakoff. This was done by setting up a moving-source model of a crack propagating at sub-Rayleigh and Supershear speeds in snow using spectral element modelling in SALVUS. The results of the simple two-layer model were evaluated to determine how the seismic signature measured by a DAS cable is influenced by the crack propagation regime. To make the physics more realistic, the cohesive zone model for mixed-mode crack propagation was set up. This was then followed by a model which translated the results of material point method simulations by Trottet et al., 2022 into seismic moment tensors. The results of this simulation were evaluated to verify effects observed in the simple model at more realistic conditions.

In this thesis, a 2D medium was set up using the spectral element solver SALVUS. Two types of domain geometry were set up: a simple snow-air case and a more complex 5-layered model consisting of the bedrock, lower and upper snow slab embedding a weak layer as well as an air layer. The materials set up were assumed to be homogeneous. In these two geometries, three types of models were set up: a simple case with a Ricker source wavelet with fixed source spacing and prescribed speed, a cohesive zone model which can account for mixed-mode cracks and a damage zone ahead of the crack, and source time functions obtained from material-point method simulations. The latter do not only account for mixed-mode cracks, but also show what signals from plastic deformation look like.

The results of these three model cases were combined with the Pyber library, which applies a gauge-length smoothing over the data. This was done so that the data obtained by the numerical models looks closer to data obtained in the field, where a DAS cable averages measured strain over a gauge length (Noe, 2025; Lindsey and Martin, 2021).

The results obtained through the 2 layer Ricker wavelet simulations showed key phenomena for crack detection and regime distinction from DAS data. Firstly, in the sub-Rayleigh results, there are two distinct wavefronts which can be observed in the strain, strain rate and velocity plots. The broad main wavefront has a slope which corresponds with the prescribed propagation speed. This is also supported by the f-k plots, where there is a clear energy band at the propagation speed. The second, faster wavefront, is believed to be the symmetric Lamb wave mode precursor S0. It travels close to the P-wave speed, because its propagation speed is limited by the plate wave speed (Rose, 2014). This precursor occurs because of lack of interference at the first and last source in the array (Madariaga, 1976), when source spacing is increased, it is emitted by every source in the array. A precursor to the main crack signal, which has close to P-wave speed has also been observed in field data by Herwijnen and Schweizer, 2011. This precursor is important for early detection for cracking, but it might be hard to see in field data because it has low amplitude compared to the main wavefront.

Contrary to the sub-Rayleigh case, the Supershear strain (rate) and velocity plots only show one main wavefront. This wavefront has the same slope as the rupture speed, which is also evident from the frequency-wavenumber plots, where all energy is concentrated around the rupture speed, the P-wave speed and the A0 and S0 wave speeds, because these are all within a couple of meters per second of each other. This means that the rupture and the S0 wave emitted by the first and last source are contained in the same wavefront, they can still be distinguished in some plots, however.

In both the Supershear and the sub-Rayleigh simple model results, the main wavefronts show polarity reversal with increasing time. This alongside animations of the polarity of the velocity wavefield are indicative of the antisymmetric Lamb wave mode A0. This is consistent with the results of the f-k plot, where the main concentration of energy lies along the A0 dispersion curve. This mode is excited when the source coupling is normal to the slab midplane (Rose, 2014), so in this case the moment tensor with a negative yy component with no counterpart in the positive yy direction. This leads to maximal excitation of the A0 mode. This is confirmed when combining the elastic wave equation with the boundaries present in the medium, which leads to the A0 dispersion equation.

Another effect which was observed in both propagation regimes is the presence of a rupture arrest structure. While this structure is present in both cases, it manifests in different ways. In the Supershear propagation regime, the rupture arrest structure is an unidirectional fan, while in the sub-Rayleigh case, it shows up as an inverted v-shape coming from the rupture arrest point. The different structures can be explained by the fact that a Supershear rupture propagates at a faster speed than the shear waves emitted by it, so the accumulated energy at the rupture arrest propagates into the undisturbed medium ahead of the crack tip (Madariaga, 1977). In the sub-Rayleigh case, the rupture never outruns the waves emitted by it, so the stopping phases can travel to all directions (Freund and Freud, 1998). The distinction between the rupture arrest signatures between the two propagation regimes could be important in identifying the type of a rupture in field data.

The last main conclusion drawn from the simple model results is that there is a significant amplitude difference between Supershear and sub-Rayleigh ruptures. In the strain plots, Supershear regimes have one power of 10 more energy than sub-Rayleigh propagation regimes. A difference in amplitude was also observed between sub-Rayleigh crack propagation at different depths: the amplitude is higher at the crack depth compared to the snow bottom. Both of these amplitude dependencies have key implications for DAS measurements. Because sub-Rayleigh cracks already emit less energy than Supershear cracks and this effect is only amplified when the cable is at the bottom of the snow, it might be possible that sub-Rayleigh crack signals are not recorded. This is not only important for avalanche detection but also when considering that there is usually a transition in propagation speed (Siron et al., 2023; Bobillier et al., 2024). This might lead to false conclusions of a pure Supershear crack propagation, because the sub-Rayleigh onset is too weak to be measured.

The propagation speed measured from the slope of the space-time strain and strain rate plots of the MPM-guided model indicates a sub-Rayleigh rupture. This means that the sub-Rayleigh effects described in the above sections were confirmed by the more physical model. The same v-shaped rupture arrest signature and the S0 precursor was observed. The same is true for the main wavefront, which in the space-time, f-k plots and the animations shows that the main wavefront consists of the antisymmetric Lamb wave mode.

In addition to the sub-Rayleigh signatures already observed in the simple model, the MPM-guided results also showed the quasi-static strain field associated with the nonzero DC-component of the source wavelet (Freund and Freud, 1998). The moment tensor source-time functions integrate to a negative value over time, which means that there is irreversible deformation in the negative direction, consistent with anticrack propagation or weak layer collapse (Heierli et al., 2003; Heierli et al., 2008; Gaume et al., 2018). Such effects are absent in the simple model case because the source wavelet, a Ricker wavelet, is symmetric and therefore integrates to zero over time. The quasi-static field has effects on the strain and velocity even before the main crack front has arrived, which means that it can also be used as a precursor to the main crack arrival. This precursor might even be more indicative than the S0, because it is stronger in amplitude.

The signatures identified in the simple model were confirmed to be geometry-independent by injecting Ricker source time functions into the five-layer MPM geometry, which produced the same wavefield character as the two-layer model.

Concluding the observations made in the results of this thesis, there are multiple implications for DAS detection of crack propagation and avalanche breakoff. Firstly, the sub-Rayleigh propagation regime might be undetectable at the snow bottom, which is where DAS cables are usually deployed. This is because the amplitude of the signal is lower at the snow bottom than at the crack depth and because sub-Rayleigh cracks emit less energy than Supershear cracks. Secondly, the rupture arrest signature can be used to distinguish between sub-Rayleigh and Supershear propagation regimes. Thirdly, if only the Supershear phase is detected, it might lead to false conclusions about the type of crack propagation, because there is usually a transition in propagation speed (Siron et al., 2023; Bobillier et al., 2024). Lastly, the quasi-static strain field observed in the MPM-guided results as well as the S0 mode emitted by the first source in all models could be used as an early warning signal for crack propagation and avalanche breakoff.

Although there are multiple insights in this thesis which could be useful for avalanche breakoff detection using DAS measurements, there are limitations which need to be considered. The results shown here are all in 2D on flat terrain. It is likely that topography in 3D will introduce additional effects which can cause the signatures described above to change. Furthermore, SALVUS solves for an elastic material, there is no attenuation or viscous effect and the presence of pore spaces in the snow is neglected. This means that the material here is highly idealized, which also becomes evident when comparing it to the snowpack in the field, which can vary throughout the day, change with temperature and compaction (Bartelt and Lehning, 2002). This change and inhomogeneity in material properties can have great effects on crack propagation in snow (Mulak and Gaume, 2019; Ancey, 2001) and might change the signals observed.

In addition to the simplifications made in the geometry and material properties, the rupture speed which was prescribed here is also constant. Multiple studies have shown that this is not the case for crack propagation in snow, as multiple factors cause crack propagation to transition from sub-Rayleigh to Supershear (Siron et al., 2023).

While the effects observed in the sub-Rayleigh simple model were validated by the MPM-guided results, there is no such validation for the effects observed in the Supershear case. Lastly, there is no validation of these effects with field data, which makes it highly likely that effects such as the S0 precursor or the quasi-static strain field cannot be observed in the field due to environmental noise in the DAS data. As demonstrated by test calculations for the Silixa iDAS cable, interrogator noise should be lower than the recorded signal.

The limitations of this thesis as well as the results presented lead to many future lines of inquiry, which have been discussed in the outlook chapter 6. As a next step, topography could be added to the models here, before they are extended in three dimensions. A three-dimensional geometry would also allow for different DAS cable orientations relative to the crack propagation direction, which is useful in understanding field data. The results presented here could be assimilated further to field data by using a transitional crack propagation speed and heterogeneous snow properties, which would also help understand how the signatures change as the snow properties change. The last step of validation would be to set up a DAS cable when performing a PST to determine which signatures can be observed under field conditions.

Appendix A

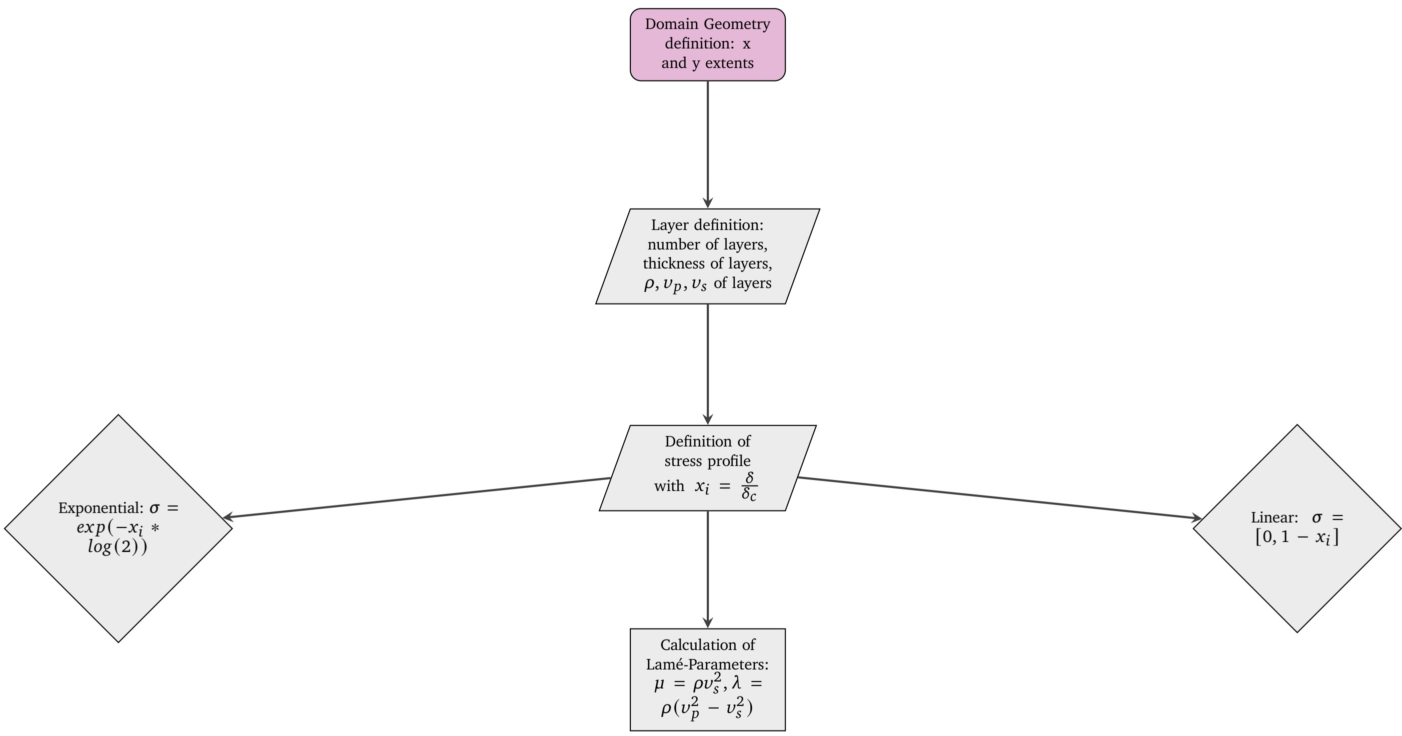
Code Repository

The complete code of all models and plots described can be found on GitHub: https://github.com/s-bachmann/Master-Thesis. Meeting slides contains slides used for intermittent meetings with supervisors, Report contains all the LaTeXdocuments and auxiliary files used to write this thesis, and Test Simulations contains all the simulation and plotting code.

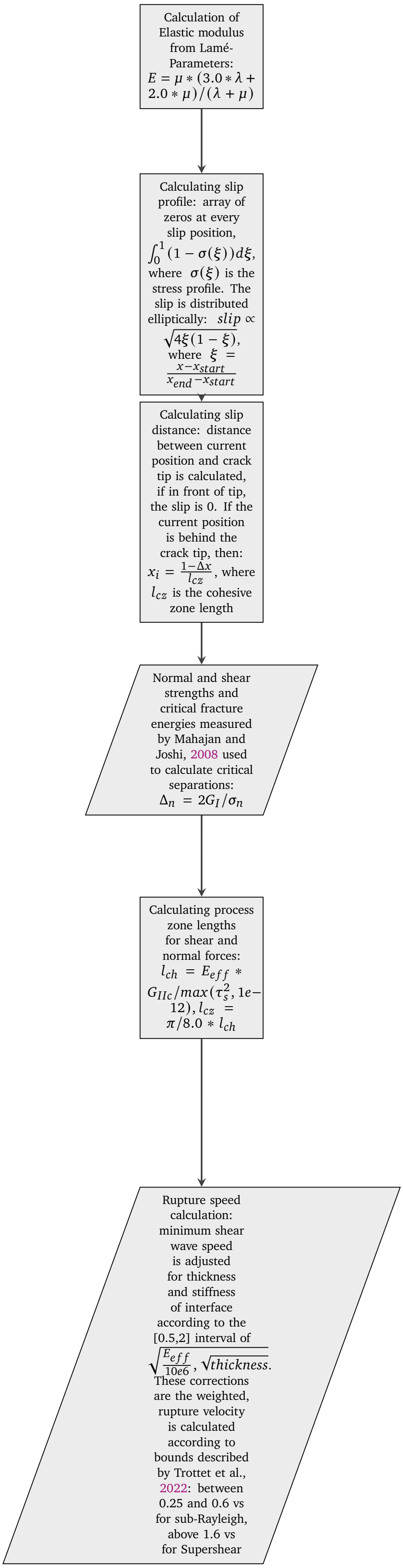
Appendix B

Code Workflow for CZM Model

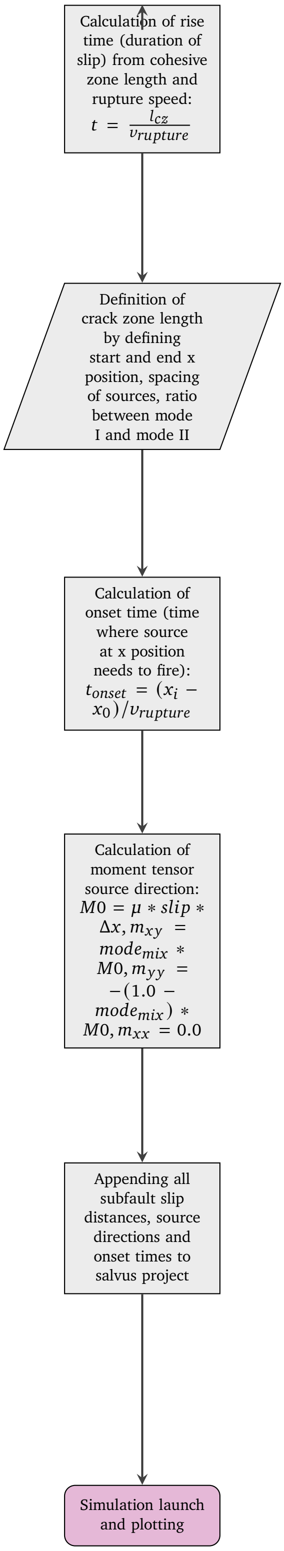
Part 1: Domain and Stress Profile Setup



Part 2: Material Properties and Process Zone Calculation

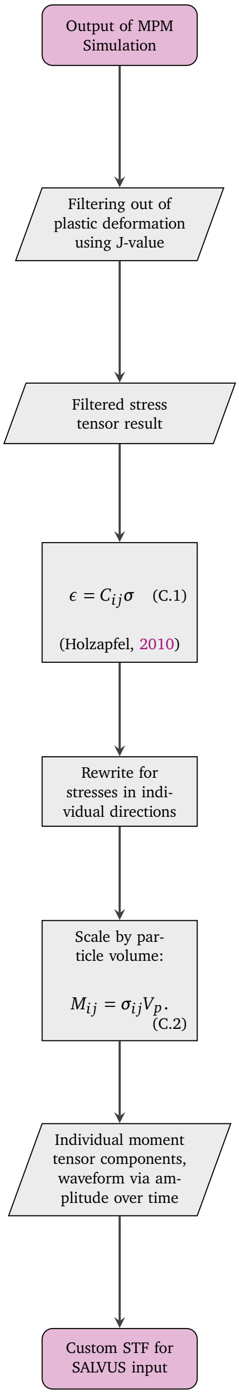


Part 3: Rupture Propagation and Source Definition



Appendix C

MPM conversion workflow



Appendix D

Parameters used in Models

Parameter	Typical value / source
σ_n	0.00184 MPa from Mahajan and Joshi, 2008
τ_s	0.004 MPa from Mahajan and Joshi, 2008
G_I, G_{II}	0.35, 0.50 J/m^s from Mahajan and Joshi, 2008
v_s, v_p, ρ	67.9366, 109.4 m/s, 250 kg/m³ from Trottet et al., 2022 (converted to μ, λ, E)
r_{mode}	User input (0 = Mode I, 1 = Mode II)

Parameter	Input Value
Maximum frequency to be resolved	300.0 Hz
Elements per wavelength	2
Side ABC Thickness	26.25m
Top ABC Thickness	1.0m
Amplitude to be tapered	600.0 Hz
Sampling Interval per Timesteps	50
Simulation Start Time	-0.2s
Simulation End Time	6.545

Table D.1: Simulation Constants used for Simple Simulation Setup.

Parameter from Trottet et al., 2022	Value
	45°
$\rho_{\text{slab}}, \rho_{\text{bed}}, \rho_{\text{wl}}$	250.0, 300.0, 100.0 kg/m^3
$E_{\text{slab}}, E_{\text{bed}}, E_{\text{wl}}$	3×10^6, 4×10^6, 1×10^6 Pa
ν	0.3 (Poisson's ratio)
Δx	0.01 m
Δt_{frame}	1/500 s
F	Deformation gradient tensor (F_{00} in xx, F_{11} in yy)
T	Stress tensor (T_{11} in xx, T_{22} in yy)
P	Particle position (x, y, z)
LogJP	Cumulative plastic damage (> 0)

Parameter	Input Value
Maximum frequency to be resolved	250.0 Hz
Elements per wavelength	4.5
Side ABC Thickness	75m
Top ABC Thickness	1.0m
Amplitude to be tapered	600.0 Hz
Sampling Interval per Timesteps	75
Simulation Start Time	-0.2s
Simulation End Time	6.545

Table D.2: Simulation Constants used for MPM Simulation Setup.

Appendix E

Wider Source Spacing Plots

The following figure shows particle velocity (vx and vy) for a sub-Rayleigh case with 10m source spacing. All other parameters apart from the spacing are the same as for the results shown in section 4.1.

The results for strain show a similar effect:

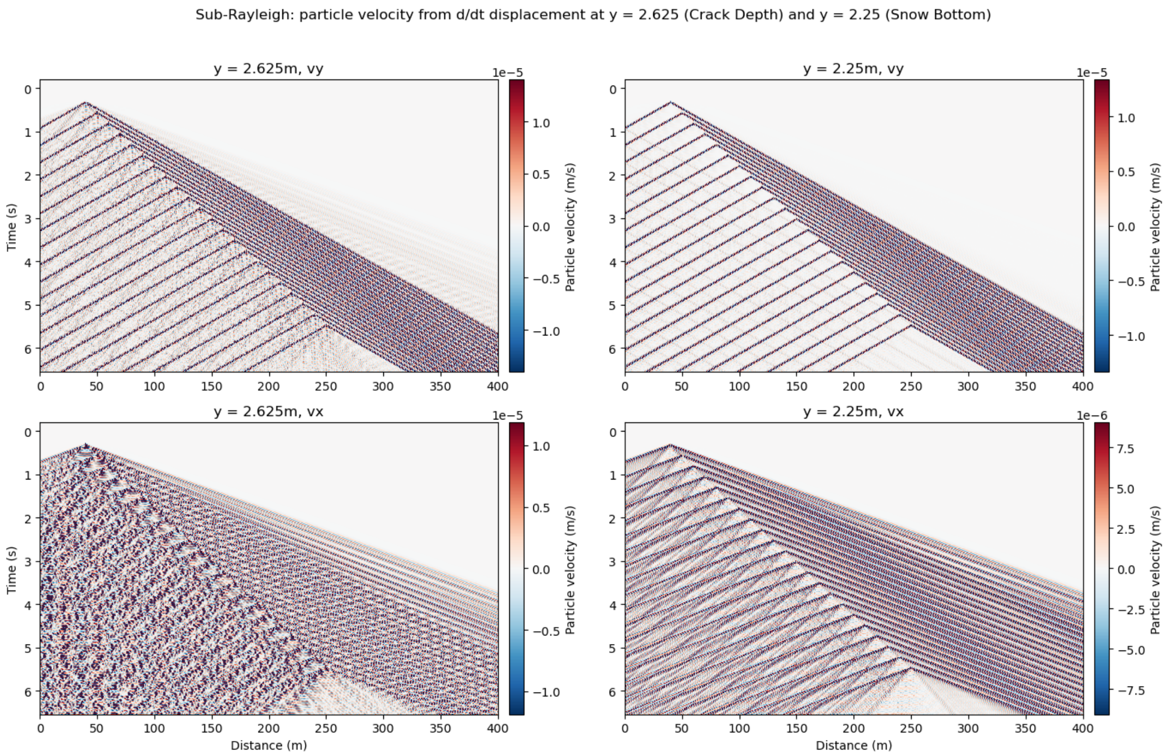


Figure E.1: Sub-Rayleigh particle velocity. Vertical velocity in top row, horizontal velocity in bottom row. Crack depth in left column, snow bottom in right column. Colorbar shows velocity (m/s). Source spacing is 10m.

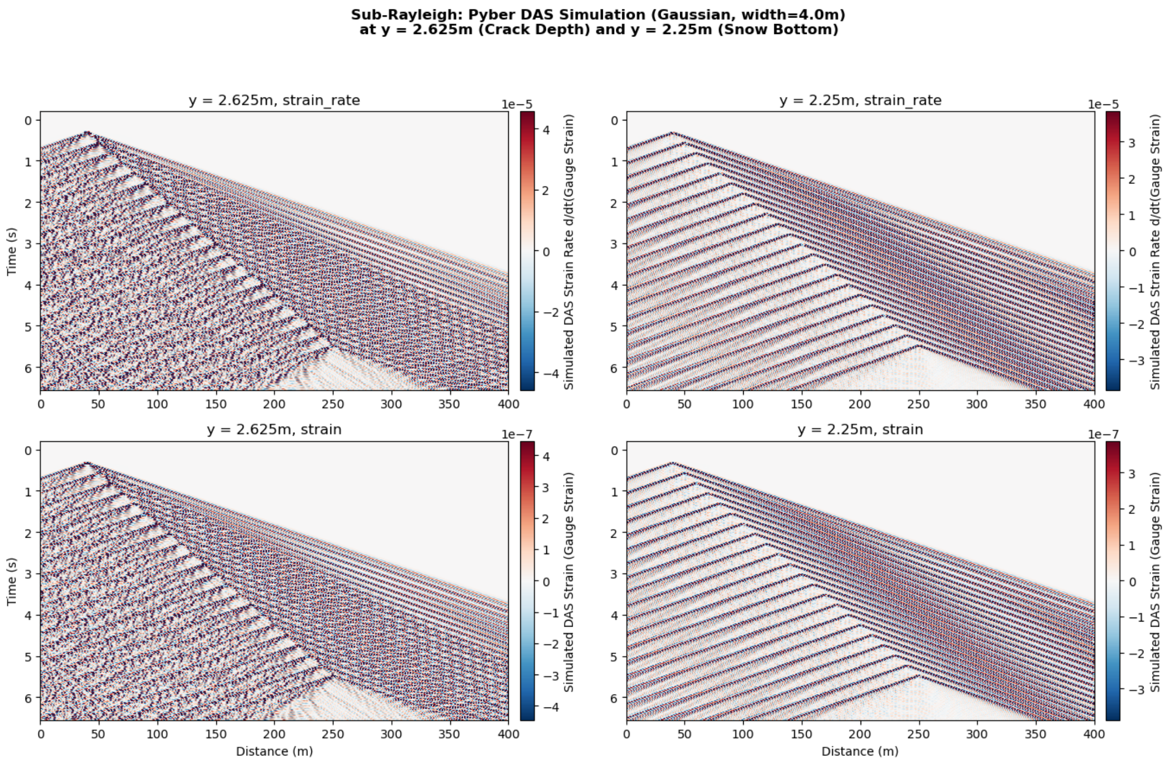


Figure E.2: Sub-Rayleigh strain and strain rate. Crack depth in left column, snow bottom in right column. Averaging over 4m gauge length, strain rate in top row, strain in bottom row. Source spacing is 10m.

Appendix F

Animations of wave propagation fields

F.0.1 Simple model

The following two animation snaphots show the velocity field for the sub-Rayleigh case in the simple model.

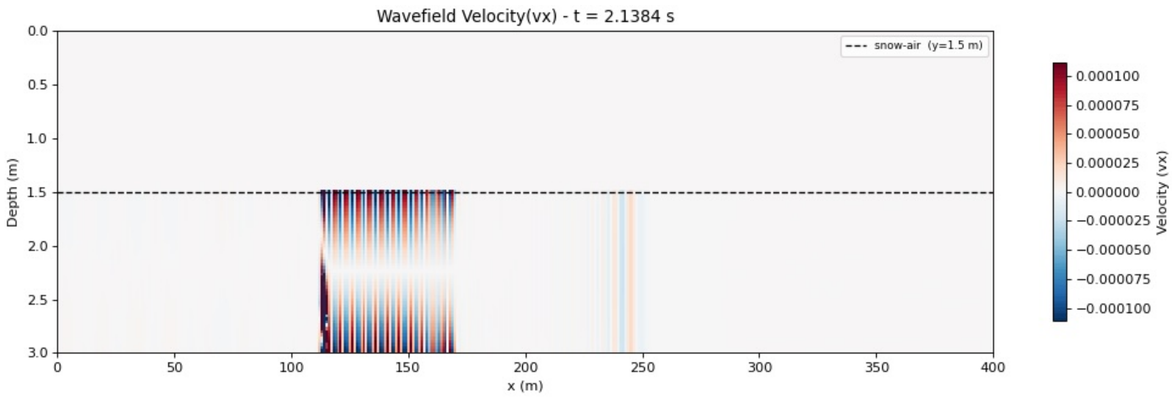


Figure F.1: Snapshot of the horizontal wavefield velocity component (v_x) evaluated at the crack location ($y = 2.625$ m).

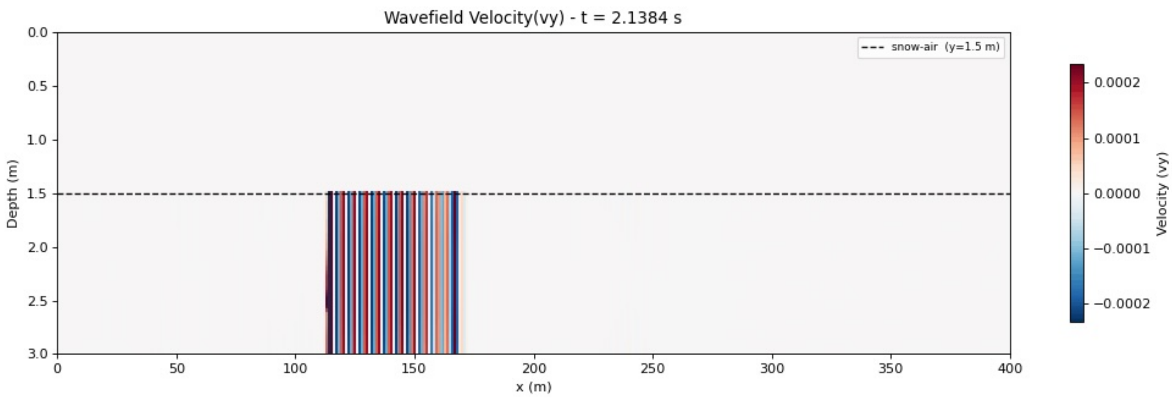


Figure F.2: Snapshot of the vertical wavefield velocity component (v_y) evaluated at the crack location ($y = 2.625$ m).

F.0.2 MPM-Guided Model

The stills of the animations of the velocity wavefield propagation for the MPM-guided simulations are shown below.

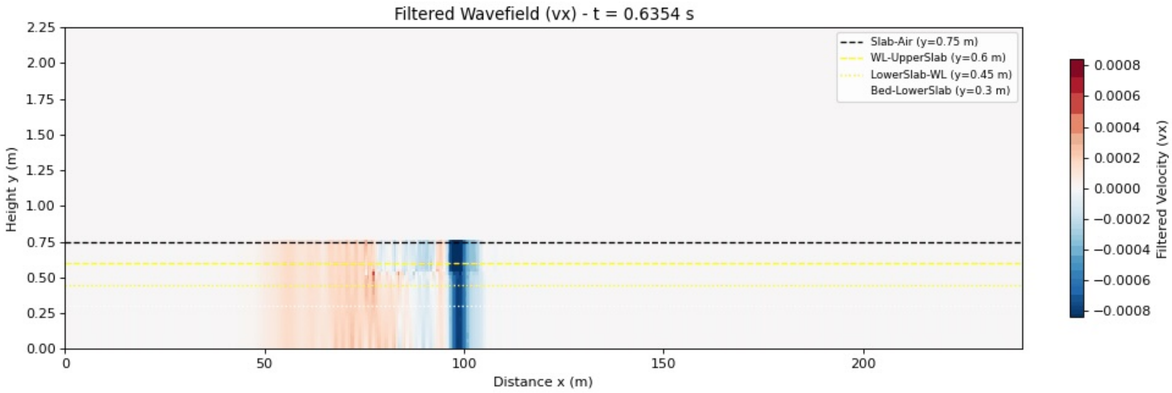


Figure F.3: Snapshot of the horizontal wavefield velocity component (v_x) for the MPM-guided model evaluated at the crack location ($y = 0.525$ m).

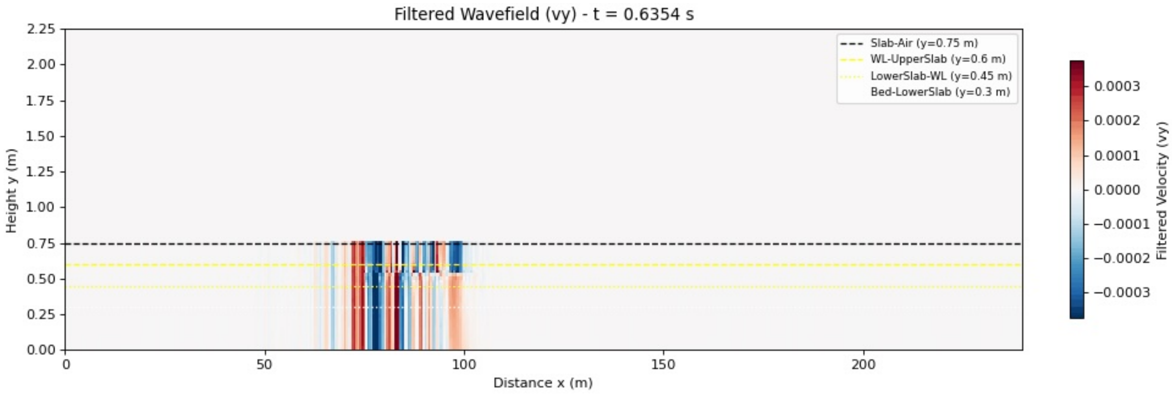


Figure F.4: Snapshot of the vertical wavefield velocity component (v_y) for the MPM-guided model evaluated at the crack location ($y = 0.525$ m).

The animations can also be found on the github repository in the folder Wave Propagation GIFS https://github.com/s-bachmann/Master-Thesis/tree/main/Wave%20Propagation%20GIFS.

Lamb wave propagation figure

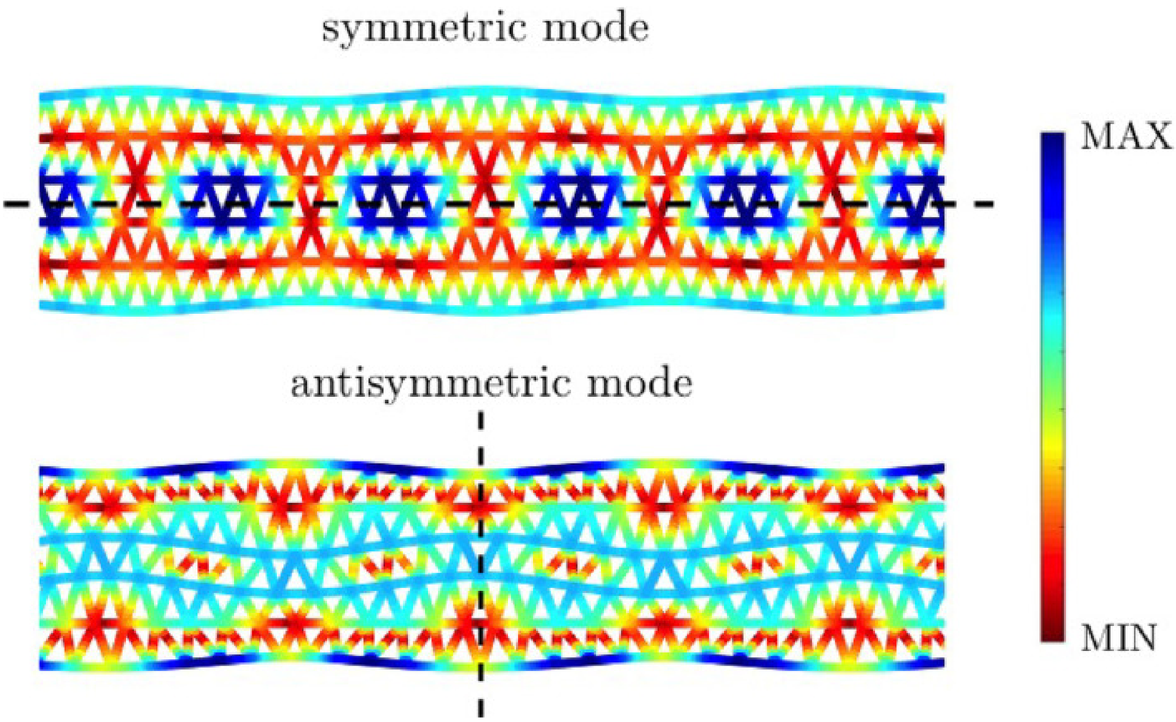


Figure G.1: Lamb wave propagation modes in plates. In the colour scale, blue (red) colour indicates larger (smaller) values of the total displacement field. Symmetric (S0) mode at top, antisymmetric (A0) mode at bottom. Figure taken from Carta et al., 2023.

Appendix H

Different DAS gauge length

H.0.1 Simple Model

The following figures show simulated DAS data of the simple model with a gauge length of 1m.

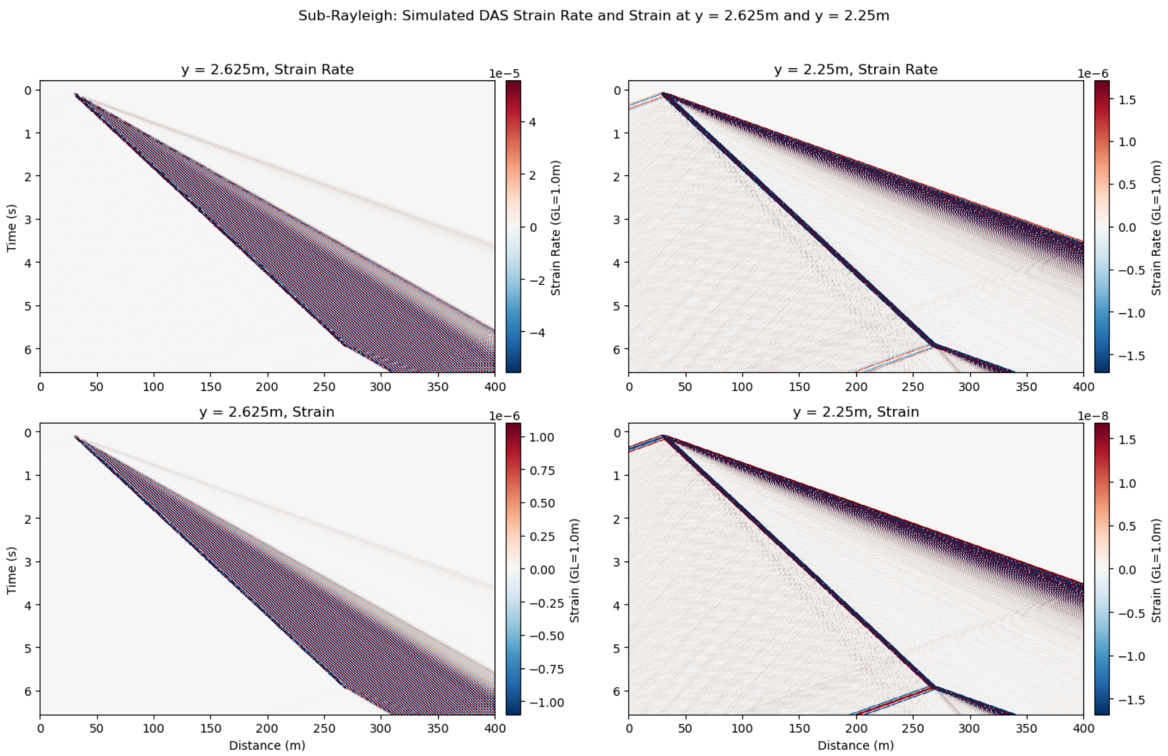


Figure H.1: Sub-Rayleigh strain and strain rate (horizontal component) averaged over 1 m gauge length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left side, snow bottom on right side. Colorbar shows magnitude of strain or strain rate, colorbars unscaled.

H.0.2 MPM-Guided Model

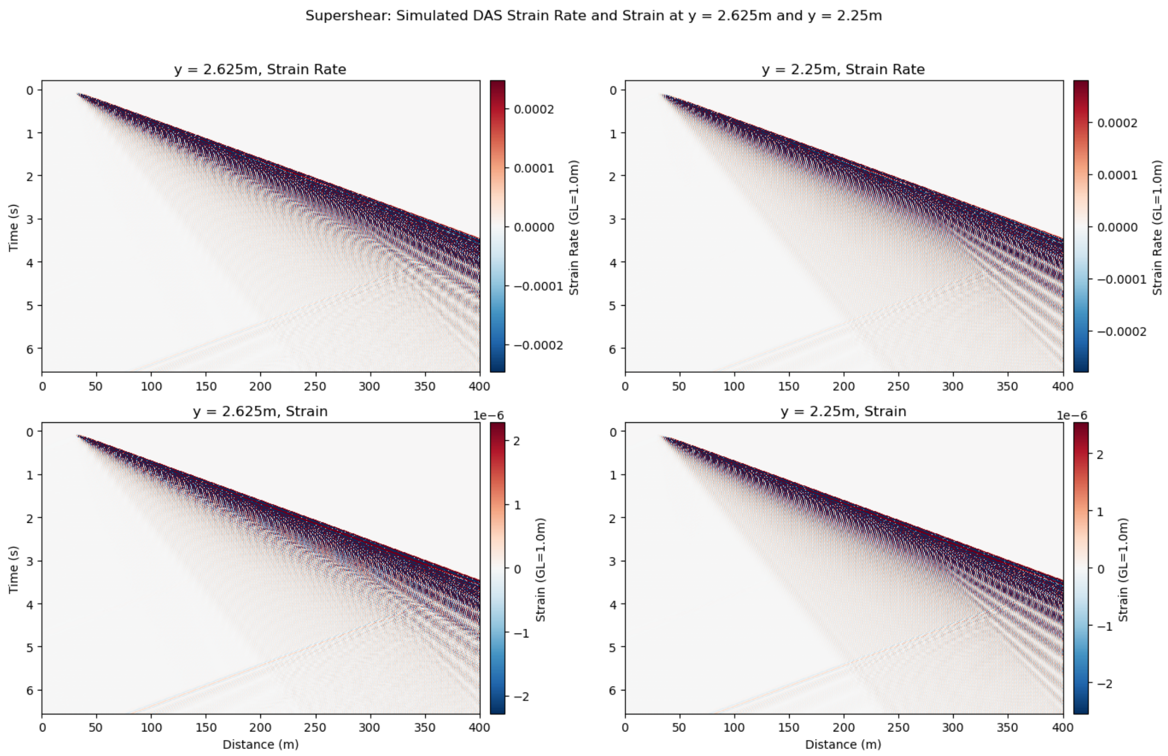


Figure H.2: Supershear strain and strain rate (horizontal component) averaged over 1 m gauge length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left side, snow bottom on right side. Colorbar shows magnitude of strain or strain rate, colorbars unscaled.

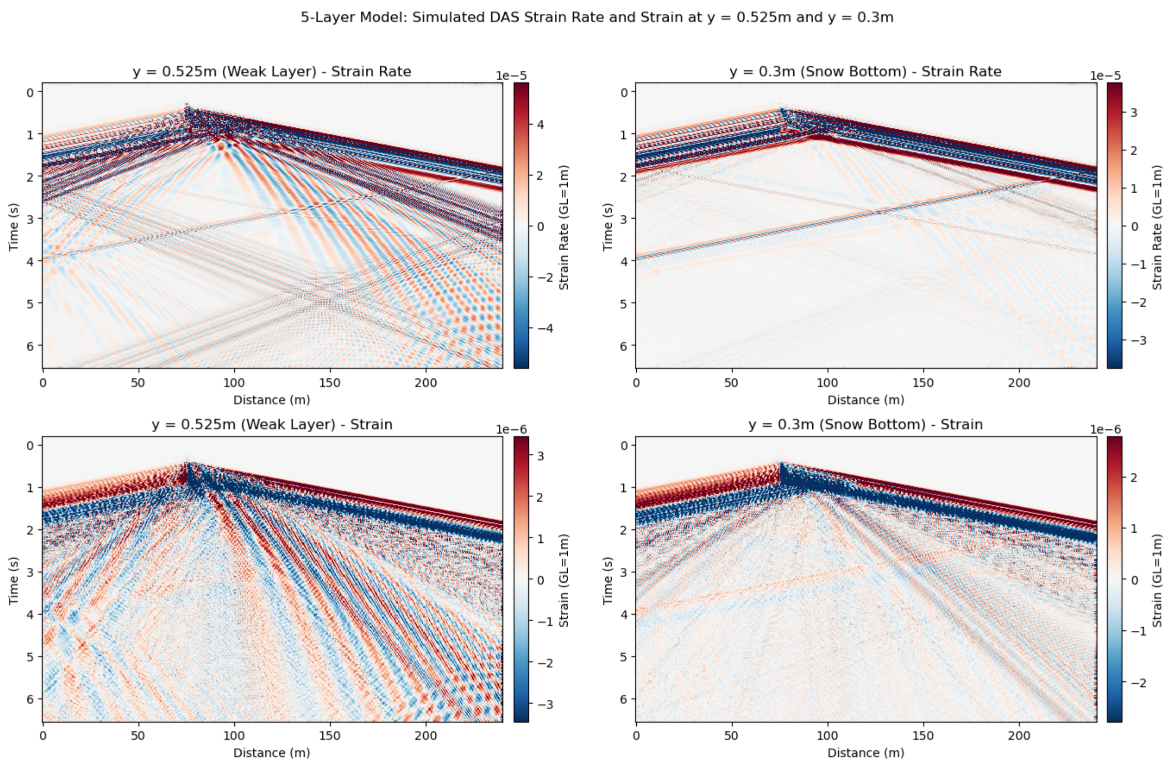


Figure H.3: MPM strain and strain rate (horizontal component) averaged over 1 m gauge length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left side, snow bottom on right side. Colorbar shows magnitude of strain or strain rate, colorbars unscaled.

Appendix I

MPM Displacement Plot

The following plot shows the displacement data extracted from the MPM simulation run. This data is shown to explain the noisiness of the velocity and strain data and to show that the output itself is not faulty.

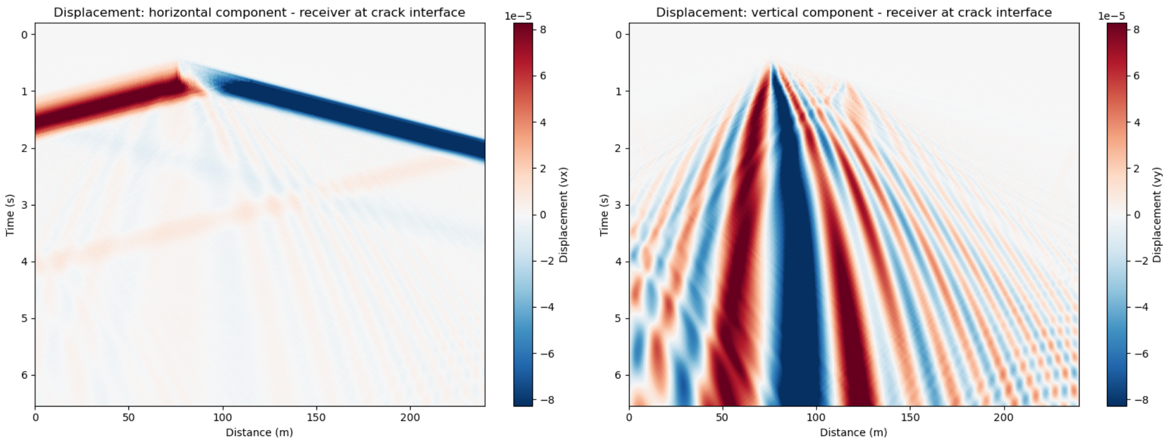


Figure I.1: Horizontal and vertical component displacement output from MPM-guided simulation, shown at crack interface. Colorbar shows magnitude of displacement.

Appendix J

Simple Source Setup in 5-Layer Model

The following figures show the simple sources (ricker wavelets with 0.1m spacing, sub-Rayleigh case) in the same 5-layer model shown in figure 3.2.

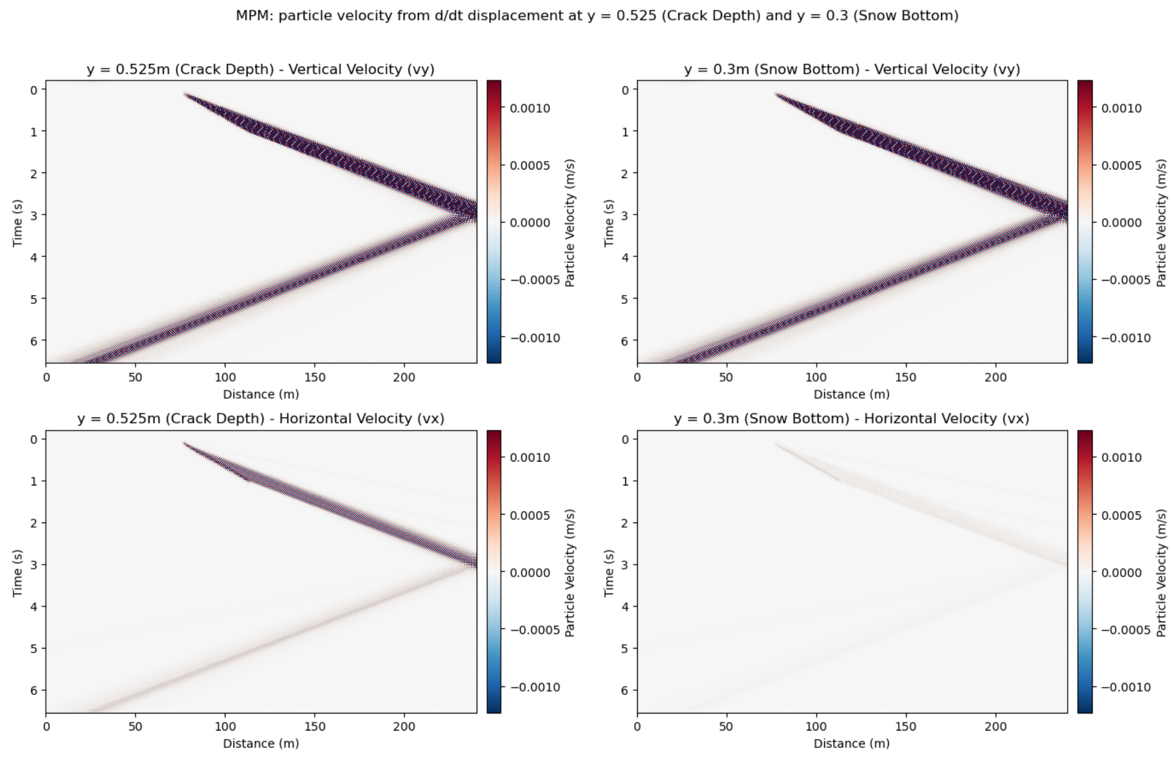


Figure J.1: Sub-Rayleigh velocity. Crack depth in left column, snow bottom in right column. Model used is 5-layer model, source spacing is 0.1m and STF is a Ricker wavelet. Colorbar shows velocity (m/s).

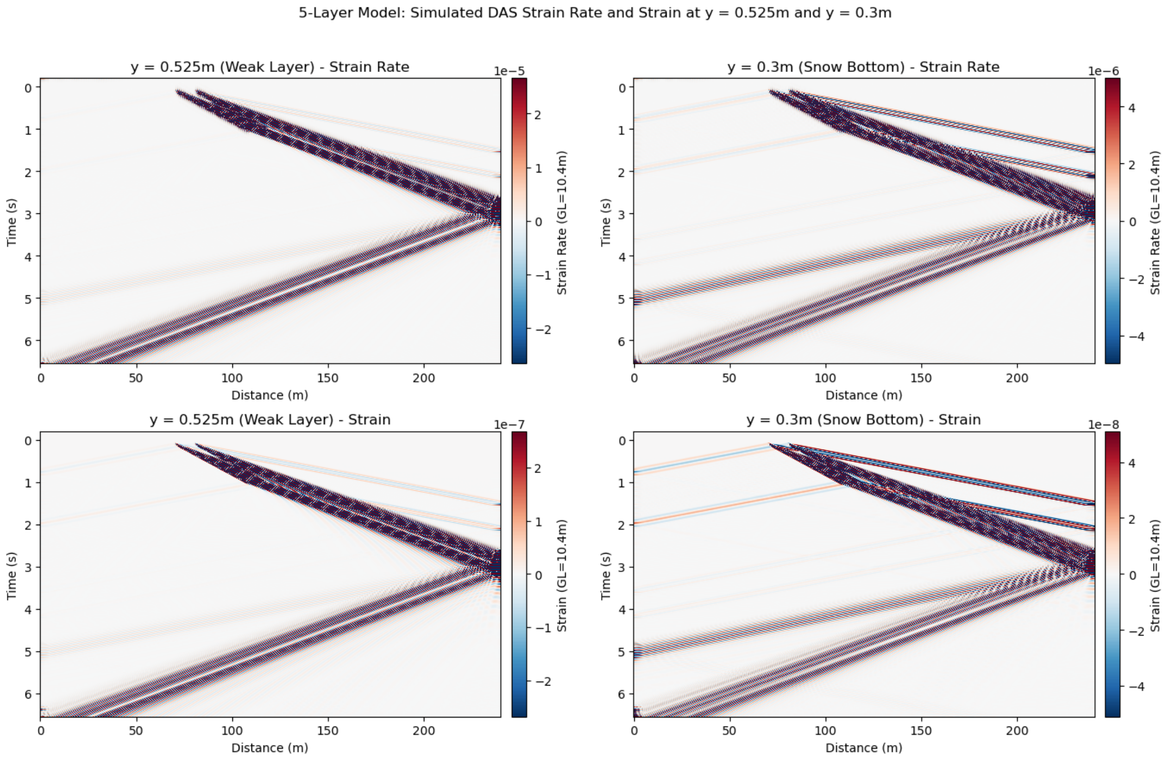


Figure J.2: Sub-Rayleigh strain and strain rate. Crack depth in left column, snow bottom in right column. Model used is 5 layer model, source spacing is 0.1m and stf is a ricker wavelet. Colorbar shows magnitude of strain or strain rate, colorbars unscaled.

Appendix K

Strain Traces

The following figures show strain and strain rate traces for the first 5 recievers after crack onset in all models. All traces are plotted at snow bottom, consistent with a cable buried in snow.

K.1 Simple Model

K.1.1 Sub-Rayleigh

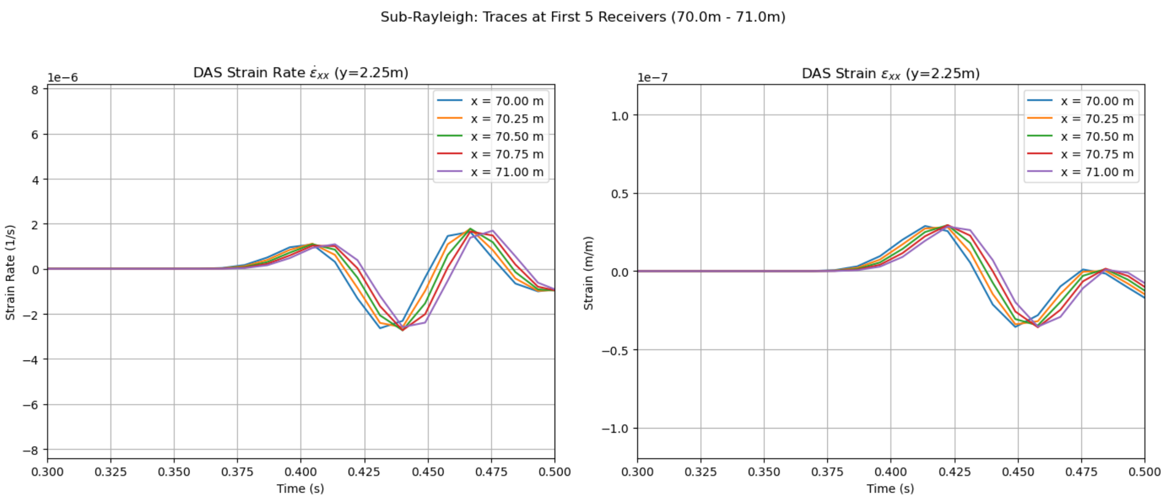


Figure K.1: Sub-Rayleigh strain and strain rate traces for the simple model at snow bottom. Strain rate on left, strain on right. X-axis zoomed to focus on first arrivals. Colors show location of traces.

K.1.2 Supershear

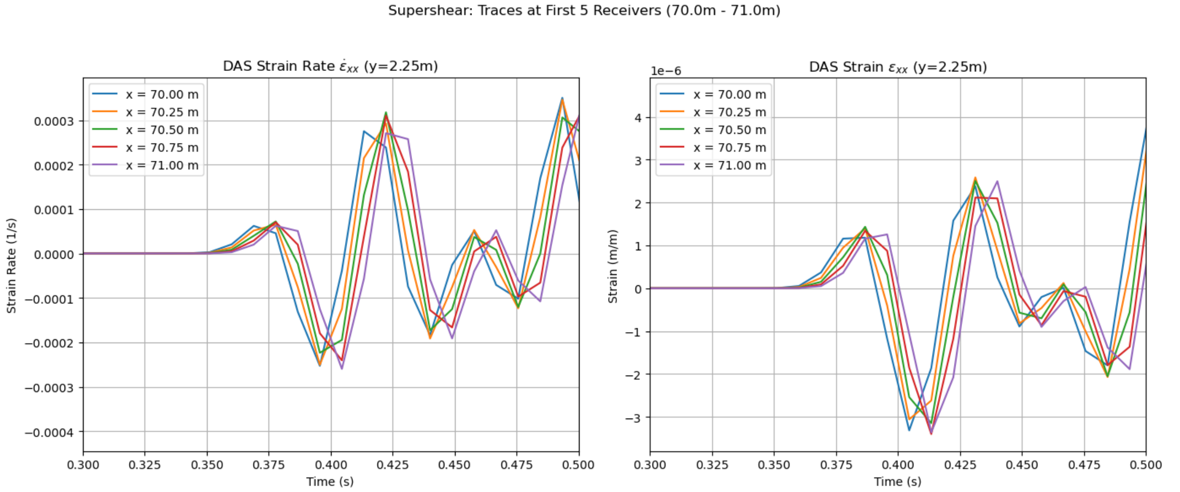


Figure K.2: Supershear strain and strain rate traces for first 5 locations in the simple model. Strain and strain rate traces for the simple model at snow bottom. Strain rate on left, strain on right. X-axis zoomed to focus on first arrivals. Colors show location of traces.

K.2 MPM-Guided

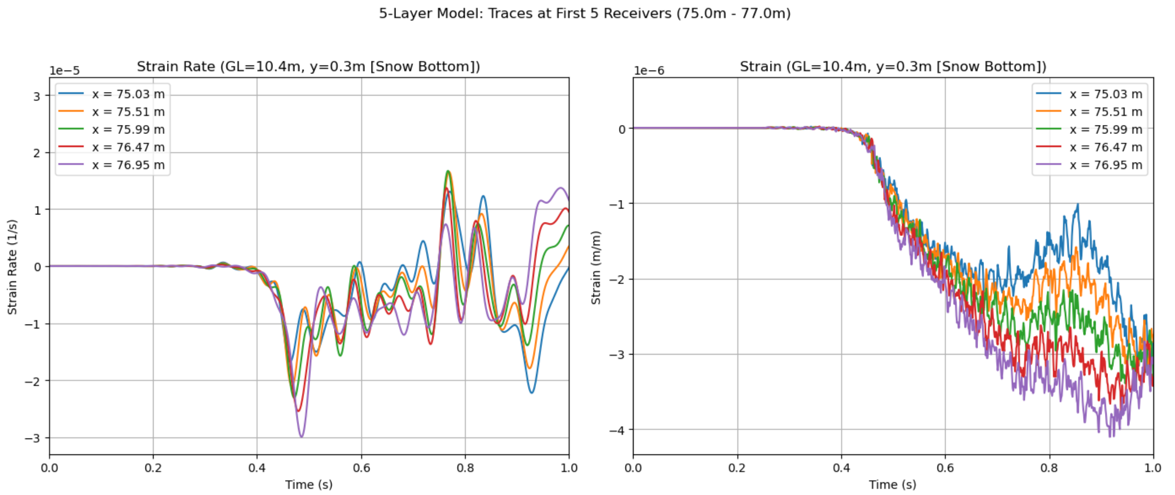


Figure K.3: MPM strain and strain rate traces for first 5 locations in the simple model. Strain and strain rate traces for the simple model at snow bottom. Strain rate on left, strain on right. X-axis zoomed to focus on first arrivals. Colors show location of traces.

Appendix L

Declaration of Orignality

All content and ideas in this thesis are original by me, unless cited otherwise.

GitHub Copilot was used to help with plotting code generation in visual studio code as well as for final spell checks in the LaTeXdocument. It was also used to help with LaTeXformatting and fixing compiling errors in the document.

Anthropic Claude Sonnet 5 was used to help proofread and clarify writing. All text written is original, AI was simply used to help identify passages that aren't written clearly or have grammar issues.

Tables Generator https://www.tablesgenerator.com was used to generate the table structures in this thesis.

Pyber Library Gauge Length Class by Noe, 2025 was used to simulate DAS data in the results. This can be found on github: https://gitlab.com/swp_ethz/public/pyber/-/blob/main/pyber/gauge_length.py?ref_type=heads.

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